

Pricing the Sensing Budget in ρ -POMDPs, with Expected Free Energy as the Canonical Information Weight

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Abstract. Information-gain utilities make sensing valuable directly, but introduce a weight that is usually tuned per task. An engineer may instead need to specify expected sensing usage. We calibrate the information weight of a ρ -POMDP policy family to a sensing target and identify the role of Expected Free Energy (EFE) within that family. We define an operational shadow price as a crossing threshold of the usage curve, distinguish it from a finite-grid bracket, and construct an endpoint mixture for targets inside usage gaps. We prove scale equivariance of the receding-horizon planner and stationary convergence of a projected controller under stated conditions. Minimizing the specified recursive, reward-based EFE objective is equivalent to maximizing reward plus information gain at $w = 1$ for observe-then-commit problems and hidden-state-preserving actions in a factored extension. The reward–information identification is exact under log scoring and otherwise depends on the declared reward convention. Usage curves collapse after rescaling, and the controller readapts empirically after a tenfold reward shift. Targets selected by a predeclared rule from calibration curves are met on held-out seeds to within 0.13 observations. Calibration can nevertheless sacrifice reward relative to spending less or choosing another mixture. Against the best mixture of sampled reference policies fitted to the same target on held-out seeds, nominal pointwise paired intervals show reward shortfalls on two of eleven distinct budgets. Replaying those fitted mixtures on fresh seeds gives two shortfalls again, and only Bandit’s gap-budget shortfall recurs. Larger gaps to a usage-capped reference mostly reflect the cost of spending the target. These intervals omit reference-selection uncertainty. The untuned $w = 1$ lies in the reward-maximizing run of sampled weights on three of five swept environments and is beaten on the other two. Usage calibration is appropriate when expected sensing usage is the design requirement. It does not maximize reward under a cap. EFE supplies an information-unit anchor under specified assumptions, while usage, reward, and success must be evaluated separately.

Keywords: Active inference · Expected free energy · ρ -POMDPs · Epistemic value · Belief-dependent utility · Partial observability · Information gain · Sensing budgets · Shadow prices

1 Introduction

An engineer designing an inspection policy may need to specify how much testing the system should perform before deciding whether to repair or release an item. A standard partially observable Markov decision process (POMDP) values a test through its effect on expected task reward. A ρ -POMDP additionally rewards properties of the agent’s belief, allowing information gain to have explicit value [2]. This introduces a practical choice: how much weight should information gain receive relative to reward?

We study a different design input, an expected sensing-usage target B . Such a target is useful when sensing is itself a deliverable, as in a coverage, monitoring, or verification requirement. It specifies the average number of tests or their expected cumulative cost. It is neither a hard limit on each episode nor an instruction to maximize reward subject to spending at most B . Those alternatives define different design problems, and our experiments measure the reward cost of meeting a target when reward alone would justify less sensing.

For a family of policies indexed by information weight w , let $U(w)$ be the expected sensing usage. We call the weight at which this curve crosses B its *operational shadow price*, $w^*(B)$. The adjective operational matters: this price is read from a usage curve, rather than obtained as the Lagrange multiplier of a usage-cap constraint. The curve can have plateaus, jumps, and non-monotone portions. A finite grid therefore estimates a crossing by a bracket. When the selected endpoint policies straddle a usage gap, a per-episode mixture can attain the target in expectation under the conditions of Definition 2. The crossing threshold, its grid bracket, and this randomized policy are distinct objects.

Figures 1 and 2 connect this construction to the engineer’s workflow. The engineer supplies a model, states an expected-usage requirement, estimates the usage curve offline, and checks usage and reward on held-out episodes. During deployment the agent chooses each test or terminal decision from its current belief. It does not perform a prescribed number of tests on every item.

The price has a useful response to a change of reward units. Multiplying all rewards and costs by $\alpha > 0$ and the information weight by the same factor preserves the receding-horizon planner’s decisions under the assumptions of Proposition 4. Observation counts are unchanged and cumulative sensing costs rescale by α . An online feedback rule can also adjust the weight when observed usage differs from the target. We prove stationary convergence for its projected, decaying-step form under explicit conditions and separately evaluate re-adaptation after a reward-scale shift.

Active inference identifies a particular point in this policy family. Its Expected Free Energy (EFE) objective combines pragmatic value with expected information gain [19, 42]. For the recursive, reward-based objective specified here, minimizing EFE is equivalent to maximizing reward plus information gain at $w = 1$. The reward–information identification is exact under log scoring [4]. For ordinary task rewards it depends on the declared reward convention and the treatment of the preference normalizer. The coefficient is therefore an untuned default under shared units, not a scale-invariant calibration. In particular, $w = 1$

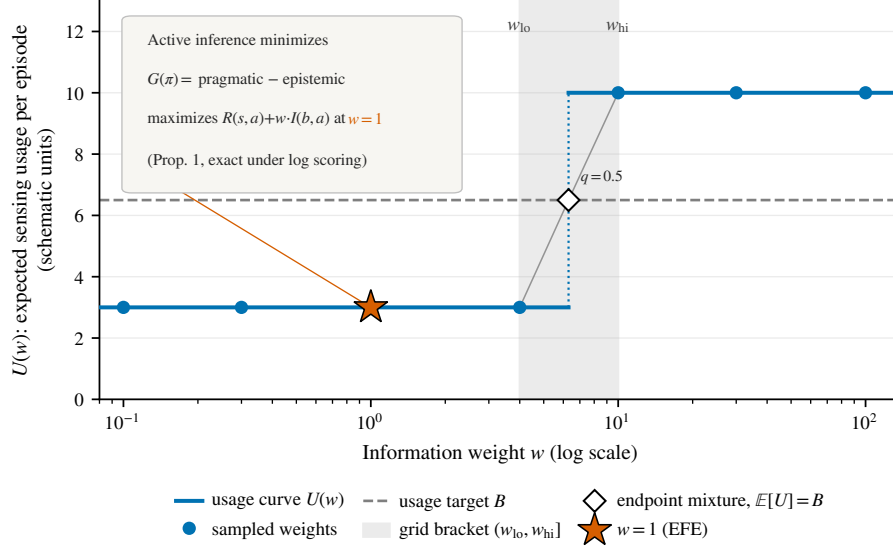


Fig. 1. An expected-usage target selects a crossing of the usage curve (schematic). Sampled grid weights bracket a jump over B . The shaded band estimates the crossing threshold, while the diamond represents a mixture of the two endpoint policies that attains B in expectation under Definition 2. Its horizontal position is schematic, not a third deterministic weight. The information-unit default $w = 1$ is one point on the curve and belongs to the level set of its own implicit budget. The equivalence shown in the inset uses the stated reward-based EFE objective, with exact reward identification under log scoring.

belongs to the level set of its implicit budget $U(1)$, but need not equal that budget’s crossing threshold.

The paper makes three contributions. First, it formulates expected-usage calibration within a ρ -POMDP policy family, distinguishing the level set, crossing threshold, grid estimate, and endpoint mixture. It establishes scale equivariance and a conditional stationary-convergence result for online control. Second, it proves the EFE equivalence for observe-then-commit problems and extends it to the hidden-state-preserving actions of factored observation POMDPs. A two-state analysis characterizes sensing onset and an interval of weights that avoids an unnecessary second observation. Third, it evaluates the calibration procedure and the information-unit default against reward-only planning, tuned information gain, POMCP, and near-optimal offline references. Code and experiment data are available at https://github.com/PatrickAllenCooper/rho_aif.

The evidence also sets limits on these contributions. Targets selected by a predeclared rule from calibration curves are met on held-out seeds to within 0.13 observations, but the selected mixtures can sacrifice reward relative to spending less or to the best sampled reference mixture meeting the same target, and need

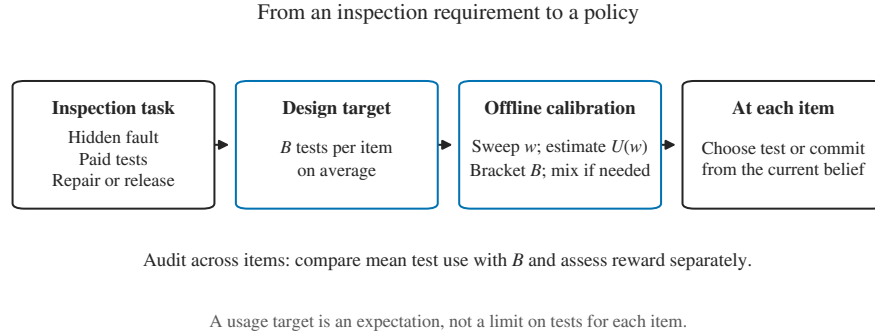


Fig. 2. From an inspection requirement to a running policy (schematic, not experimental data). For a state-preserving task with a hidden fault and paid tests, the engineer states a target B in expected tests per item. An offline sweep estimates usage $U(w)$ and brackets B . When a usage gap lies between the selected endpoint policies, mixing them across items can meet the target under Definition 2’s conditions. The deployed agent chooses tests and terminal actions from its current belief. Usage and reward are assessed separately: meeting B does not establish reward optimality under a sensing cap.

not be the best mixtures in the weight family. The untuned $w = 1$ lies in the reward-maximizing run of sampled weights on three of five swept environments and is beaten on the other two. Its advantage over reward-only planning depends on the observation structure and available planning depth, rather than on state-space size alone. Ordinary information gain can spend the budget on uncertainty unrelated to reward, and the stated equivalence does not cover observation actions that change the hidden state. These limitations remain part of the main argument, with full controls and counterexamples in the appendices.

An abridged version was accepted at IWAI 2026 [10]. It contains the EFE equivalence results and their benchmark evaluation. The budgeted formulation, scale-equivariance and controller results, budget experiments, and distractor and destructive-sensing analyses are extensions. We also revise the workshop version’s claim that the advantage grows with state-space size and observation-action count, which RockSample[11,11] does not support.

Section 3 defines the model and EFE objective. Section 4 develops usage calibration, and Section 5 specifies the evaluation protocol. Sections 6 and 7 report the budget and EFE results. Section 8 explains their practical scope. Appendices retain proofs, complete tables, robustness studies, and implementation details.

2 Related Work

POMDP planning and belief-dependent rewards. POMDPs formalize decision-making under state uncertainty [29, 57]. Point-based offline solvers approximate the value function on reachable beliefs, including PBVI, HSVI, and SARSOP [32, 45, 54, 58]. Online solvers plan from the current belief. POMCP uses Monte Carlo tree search with UCB1 action selection and rollout evaluation [55], DESPOT searches regularized sparse belief trees [68], and POMCPOW uses progressive widening for continuous spaces [61]. Value of Information Monte Carlo planning selectively skips observation branches with low expected decision value [34]. Our planner instead adds expected information gain to the objective. These are different uses of information within planning, and our comparisons do not establish superiority over state-of-the-art POMDP solvers generally.

Araya-López et al. [2] introduced ρ -POMDPs with belief-dependent rewards. Their ρ denotes the whole reward, whereas we reserve it for an information term added to the ordinary task reward. Convex belief rewards support convex value functions and piecewise-linear convex approximations. Expected information gain is concave in the belief, so that machinery does not apply directly. Fehr et al. [15] treat non-convex rewards under a Lipschitz hypothesis, and Benchetrit et al. [5] develop an anytime planner for continuous spaces. For a finite observation model, strictly positive likelihoods make expected information gain Lipschitz on the belief simplex. Zero likelihoods, including noiseless sensing, can violate this condition. Our exact tree recursion does not rely on that approximation guarantee.

POMDPs with information rewards were developed for active cooperative perception [59], and their relation to ρ -POMDPs is discussed by Satsangi et al. [51]. Earlier preference-elicitation work selected queries by sequential value of information while retaining state-dependent rewards [7]. These approaches establish the motivation for valuing information explicitly. Our focus is the coefficient on that value and its relation to expected sensing usage.

Information value and experimental design. Classical value of information measures the improvement an observation can bring to a decision [28]. Bayesian experimental design instead commonly selects observations by expected information gain [16, 36]. The criteria coincide under a particular reward model: when the reward is a proper log score, the expected improvement in score is the expected information gain [4]. This supplies the exact reward–information identification used here. Sequential experimental design is also the closest structural analogue to our observe-then-commit setting.

Information-directed sampling (IDS) trades instantaneous regret against information gain through their ratio, rather than through a fixed additive coefficient [48]. We include an observe-then-commit IDS baseline in Appendix N. Walraven et al. [66] connect active inference with information gathering in POMDPs. Our contribution within this setting is to define and evaluate a calibration procedure for the weight in a specified ρ -POMDP policy family.

Active inference and the EFE formulation. Active inference casts perception and action as variational inference [18, 43]. EFE combines pragmatic and epistemic terms [11, 19, 42]. Its recursive sophisticated-inference form conditions future choices on future beliefs [20]. Da Costa et al. [12] prove reward-Bellman optimality for finite-horizon recursive inference in the limit of precise reward-encoding preferences, with exact state estimation. That limiting result does not imply that a fixed finite information weight is reward-optimal.

Several distinct constructions occur under the EFE name. Millidge et al. [40] distinguish EFE from a naive future extension of variational free energy and propose Free Energy of the Expected Future (FEEF). Champion et al. [9] examine which EFE expressions follow from common root definitions and the restrictions those roots impose on prior preferences. de Vries et al. [13] derive EFE-based planning from a model augmented with preference and epistemic priors. The generalised free energy of Parr and Friston [42] is another distinct functional. Our equivalence concerns only the recursive, reward-based objective defined in Section 3, with expected information gain about the hidden state as its epistemic term. It does not assert an unchanged coefficient for these alternative constructions. Broader comparisons with decision-theoretic objectives include an EFE optimality-gap analysis [67] and a survey of formal equivalences [62].

Scaling work combines active inference with deep models, reinforcement-learning objectives, and Monte Carlo tree search [17, 37, 65]. Our MCTS-EFE variant follows this direction. Its controls test exploration constants, rollout choices, and two internal components, while leaving exact beliefs, exact commit values, and EFE leaf evaluation coupled (Appendix S).

Exploration and control as inference. Control-as-inference relates maximum-entropy control to probabilistic inference [35, 63], with soft actor-critic and KL-minimizing control as algorithmic instances [24, 46]. Its treatment of value differs from active inference [39]. EFE can produce intrinsic-motivation behavior [50], while curiosity, count-based exploration, VIME, and random network distillation use weighted exploration bonuses [3, 8, 27, 44, 52]. Oudeyer and Kaplan [41] survey these intrinsic signals. Bayes-adaptive planning and Bayesian reinforcement learning primarily address uncertainty about the model [14, 21, 23]. We study uncertainty about state under a known model, and retain dependence on a shared reward convention when describing $w = 1$ as untuned.

Budgets and dual control. Rational inattention constrains or prices information capacity [38, 56]. Constrained MDPs use Lagrange multipliers on expected resource costs [1], and constrained POMDP methods track admissible cost alongside belief [30]. Automatic temperature tuning in soft actor-critic drives policy entropy toward a target [25]. These provide close conceptual precedents for resource control. Our weight multiplies information gain, however, while the target concerns observation count or cost. It is not generally the multiplier of a sensing-cap constraint. Section 4.6 gives the detailed distinctions, including the difference from a dual formulation of EFE itself [31]. The contribution is an operational

calibration for this policy family, its exact scale transformation under stated assumptions, and explicit treatment of finite-grid crossings and usage gaps.

3 Methodology

We first specify the policy family whose information weight will be calibrated to a sensing target. The EFE connection identifies one member of that family under explicit reward and information-unit conventions. It does not establish reward optimality.

3.1 The ρ -POMDP Framework

A ρ -POMDP adds a belief-dependent utility $\rho : \Delta(S) \times \mathcal{A} \rightarrow \mathbb{R}$ to task reward, where $\Delta(S)$ is the probability simplex over hidden states S . Its objective is

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^H \gamma^t (R(s_t, a_t) + \rho(b_t, a_t)) \right]. \quad (1)$$

Here S is the finite hidden-state space, $\mathcal{A}$ the action set, $b_t \in \Delta(S)$ the Bayesian belief, H the planning horizon, and γ the discount factor. The expectation runs over the initial state, state transitions, observations, and resulting belief updates under policy π . Setting $\rho = 0$ recovers a standard POMDP. Choosing information gain as ρ explicitly rewards uncertainty reduction. The equivalence below concerns $\gamma = 1$.

In the observe-then-commit class, $\mathcal{A}$ partitions into observation actions $\mathcal{A}_{\text{obs}}$, which leave the hidden state fixed and update the belief at known cost, and terminal commit actions $\mathcal{A}_{\text{com}}$, which yield state-dependent reward and emit no observation. Rewards are not observations: beliefs are conditioned on observation symbols alone. Each episode consists of observations followed by one commit. This structure isolates the information-weight question and permits analytic thresholds and near-optimal offline references. Section 3.3 extends the reduction to interleaved sensing and navigation. Appendix U.1 includes a schematic of the basic decision loop.

3.2 Expected Free Energy as ρ

At future time τ , the EFE formulation considered here combines expected negative log-preference with negative expected information gain,

$$\mathcal{G}(\pi) = -\mathbb{E}_{Q(o_{\tau}|\pi)}[\ln P(o_{\tau} | C)] - \mathbb{E}_{Q(o_{\tau}|\pi)}[D_{\text{KL}}[Q(s_{\tau} | o_{\tau}, \pi) \| Q(s_{\tau} | \pi)]] . \quad (2)$$

$Q(o_{\tau} | \pi)$ predicts observations under policy π , while $Q(s_{\tau} | \pi)$ and $Q(s_{\tau} | o_{\tau}, \pi)$ are the corresponding prior and posterior beliefs. The parameters C specify the preference distribution $P(\cdot | C)$, which is distinct from the observation model conditioned on state and action. Minimization favors preferred outcomes and

informative observations. Following the recursive, closed-loop formulation associated with sophisticated inference [12, 20], we evaluate actions from the current belief and replan after every observation.

We identify the pragmatic term with negative task reward under the reward-based convention detailed below. A terminal commit has value $\mathcal{G}(\text{commit}_i \mid b) = -\mathbb{E}_b[R_i]$, where $R_i = R(s, \text{commit}_i)$. It has neither an epistemic term nor a continuation term, because it emits no observation and ends the episode. An observation action k costs $c_k > 0$ and supplies expected information gain

$$I_k(b) = H(b) - \mathbb{E}_{o \mid \text{obs}_k}[H(b'_o)],$$

where b'_o is the Bayesian posterior and $H(b) = -\sum_s b(s) \log b(s)$ is entropy. The entropy functional $H(\cdot)$ is distinct from the planning horizon H . The recursion is

$$\mathcal{G}(\text{observe}_k \mid b) = c_k - I_k(b) + \mathbb{E}_o \left[\min_a \mathcal{G}(a \mid b'_o) \right]. \quad (3)$$

At depth H , only commit actions are available. The agent selects a minimizing action using a fixed tie-breaking rule. As policy precision $\kappa \rightarrow \infty$, the softmax $Q(\pi) \propto \exp(-\kappa \mathcal{G}(\pi))$ concentrates on policies minimizing $\mathcal{G}$. The implementation makes a deterministic selection among tied minimizers. Policy precision κ is distinct from the reward inverse temperature β below. A separately normalized preference distribution does not generally give this recursion on variable-length episodes, because its omitted normalizer can affect action choice.

3.3 Formal Equivalence with ρ -POMDPs

Writing d for search depth and setting $V = -\mathcal{G}$ gives

$$V(\text{observe}_k, b, d) = -c_k + I_k(b) + \mathbb{E}_o \left[\max_a V(a, b'_o, d+1) \right], \quad (4)$$

$$V(\text{commit}_i, b, d) = \mathbb{E}_b[R_i]. \quad (5)$$

At $d = H$, the value is the best commit value. Write $R(b, a) := \sum_s b(s) R(s, a)$ for expected immediate reward and $V^*(b, d) := \max_a V(a, b, d)$ for the optimal belief-state value. This is a ρ -POMDP Bellman recursion with an action-dependent belief utility, as permitted by Araya-López et al. [2].

Proposition 1. *Define $\rho_{\text{EFE}}(b, a)$ as*

$$\rho_{\text{EFE}}(b, a) = \begin{cases} I_a(b) & \text{if } a \text{ is an observation action} \\ 0 & \text{if } a \text{ is a commit action} \end{cases}$$

where $I_a(b) = H(b) - \mathbb{E}_{o \mid a}[H(b'_o)]$ is the expected information gain from observation action a at belief b . Assume (i) pragmatic value is identified with task reward at $\beta=1$ per reward unit, (ii) beliefs are updated by exact Bayesian conditioning, (iii) information is measured in the same units on both terms, and (iv) both recursions select actions deterministically with the same fixed tie-breaking

rule. Commit actions carry $\rho_{\text{EFE}}=0$ not by stipulation but because the episode terminates on commitment, so no further observation is emitted and the episodic term has no successor belief to score. Then for undiscounted finite horizon ($\gamma=1$), minimizing recursive EFE (Eq. 3) over horizon H produces the same policy as solving the ρ -POMDP Bellman recursion

$$V^*(b, d) = \max_a \left\{ R(b, a) + \rho_{\text{EFE}}(b, a) + \mathbb{E}_o[V^*(b'_o, d + 1)] \right\}$$

over the same horizon, with terminal commit actions carrying no continuation term and only commit actions available at $d = H$.

Negating the objective turns each EFE minimization into the corresponding Bellman maximization. Appendix A gives the proof. This is a notational bridge, not a new optimality theorem. It identifies the specified recursive EFE agent with Planning+IG at $w = 1$ under the stated convention. Unlike the $\beta \rightarrow \infty$ reward-dominant limit of Da Costa et al. [12], it retains information gain at full weight. The claim covers the information-gain and pragmatic-value formulation among the four that Champion et al. [9] relate, evaluated recursively. It does not establish the same coefficient for other formulations of free energy. Appendix U.1 records those distinctions. Because information gain is concave in the belief, the convexity and PWLC results of Araya-López et al. [2] do not apply. The non-convex guarantee of Fehr et al. [15] requires Lipschitz rewards. Expected information gain meets that hypothesis for strictly positive observation likelihoods, but need not when likelihoods contain zeros. Our exact enumeration does not rely on that guarantee.

Reward and information units. The identification in assumption (i) fixes an exchange rate rather than deriving one. With $P(o | C) \propto \exp(\beta R(o))$,

$$\ln P(o | C) = \beta R(o) - \ln Z, \quad Z = \sum_{o'} \exp(\beta R(o')).$$

Setting $\beta = 1$ assigns one nat to each reward unit. A path scoring T outcomes also accumulates $-T \ln Z$ in log-preference. Since observe-then-commit paths have different lengths, this term can be policy-dependent. It vanishes for a log-scoring reward with $Z = 1$ [4], or cancels when every path scores the same number of outcomes under a common preference distribution. We do not impose that fixed-length construction. Proposition 1 therefore proves equivalence for the reward-based recursion as written, conditional on its pragmatic identification, rather than deriving it from general normalized preferences.

The empirical convention is different again: the implementation computes entropy in bits and runs the reported EFE agent at one reward unit per bit. Since one bit is $\ln 2$ nats, implemented $w = 1$ corresponds to $1/\ln 2 \approx 1.44$ reward units per nat. The exactly nat-canonical agent uses $w = \ln 2 \approx 0.69$ in implementation units. Appendix R compares these directly. Their reward,

success, and observation count are bit-identical on Tiger, Diagnosis, Bandit, and Testbed. Tileworld changes policy, and $w = 1$ remains Pareto-dominated there. Thus the exact log-score case, the conditional recursion equivalence, and the empirical bit convention are separate claims.

The weight is also reward-scale dependent. Multiplying all rewards and costs by $k > 0$ at fixed w is behaviorally equivalent to using w/k before rescaling. The reward-maximizing weight scales with k (Appendix Q). Proposition 4 gives the corresponding usage identities. Whether $w = 1$ is a useful default at a fixed scale remains an empirical question.

For the simplest two-state model, the following thresholds distinguish under-observation from paying for an instrumentally unprofitable second observation. The two parts concern separate horizons.

Proposition 2. *Consider a two-state observe-then-commit p -POMDP with uniform prior $b(s_0)=b(s_1)=\frac{1}{2}$, a single observation action (accuracy $\frac{1}{2} < p \leq 1$ in Part (a) and $\frac{1}{2} < p < 1$ in Part (b), cost $c > 0$), and two commit actions (correct reward $R^+ > 0$, incorrect penalty $R^- < 0$ with $|R^-| > R^+$). Define the reward asymmetry ratio $\alpha = |R^-|/R^+$ and the informativeness ratio $\eta = I_{\max}/c$. Here $I_{\max} = \ln 2 - H_{\text{post}}(p)$ is the maximum expected information gain in nats, and $H_{\text{post}}(p) = -p \ln p - (1-p) \ln(1-p)$ is the entropy in nats of the post-observation belief $(p, 1-p)$. Thus $I_{\max}$ is the uniform prior's entropy $\ln 2$ less the entropy remaining after one observation of accuracy p . This proposition states two separate results at two separate horizons. Part (a) is a statement about $H=1$, a single decision to observe once or not and then commit, and Part (b) is a statement about $H=2$ (whether a second observation is also worthwhile). They share notation but are logically independent claims, proved separately in Appendix U.2.*

Part (a) ($H=1$ lower threshold). *The minimum weight w_{thresh}^* above which the Planning+IG objective, expected reward plus w times expected information gain, prefers observing once to committing immediately is*

$$w_{\text{thresh}}^* = \frac{c - (p - \frac{1}{2})(R^+ - R^-)}{I_{\max}} = \frac{c - (p - \frac{1}{2})(1 + \alpha)R^+}{I_{\max}}$$

For any $w > w_{\text{thresh}}^$ the agent observes before committing at $H=1$. The threshold w_{thresh}^* is negative, making $w=1$ sufficient for at least one observation, whenever $\alpha > c/[(p - \frac{1}{2})R^+] - 1$. In particular, $w_{\text{thresh}}^* \rightarrow -\infty$ as $\alpha \rightarrow \infty$ for any fixed $p > \frac{1}{2}$ and $c > 0$.*

Part (b) ($H=2$ near-optimality interval). *At the post-first-observation belief $b=(p, 1-p)$, define the instrumental value of a second observation $\text{VoI}_2(b)$ and the expected information gain $I(b)$ of one further observation. The upper (over-observation) threshold is*

$$w_{\text{over}}^* = \begin{cases} (c - \text{VoI}_2(b))/I(b) & \text{if } \text{VoI}_2(b) < c, \\ +\infty & \text{otherwise.} \end{cases}$$

At $H=2$ for this two-state case, when $w_{\text{thresh}}^ \leq 0$, so that the first observation is instrumentally justified, every $w \in (w_{\text{thresh}}^*, w_{\text{over}}^*) \cap [0, \infty)$ reproduces the*

instrumentally optimal single observation, and we call this set the near-optimality interval. When $\text{VoI}_2(b) \geq c$ a second observation is instrumentally justified at any weight, so no over-observation threshold exists and the interval is unbounded above. In this uniform-prior symmetric two-state model $\text{VoI}_2(b) = 0$ identically, since the disconfirming second observation returns the belief exactly to the tie point, so the finite branch always applies and all four rows of Table 31 fall in it. The interpretation depends on the sign of w_{thresh}^* . When $w_{\text{thresh}}^* \leq 0$, weights inside the interval reproduce the instrumentally optimal single observation, while weights above a finite w_{over}^* buy a second observation whose instrumental VoI is below cost. When $w_{\text{thresh}}^* > 0$ committing immediately is reward-optimal, weights below the threshold are exactly reward-optimal, and weights inside the interval buy one instrumentally unprofitable observation at a reward loss of $c - (p - \frac{1}{2})(R^+ - R^-)$, positive in this regime and bounded above by the observation cost c . Part (b) presupposes Part (a)’s threshold as its lower endpoint but is otherwise a distinct claim. Perfect sensing, $p=1$, is excluded from Part (b), since a second observation then carries no information, $I(b)=0$, and the over-observation ratio is undefined, while Part (a) holds at $p=1$ with $H_{\text{post}}(1) = 0$ under the convention $0 \ln 0 = 0$, so that $I_{\text{max}} = \ln 2$.

The proof and descriptive threshold comparisons are in Appendix U.2. Only Tiger among those comparisons satisfies the proposition as stated. Diagnosis and Tileworld use unproved two-state reductions, and Testbed lies at the boundary $|R^-| = R^+$. The $H = 2$ interval is not a general guarantee for longer receding-horizon episodes. Discounting preserves the algebraic reduction when the same discount is applied to both recursions. Its behavioral effects are reported in Appendix K.

Extension to factored observation POMDPs. The reduction extends to interleaved sensing and navigation when both preserve the hidden state.

Definition 1 (Factored observation POMDP). *A POMDP is a factored observation POMDP if its state decomposes as $s = (s_{\text{vis}}, s_{\text{hid}})$ where s_{vis} is fully observable and s_{hid} is hidden, and the action set partitions into (i) observation actions $\mathcal{A}_{\text{obs}}$ that produce observations about s_{hid} without changing s_{hid} (though they may change s_{vis}), (ii) navigation actions $\mathcal{A}_{\text{nav}}$ that change s_{vis} deterministically without changing s_{hid} and produce no informative observation, and (iii) exploitation actions $\mathcal{A}_{\text{exp}}$ that yield reward that may depend on s_{hid} and emit no informative observation. Throughout, rewards are not observations. The belief is conditioned on the observation symbol alone, so an exploitation action’s payoff, however informative about s_{hid} it would be to an agent that conditioned on it, does not update the belief.*

Observe-then-commit is the special case with no navigation and a trivial visible state. Our RockSample variant [58] holds rock qualities fixed, makes position and sampling status visible, and classifies checks, moves, and sample/exit actions as observation, navigation, and exploitation respectively. The original

RockSample changes a good rock to bad on sampling. Our variant differs in that respect.

To state the boundary precisely, write s, s' for the hidden components, b for their belief, and O for the observation likelihood. Let

$$b'_o(s) \propto O(o \mid s, a)b(s), \quad b'_{o,T}(s') \propto \sum_s T_{\text{hid}}(s' \mid s, a)O(o \mid s, a)b(s).$$

Here the observation precedes the transition, and the known visible state is suppressed. Interpret b'_o on the post-transition state space through the identity coupling $s' = s$. The transition–observation diagnostic is

$$\Delta_T(b, a) := \mathbb{E}_o D_{\text{KL}}[b'_{o,T} \parallel b'_o].$$

It compares the correct posterior with the one obtained by assuming preservation.

Proposition 3. *In a factored observation POMDP (Definition 1), let b denote the belief over s_{hid} . For any action a that preserves s_{hid} (i.e., $a \in \mathcal{A}_{\text{obs}} \cup \mathcal{A}_{\text{nav}}$), the transition–observation coupling term vanishes, $\Delta_T(b, a) = 0$, and $\rho_{\text{EFE}}(b, a) = I_a(b)$ for observation actions, $\rho_{\text{EFE}}(b, a) = 0$ for navigation actions.*

Under preservation, the two posteriors coincide, which proves the claim (Appendix U.3). Terminal exploitation has no continuation or epistemic term. Non-terminal exploitation carries no information gain here because its observation is uninformative and its payoff is not used to update the belief. A model that conditions on exploitation rewards must include the resulting information gain. Proposition 3 covers observation and navigation actions, not arbitrary exploitation transitions.

State preservation is an exact hypothesis. Slow drift alone does not ensure a small diagnostic: a perfect sensor gives a point-mass posterior, and an arbitrarily small probability of a state change can place mass outside its support, making Δ_T infinite. Example 1 shows a complete action-ranking reversal from using obsolete information.

Example 1 (Destructive sensing can make naive information gain actively harmful). Let $s \in \{0, 1\}$ with uniform prior $b = (0.5, 0.5)$ and the two-state commit reward $r(\text{Commit}_i, s) = +1$ if $i = s$, -1 otherwise. Add one destructive test action a_D with cost $c > 0$ that perfectly reveals the pre-transition state, $O(o=s \mid s, a_D) = 1$, but always leaves the hidden state depleted afterward, $T(s'=0 \mid s, a_D) = 1$ for every s . Drilling a core sample destroys whatever ore was there, rich or poor.

The state-preserving assumption behind Propositions 1–3 treats a_D as if it left s untouched, giving the naive posterior $b'_o \propto O(o \mid s)b(s)$ (a point mass at the revealed value o) and the naive value

$$V_{\text{naive}}(a_D) = -c + w I(b) + \mathbb{E}_o \left[\max_i r(\text{Commit}_i, o) \right] = -c + w \cdot 1 \text{ bit} + 1,$$

which at $w=1$ is $2 - c$, exceeding $V_{\text{naive}}(\text{Commit}_i) = \mathbb{E}_b[r(\text{Commit}_i, s)] = 0$ for any $c < 2$.

The correct joint transition-observation posterior, $b'_{o,T}(s') \propto \sum_s T(s' | s, a_D) O(o | s, a_D) b(s)$, evaluates to $b'_{o,T} = (1, 0)$ for either observed outcome. The observation describes ore that drilling has already destroyed, so the correct post-action belief is certain the state is now 0 no matter what was observed. The coupling term is not merely nonzero but maximal here. The posteriors $b'_{o,T}$ and b'_o are disjoint point masses whenever $o=1$, so $D_{\text{KL}}[b'_{o,T} \| b'_o] = \infty$. The correct Bellman value is

$$V_{\text{true}}(a_D) = -c + \max_i r(\text{Commit}_i, 0) = 1 - c.$$

For $w = 1$ reward unit per bit, this is one reward unit below the naive estimate, for either outcome.

A transition-aware EFE evaluation, one that uses the correct joint posterior $b'_{o,T}$ above, assigns the drill zero epistemic value about the post-drill state and recovers V_{true} .

For $c \in (1, 2)$ the two evaluations disagree in sign. $V_{\text{naive}}(a_D) = 2 - c > 0 = V_{\text{naive}}(\text{Commit}_i)$ recommends drilling, while $V_{\text{true}}(a_D) = 1 - c < 0 = V_{\text{true}}(\text{Commit}_i)$ recommends committing immediately. That is a full ranking flip from a two-state, one-step example. The failure is worse than a mispriced tradeoff. An agent that drills and then acts on its now-obsolete observation realizes expected reward $-c + 0.5 r(\text{Commit}_0, 0) + 0.5 r(\text{Commit}_1, 0) = -c$ in the true environment, strictly below the immediate-commit value of 0 for every $c > 0$, not only within the sign-flip range. Verified numerically in `tests/test_destructive_boundary.py`.

Figure 3 isolates the timing distinction behind Example 1.

This example limits the state-preserving reduction, not transition-aware EFE. Recomputing predictive and posterior information about the post-transition state removes its illusory credit. We have not established Δ_T as an additive value correction or a decision-error bound, and extending the budget machinery to state-changing sensing remains open. Appendix U.3 gives the proof of Proposition 3 and the fuller scope discussion.

3.4 Agents

The six core agents in Table 1 share a known generative model and exact Bayesian updates. Planning, Planning+IG, and EFE use the same recursive search, so the comparison separates the objective from planning depth. Every $H > 1$ agent replans a full depth- H tree after each real observation and executes only its root action.

Planning+IG is the central baseline and represents the familiar information-reward construction in POMDP-IR and ρ -POMDPs [2, 51, 59]. By Proposition 1, its difference from EFE is the chosen weight. The `pymdp` check in Appendix H tests qualitative consistency of the epistemic term, not certified numerical equivalence [26].

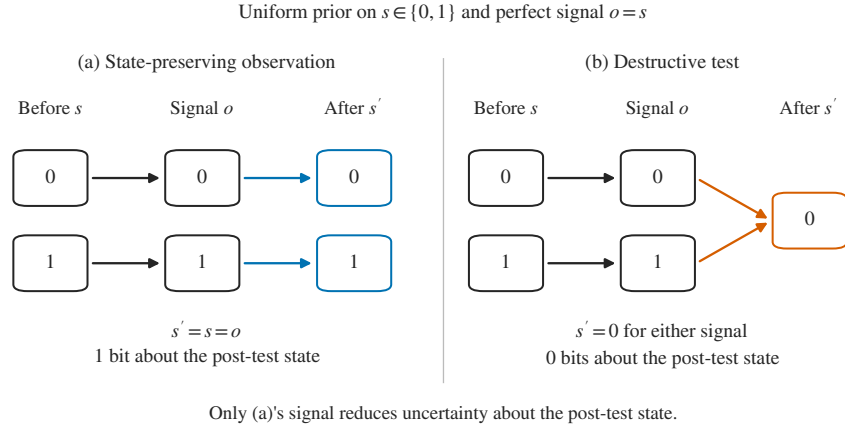


Fig. 3. An observation can describe a state that no longer exists. These schematic timelines use a perfect sensor to contrast state-preserving sensing with the destructive test of Example 1. The sensor reports the pre-action state in both cases. Under preservation that report identifies the state used by the next decision. Under destruction both possible initial states become 0, so the correct post-action belief is $(1, 0)$ for either report. Treating a report of 1 as the current state gives an obsolete posterior. The example limits the state-preserving reduction, not transition-aware EFE.

Epistemic-only sets commit EFE to zero, removing reward from commit timing while retaining sensing costs and information gain. On commitment it chooses the most probable state. Myopic has no information bonus but can still observe when one-step lookahead reveals positive instrumental value of information. Additional baselines are Greedy without sensing, information-directed sampling (Appendix N), and a posterior-vote rule. The last draws 100 states and takes the majority of their individually optimal actions. It is not Thompson sampling, although the committed CSVs label it `ThompsonSamplingAgent`. Appendix U.4 gives the full specifications and ablation interpretation.

Info Gain and Planning+IG use success-maximizing weights w_{succ}^* in the main comparison tables, selected from $\{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100\}$. Each candidate uses a common tuning stream at seed 7, reset between candidates, with 200 episodes (100 on Tileworld). Ties favor the smallest maximizing weight. This stream is disjoint from the five evaluation seeds, and the selected weights retain its sampling noise. Seed 7 also appears in the separate frontier battery, which evaluates no tuned baseline. The Pareto study instead reports reward-maximizing weights w_{ret}^* from its wider eleven-point grid (Section 7.2). Neither is the usage-calibrated price $w^*(B)$ defined next.

Table 1. Agent specifications. All use exact Bayesian belief updates over a known generative model.

Agent	ρ function	Horizon	Controls for
Myopic	$\rho = 0$	$H=1$	Weakest baseline
Planning	$\rho = 0$	$H > 1$	Planning depth
Info Gain	$w \cdot I(b)$	$H=1$	Epistemic bonus (myopic)
Planning+IG	$w \cdot I(b)$	$H > 1$	IG + planning depth
EFE	$I_a(b)$ via EFE	$H > 1$	Joint objective (Prop. 1)
Epistemic-only	$I_a(b)$ only	$H > 1$	Ablation: no pragmatic term

4 Budgeted ρ -POMDPs and the Price of Information

We now choose an information weight from a target for expected sensing usage. If sensing already has a known economic cost in reward units, that cost belongs directly in R . A usage target is useful when the reward scale is conventional or the design requirement is naturally stated as an average number or cumulative cost of tests. It controls sensing effort. It does not by itself maximize reward.

4.1 The Budgeted Problem and the Usage Curve

Let $R(\pi)$ and $U(\pi)$ denote expected episodic return and sensing usage. The calibration problem asks for a member of the receding-horizon Planning+IG family, or a per-episode mixture of two members, whose expected usage equals B . The separate constrained problem is

$$\max_{\pi} R(\pi) \quad \text{subject to} \quad U(\pi) \leq B. \quad (6)$$

The policy class may be the fixed family Π or the broader class used by a constrained-POMDP reference. We use those references to measure the reward cost of calibration (Section 6.8). Their cap may be slack at the reward optimum.

The literal Lagrangian of (6) penalizes usage through $R(\pi) - \lambda U(\pi)$, $\lambda \geq 0$. Planning+IG instead uses $R(\pi) + wI(\pi)$, where $I(\pi)$ is expected cumulative information gain. This is an information-floor relaxation. For $w \geq 0$, an exact maximizer π_w over a fixed class solves $\max R(\pi)$ subject to $I(\pi) \geq I(\pi_w)$: optimality gives

$$R(\pi) \leq R(\pi_w) + w(I(\pi_w) - I(\pi)) \leq R(\pi_w)$$

for every floor-feasible π . This statement concerns a one-shot exact maximizer, not the implemented receding-horizon agent.

For that implemented family, let π_w^* denote the closed-loop policy using weight w and define its usage curve by $U(w) := U(\pi_w^*)$. The star here denotes the policy produced by the finite-horizon planner, not an optimizer of total episodic return. Its operational shadow price $w^*(B)$ locates a crossing of the target by this curve. It is not the usage-cap multiplier $\lambda^*(B)$, and meeting the target neither

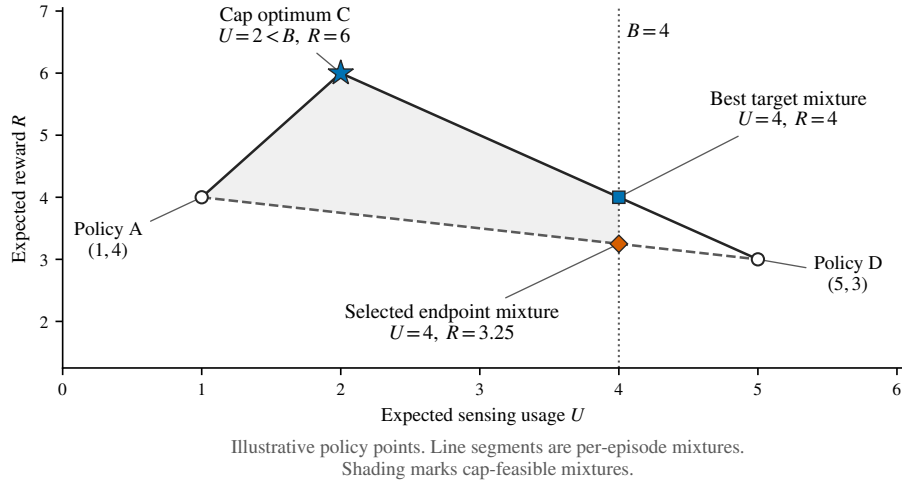


Fig. 4. An expected-usage target and a usage cap answer different questions. This schematic uses three illustrative policies and their per-episode mixtures, not benchmark data. The target $U = B$ is the vertical line, while the shaded portion of the mixture hull satisfies the cap $U \leq B$. Suppose increasing grid weights select policies C, A, then D, so A and D form the selected crossing pair. A selected endpoint mixture can meet the target yet earn less than another mixture at the same usage. The highest-return cap-feasible policy can also leave part of the budget unused. These possibilities explain why calibration alone supplies neither an equality-constrained nor a cap-constrained reward optimum. They do not assert that every instance has either shortfall.

solves (6) nor guarantees the best reward among mixtures at the same usage. Figure 4 distinguishes these possibilities.

Discrete policy changes produce usage gaps, so a single weight need not attain B . The curve may also be nonmonotone. We estimate it on a finite grid and use the following four distinct objects to avoid treating a grid estimate as an exact inverse.

Definition 2 (Operational shadow price, crossing bracket, and endpoint mixture). *Fix the usage curve $U(w)$ of the implemented family on $w \geq 0$ and a budget B . Four objects are involved, and they are distinct. First, the level set $U^{-1}(B) = \{w \geq 0 : U(w) = B\}$ is the set of weights at which usage equals the budget exactly. It is empty when B falls inside a usage gap, and it can be a nondegenerate set when U is flat at level B . Second, for a budget that the curve crosses from below, the crossing threshold $w^\dagger(B)$ is the single weight at which the curve last passes from below B to at or above B , namely the supremum of the weights w with $U(w) < B$ that precede a weight with usage at or above B . It is undefined for a budget the curve never crosses from below, whether because the curve sits at or above B at every weight, as Tiger’s does at $B=4$ in Table 33, or because it never reaches B , and the solver then reports a non-straddling fallback flagged as unbracketed. Under the local non-monotonicity documented in Section 4.3 the curve can cross B more than once, and taking the last crossing is a convention adopted here, not a property of the problem. At a jump of the curve the threshold is a single number, the location of the jump, even though the level set is empty there. The operational shadow price of this article is this threshold, $w^*(B) := w^\dagger(B)$, defined wherever the threshold is. Third, the crossing bracket $(w_{lo}, w_{hi}]$ is the finite-grid estimate of the threshold that the released solver reports (`rho_aif/budget.py`). Its lower endpoint is the largest grid weight with $U(w_{lo}) < B$ and its upper endpoint is the first subsequent grid weight with $U(w_{hi}) \geq B$. The bracket therefore straddles the budget on the grid whenever a grid point at or above B follows the last grid weight with usage below B . Its closed enclosure $[w_{lo}, w_{hi}]$ contains the threshold whenever the curve has no crossing of B that the grid fails to resolve, between grid weights or beyond the last one. A curve whose final grid point falls back below B would instead yield a non-straddling fallback flagged as such even though its threshold can exist. No swept curve exhibits that shape. The enclosure is stated closed because the threshold can coincide with w_{lo} itself, when the curve rises immediately after that grid weight, and the half-open reported bracket excludes that one point by construction. The width of the bracket is grid resolution. The weights inside it do not each attain the budget, and the bracket is not a set of exact prices. Offline solving additionally accepts the nearest grid point as achieving the budget when $|U(w) - B| \leq \max(0.3, 0.1B)$, a numerical tolerance that affects achievability calls only, never the bracket. Fourth, the endpoint mixture is the randomized policy that, when B falls inside a usage gap so that no single weight attains it, plays the policy at w_{hi} with probability $q = (B - U(w_{lo})) / (U(w_{hi}) - U(w_{lo}))$ and the policy at w_{lo} otherwise, choosing afresh each episode. By linearity its expected usage is exactly B . This is one randomized policy attaining the bud-*

get in expectation, not an interval of scalar prices, and it mirrors the classical constrained-MDP fact that optimal budget-constrained policies may need to be randomized [1, 30].

A budget is attainable by some per-episode mixture of two members of the family iff $\inf_{w \geq 0} U(w) \leq B \leq \sup_{w \geq 0} U(w)$, up to the closure of the attainable usage set, since mixing two members whose usages straddle B attains it in expectation. The endpoint mixture of the fourth object attains it only when the budget is straddled by a last crossing from below, that is, when the solver returns a straddling bracket, which holds at every budget for which this article reports a mixture, while the unbracketed cases (Tiger at $B=4$ in Table 33 and the declared-unattainable budgets of Section 6.8) are reported as such. A curve that only descends through B , with usage 2 below some weight and 1 above it and $B=1.5$, is attainable by mixing the two plateaus but has no crossing from below, so its crossing bracket and endpoint mixture are undefined and the solver flags the fallback. The lower endpoint need not be $U(0)$. Usage is not monotone in w , and the implemented family can spend less at a positive weight than at $w=0$ (on Tileworld 6×6 , reward-only Planning at $w=0$ scans 15.64 times per episode while $w=1$ scans 14.75, Section 7.3). Within the suite, $U(0) > 0$ reflects instrumental sensing (observing pays for itself on reward alone at some instances) and the upper endpoint reflects belief saturation and the episode step cap.

4.2 Exact Scale Equivariance

Let an α -rescaled environment multiply every pragmatic quantity, including observation costs, by $\alpha > 0$, while leaving dynamics, observations, and belief updates unchanged. Scaling w by the same factor then multiplies every candidate value by α .

Proposition 4 (Exact scale equivariance). *For every $\alpha > 0$ and $w \geq 0$, the receding-horizon Planning+IG agent facing the α -scaled environment with weight αw selects the same action as the agent facing the unscaled environment with weight w , at every belief, provided ties are broken by a scale-independent rule. Consequently the closed-loop policies coincide, and so do the induced distributions of every quantity that depends on the actions alone. For usage denominated in observation count this gives*

$$U_{\text{count},\alpha}(\alpha w) = U_{\text{count}}(w), \quad (7)$$

and for usage denominated in cumulative sensing cost, which sums the rescaled costs of the same actions, it gives $U_{\text{cost},\alpha}(\alpha w) = \alpha U_{\text{cost}}(w)$.

Proof. For any candidate plan, the α -scaled value is $\alpha \cdot (\text{pragmatic sum}) + \alpha w \cdot (\text{IG sum}) = \alpha \cdot [\text{pragmatic sum} + w \cdot \text{IG sum}]$, i.e. exactly α times the unscaled value. Multiplying every candidate's value by the same positive constant preserves the ordering of the candidates, and the scale-independent rule compares values and a tolerance that have both been multiplied by α , so it returns the same action. By induction over decision points, the closed-loop policies agree at

every belief. Since the environment’s stochastic dynamics are otherwise identical, the induced trajectory distributions agree, and with them the distribution of the observation count. Each action’s cost is α times its unscaled cost, so the cumulative sensing cost of every trajectory is α times its unscaled value, which gives the cost identity.

The implementation’s comparison tolerance is relative, 10^{-12} times the larger candidate magnitude, and favors the incumbent commit action. This rule scales with value. An absolute tolerance would not. Floating-point rounding need not be exactly homogeneous, so bit-identical collapse in the experiments is a measurement rather than a mathematical guarantee.

For count usage, plotting against w/α gives the same curve. Cost usage must also be divided by α . Wherever the threshold exists,

$$w_{\text{count}}^*(B; \alpha) = \alpha w_{\text{count}}^*(B; 1), \quad w_{\text{cost}}^*(B; \alpha) = \alpha w_{\text{cost}}^*(B/\alpha; 1).$$

The grid bracket rescales only when the grid itself is multiplied by α . A fixed $w = 1$ at scale α instead behaves as $w = 1/\alpha$ at scale 1, with implicit count budget $B_{\text{EFE}}(\alpha) := U(1/\alpha)$. Equivariance applies equally to a directly tuned weight on the correspondingly rescaled grid.

4.3 Monotone Comparative Statics and the Usage Staircase

Scale equivariance does not imply that usage increases with w . The general monotonicity result concerns expected information gain under exact maximization over a fixed plan set.

Proposition 5 (Monotone comparative statics for exact maximizers).

Let Π be any fixed set of plans and, for $w \geq 0$, let $\pi_w \in \arg \max_{\pi \in \Pi} R(\pi) + wI(\pi)$. Then for $w_2 > w_1 \geq 0$,

$$I(\pi_{w_2}) \geq I(\pi_{w_1}) \quad \text{and} \quad R(\pi_{w_2}) \leq R(\pi_{w_1}). \quad (8)$$

Proof. Optimality of π_{w_1} at w_1 gives $R(\pi_{w_1}) + w_1 I(\pi_{w_1}) \geq R(\pi_{w_2}) + w_1 I(\pi_{w_2})$, and optimality of π_{w_2} at w_2 gives $R(\pi_{w_2}) + w_2 I(\pi_{w_2}) \geq R(\pi_{w_1}) + w_2 I(\pi_{w_1})$. Adding the two inequalities and simplifying yields $(w_2 - w_1)(I(\pi_{w_2}) - I(\pi_{w_1})) \geq 0$, and since $w_2 > w_1$, $I(\pi_{w_2}) \geq I(\pi_{w_1})$. Substituting this back into the first inequality gives $R(\pi_{w_1}) - R(\pi_{w_2}) \geq w_1(I(\pi_{w_2}) - I(\pi_{w_1})) \geq 0$.

These inequalities do not imply monotone observation count. A policy with fewer, more informative observations can collect more information. Moreover, a receding-horizon agent maximizes again at each reached belief, so its complete trajectory is not a single optimizer over the fixed plan set in the proposition. The empirical usage staircases can therefore dip, as they do on Bandit and Tileworld (Section 6.5).

The single observe-or-commit decision does admit an exact usage threshold. In this corollary and its onset checks, we write w_{thresh} for the w_{thresh}^* of Proposition 2, dropping the star to distinguish it from $w^*(B)$.

Corollary 1 (Proposition 2’s onset thresholds are usage-staircase knots). *In the two-state $H=1$ setting of Proposition 2 with $w_{\text{thresh}} \geq 0$, equivalently $c \geq (p - \frac{1}{2})(R^+ - R^-)$, for any budget $B \in (0, 1)$ the crossing threshold of Definition 2 is exactly w_{thresh} , the weight at which the staircase leaves zero, and the closed enclosure $[w_{\text{lo}}, w_{\text{hi}}]$ of a finite-grid crossing bracket contains that threshold whenever the sampled usages straddle B , locating it to grid resolution rather than equalling it. When $w_{\text{thresh}} < 0$ every nonnegative weight already observes, so $U \equiv 1$ on $[0, \infty)$ and no onset knot lies in the nonnegative range. The onset boundary of the budgeted problem recovers Proposition 2’s closed form as the first knot of the usage staircase.*

Its proof is the single-step curve $U(w) = \mathbf{1}[w > w_{\text{thresh}}]$ under the committing tie rule, where $\mathbf{1}[A]$ is 1 when A holds and 0 otherwise. The full argument and implementation check are in Appendix U.5. That check runs whole receding-horizon episodes and verifies the onset, not a claim that every such episode takes exactly one observation.

4.4 Solving for the Shadow Price

We estimate $U(w)$ by multi-seed Monte Carlo on a log-spaced grid: five seeds and 100 episodes per cell on the discrete observe-then-commit domains, 40 on Tileworld- 6×6 , and lighter settings on tree-search domains (Section 6). Per-seed means give standard errors. We report the crossing bracket and the grid weight $\hat{w}(B)$ minimizing $|U - B|$, rather than a bisection root. This plotted point can lie on the lower bracket edge or, at a slack budget, below it. Count and cost usage are distinct. When all sensing actions share a cost, the two differ by a fixed conversion (Section 6.3).

4.5 Online Dual Control of the Shadow Price

A reward-scale change moves the count-budget crossing. To adjust without a fresh grid sweep, we adapt SAC’s automatic temperature update [25], replacing the policy-entropy target by a sensing target. After episode t with observed usage U_t , update

$$w_{t+1} = \text{Proj}_{[0, w_{\text{max}}]}(w_t + a_t(B - U_t)), \quad a_t = \frac{\eta_0}{1 + \delta t}, \quad (9)$$

where η_0 is the initial step, δ its decay rate, and projection clips the weight to $[0, w_{\text{max}}]$. If $\mathbb{E}[U_t \mid w_t] = U(w_t)$, usage is a noisy observation of the curve. Underspending raises w and overspending lowers it. This sign corrects usage only under suitable crossing conditions.

The proposition covers the projected decaying-step controller, with finite w_{max} and $\delta > 0$. Every reported run uses $w_{\text{max}} = 100$ and $\delta = 0.02$. No iterate exceeded 3.1, so the upper projection was present but inactive. It does not cover the released library default, which has a constant step and no upper projection, or the reset-on-shift heuristic described below.

Proposition 6 (Stationary convergence of the dual controller). *Consider the projected decaying-step controller, the recursion (9) with a finite $w_{\max}$ and step sizes satisfying $\sum_t a_t = \infty$ and $\sum_t a_t^2 < \infty$ (the schedule in (9) with $\delta > 0$). Suppose four conditions hold. (i) The environment, agent family, and budget B are stationary, so that $\mathbb{E}[U_t | w_t] = U(w_t)$ for a fixed function U . (ii) Per-episode usage is bounded, which holds here because usage is bounded by the episode step cap, and each episode is run on an environment reset and freshly seeded independently of the past, so that the noise $U_t - U(w_t)$ is a martingale difference with bounded conditional second moment. (iii) There is a single crossing $w^* \in (0, w_{\max})$ with $U(w) < B$ for $w < w^*$ and $U(w) > B$ for $w > w^*$. (iv) The curve stays away from the budget outside every neighborhood of the crossing, that is, for every $0 < \epsilon_1 < \epsilon_2$, $\inf\{|U(w) - B| : \epsilon_1 \leq |w - w^*| \leq \epsilon_2, w \in [0, w_{\max}]\} > 0$. Then $w_t \rightarrow w^*$ with probability one, and hence in probability. Condition (iii) does not require $U(w^*) = B$, and condition (iv) does not require continuity, so the conclusion covers a gap budget in the sense of Definition 2, where U jumps over B at w^* without touching it. What converges in that case is the weight. The usage at the limit is $U(w^*)$, one side of the jump, so the expected usage at w_t need not converge to B . The running average of observed usage under the schedule in (9) does converge to B , $\frac{1}{n} \sum_{t < n} U_t \rightarrow B$ with probability one, gap budgets included, as Appendix U.6 proves. Attaining B in expectation with one stationary policy at a gap budget is the job of the endpoint mixture of Definition 2, not of this recursion.*

The update is a projected Robbins–Monro recursion [47], with classical almost-sure extensions due to Blum [6] and constrained treatments in Kushner and Clark [33]. Appendix U.6 proves this bounded scalar case directly by a supermartingale argument, including discontinuous curves. Kronecker’s lemma gives the running-average result for the harmonic schedule above. The theorem is conditional on one sign-consistent crossing and does not cover an arbitrary nonmonotone usage curve.

The reset-on-shift variant detects a persistent recent usage mismatch once the step size has fallen below half its initial value, then restarts the decay clock at zero. These resets break the single diminishing-step schedule, so the proposition does not justify their convergence. Section 6.2 evaluates both decay-only and reset-on-shift recovery empirically. Constant-step tracking would require separate drift and noise arguments and is not claimed here.

4.6 Positioning against Prior Art

Adjusting a price to meet a resource target is classical in constrained MDPs [1], rational inattention [38, 56], and SAC [25]. Our contribution is its operational use for sensing within the Planning+IG family, with separate count and cost equivariance, finite-grid crossing estimates, endpoint mixtures, and a conditional online convergence result. Rational inattention constrains information processing. Here the design input is sensing usage, while w prices information gain.

An exact constrained-POMDP solver can randomize belief-dependently and outperform a family that runs one weight throughout each episode [30]. Our endpoint mixture only randomizes that weight between episodes. The references in Section 6.7 are mixtures selected as cap-feasible from simulated usage estimates for policies in a SARSOP usage-penalty sweep. They are not certified constrained optima. Sampling error and selection over noisy estimates qualify all measured reference gaps. The shallow RockSample reference fails to reach EFE’s operating region. Appendix U.7 reports that limitation rather than treating it as optimality evidence.

Sequential Bayesian experimental design selects tests for information gain about a latent parameter [16, 36]. Here reward and sensing cost remain in the objective. Reward at the calibrated weight is measured rather than optimized by the usage calibration. The large-weight limit prioritizes information gain. Closely related work gives a Bethe-Lagrangian account of EFE with an internal information constraint [31], and recursive per-node dual ascent inside constrained POMDP tree search [60]. These differ from controlling one information weight across episodes toward a sensing target. Appendix U.7 develops the comparisons.

Offline calibration does not save the sampling needed for a weight sweep. Its benefit is a design requirement expressed in observable sensing units. Under its hypotheses, the online controller can locate a crossing from interaction without that sweep. Scale equivariance transfers both calibrated and directly tuned weights across pure reward-and-cost rescalings, so transfer is not a unique advantage of budget calibration.

5 Experiments

We evaluate two claims separately. The budget experiments ask whether an information-weighted policy family can meet an expected sensing-usage target, transfer that calibration across reward scales, and re-adapt after a scale change. The EFE experiments ask where the canonical $w=1$ policy lies relative to reward-only planning, tuned information gain, and established POMDP solvers. Meeting a usage target and maximizing reward under a usage cap are different objectives, so the first battery measures both usage error and reward relative to a constrained reference.

The six main environments form a progression in sensing decisions. Tiger has one sensor and asks whether to listen again. Diagnosis and Bandit offer several tests and ask which observation to buy. Tileworld makes those tests spatial partitions of a grid. RockSample and Structural Inspection interleave sensing with movement and exploitation. The first four have a fixed hidden state and an observe-then-commit structure. The last two satisfy the factored-observation assumptions of Proposition 3. Complete specifications and protocols appear in Appendices E and V.

Tiger [29] has two hidden states, one listen action of accuracy 0.85 and cost 1, and terminal door choices paying +10 or −100. Diagnosis has four possible conditions, two binary tests of accuracy 0.80 and cost 1, and a terminal diagnosis

paying +10 or −50. Each test partitions the conditions differently. Bandit has four arms and a single hidden variable identifying the best arm. An inspection targets one arm, has accuracy 0.80, and costs 0.5. A terminal pull pays +10 for the best arm and +1 otherwise. Tileworld hides one target on a 6×6 grid. Its six noisy binary scans query bits of the target’s row or column index, each with accuracy 0.80 and cost 1. Declaring a cell pays +10 or −50.

Our RockSample variant [58] has known rock positions and hidden binary qualities. Checking is noisy, movement changes sensor range, and sampling and exit yield reward. It differs materially from the original benchmark. Every action costs 0.5, maps are our own, returns are undiscounted with an episode cap, exit is available from any cell, hidden rock qualities remain fixed after sampling, and sensor accuracy starts at 0.95 and reaches chance at finite range. Sample history is observable and prevents a collected rock from yielding a second reward. These choices support the fixed-hidden-state analysis and make sensing use directly interpretable, but absolute returns are not comparable to published RockSample values. Appendix V gives the six differences in full. We evaluate RS[5,3], RS[7,4], RS[7,8], and RS[11,11], where the pair denotes grid width and rock count. Hidden-state counts exclude observable position.

Structural Inspection places N components with fixed hidden nominal/faulty states on a grid. The agent moves, chooses a visual test (accuracy 0.70, cost 0.5) or detailed test (accuracy 0.90, cost 2), and makes an irreversible declaration for each component. Missing a fault costs 50, while correct declarations pay 2 or 5. We evaluate $N=8$ and $N=16$, the latter with 65,536 hidden configurations. This is a synthetic benchmark with a known model and a shared hand-coded search-leaf rule, not a deployed inspection study. Two additional boundary cases, the mild-penalty two-state Testbed and Navigation without separate observation actions, are reported in Appendices C and F.

Unless a battery states otherwise, the canonical seeds are {42, 123, 456, 789, 1024}. The main EFE comparison uses 1,000 episodes per seed on Tiger, Diagnosis, and Bandit, and 500 on Tileworld and Inspection. RockSample protocols appear with their tables. Means aggregate episodes across seeds, and main-table uncertainty is the SE of per-seed means. Pairwise comparisons use seed-level Welch t -tests with Holm–Bonferroni correction for the stated comparison families. Pooled episode-level statistics and bootstrap intervals are supplementary (Appendix I). A failure to reject a difference is not called equivalence. Equivalence is tested separately with fixed-margin TOST below.

The budget battery generally uses 100 episodes per evaluation cell, with each cell defined by one weight and one seed. Tileworld uses 40 episodes per usage-curve cell and 30 per scale-collapse cell, and Inspection- $N=8$ uses 30 per usage-curve cell. Different batteries have different episode counts and seeding rules, so their estimates are not interchangeable. The frontier study below explicitly addresses a consequential cross-battery difference. The JAIR version’s reproducibility checklist records all exceptions and artifact locations.

6 Experiments on Shadow Prices and Sensing Budgets

We first test scale transfer and online re-adaptation, then examine cost-denominated budgets, interleaved settings, and the measured staircases. Comparisons with unconstrained and usage-constrained references lead to the held-out calibration study. A distractor experiment then separates the amount of sensing from its relevance to reward. Additional onset and operating-point checks are summarized with appendix pointers at the end of the section.

6.1 Curve Collapse across Reward Scales

Proposition 4 predicts that rescaling rewards and sensing costs by α preserves a count-usage curve when the information weight is also multiplied by α . We evaluate reward scales $\alpha \in \{0.1, 1, 10\}$ on corresponding scaled weight grids. Figure 5 shows the result in raw and normalized coordinates. Every matched w/α point agrees bit-exactly on Diagnosis, Bandit, Tiger, and Tileworld. Shared per-episode random streams make exact empirical agreement possible when actions coincide. These CSVs are byte-identical under an absolute comparison tolerance in place of the relative one of Section 4.2, which does not bind at these points.

The normalized grid bracket is $(0.141, 0.323]$ for Diagnosis at $B=8$, $(3.84, 8.76]$ for Bandit at $B=8$, and $(5.80, 20.0]$ for Tileworld at $B=15$, unchanged across scales. Tiger is a necessary negative control. Its usage is constant at 4.37 across the tested normalized range $[0, 20]$, so collapse is trivial and $B=4$ is unbracketed. Equivariance preserves a curve but does not ensure that it offers a useful sensing dial. Table 33 and Appendix W.1 give the complete breadth check.

6.2 Online Re-adaptation after a Reward Rescale

We compare the projected decaying-step controller with its reset-on-shift extension on Diagnosis at target $B=8$. Both use $\eta_0=0.05$, decay $\delta=0.02$, and a tenfold reward-and-cost rescale at episode 200 of a 400-episode run. Ten shared controller seeds support a paired comparison. The extension watches a 20-episode usage window. Once the step has decayed below half its initial value, a deviation exceeding both three window standard errors and 0.5 observations restarts its decay clock. Appendix W.3 specifies the detector and steady-state summaries.

Recovery occurs when a rolling mean computed only from post-rescale episodes stays within ± 1 of B for 20 consecutive episodes. The comparison uses the restricted time $\min(T, 180)$ for every seed, where T is the zero-based post-rescale index at the start of the first qualifying hold and 180 is the last admissible start index. A seed that never completes it receives 180. Decay-only averages 157.3 episodes and reset-on-shift 51.2, a descriptive ratio of 3.07. The paired difference is 106.1 episodes with a Student- t 95% interval $[96.8, 115.4]$. All ten paired differences favor resetting. Reset-on-shift recovers on 10/10 seeds and decay-only on 9/10, so the latter’s unrestricted recovery distribution is

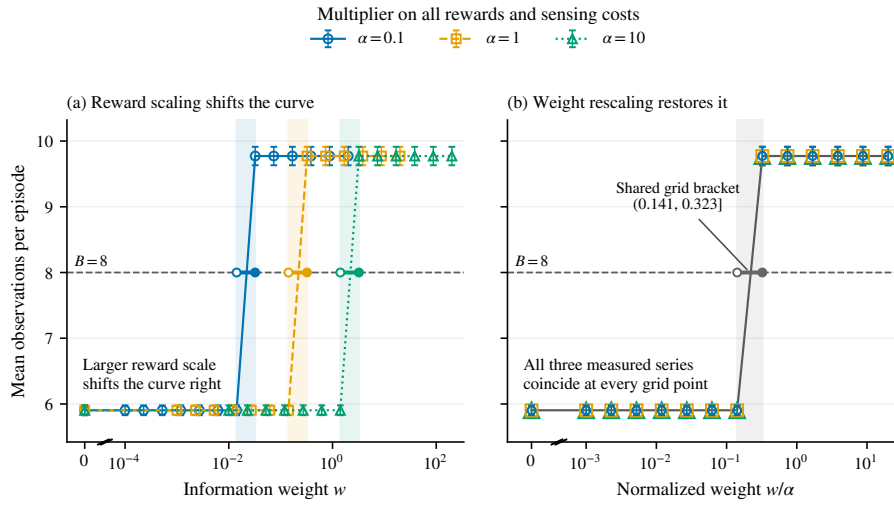


Fig. 5. Reward-scale transfer on Diagnosis. In (a), separately evaluated usage curves shift when rewards and sensing costs are multiplied by $\alpha \in \{0.1, 1, 10\}$. In (b), plotting against w/α collapses them exactly, including the $B=8$ grid bracket $(0.141, 0.323]$. Points and error bars are means ± 1 seed-level SE over five seeds and 100 episodes per seed, with shared per-episode streams. Lines guide the eye. Bands and open-closed endpoints mark grid brackets, not uncertainty intervals or policies attaining the target. The crossing between sampled points remains unresolved.

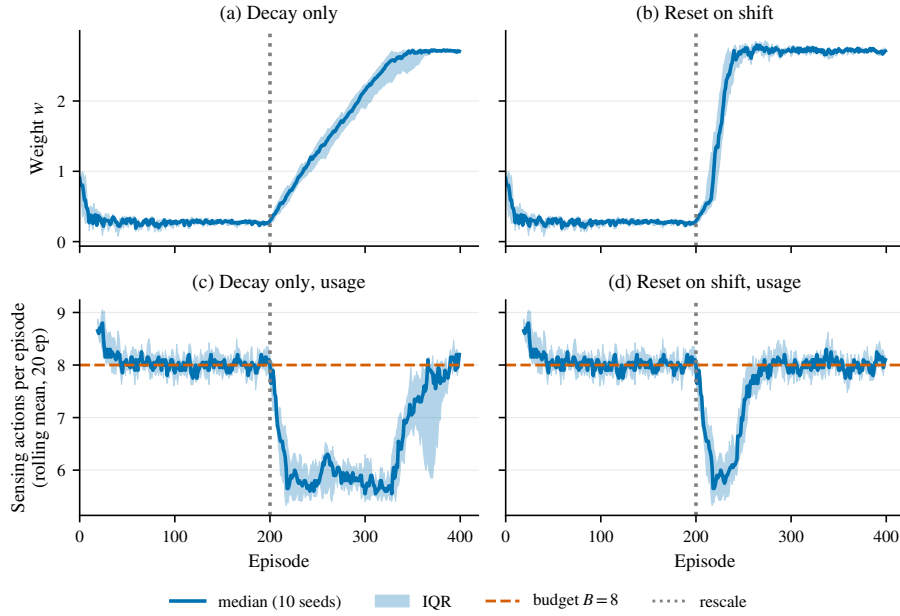


Fig. 6. Online control on Diagnosis through a tenfold reward-and-cost rescale at episode 200. Columns compare decay-only and reset-on-shift. Top panels show weight, as median and interquartile range over ten seeds. Bottom panels show rolling usage with window 20 and target $B=8$. Recovery uses post-shift-only windows and a 20-episode hold within ± 1 of the target. Restricted recovery time $\min(T, 180)$ averages 157.3 versus 51.2 episodes, with paired difference 106.1 and Student- t 95% interval [96.8, 115.4]. This nonstationary experiment is outside Proposition 6.

right-censored. The conditional mean among its nine recovered seeds is not the comparison estimand.

These measurements come from the per-episode-seeded runner, which also seeds the environment constructed at the rescale. Decay-only is slower than reset-on-shift on every seed, and on one of the ten it does not recover within the window. Figure 6 plots the trajectories. The mid-run rescale is nonstationary, and the reset variant restarts its step schedule, so neither experiment is covered by Proposition 6’s stationary convergence guarantee. This is empirical re-adaptation evidence on one shift scenario.

6.3 Cost-Denominated Budgets

The count-usage curves of Section 6.1 treat every observation as one unit regardless of price. Section 4.4 noted that usage can instead be denominated in cumulative sensing cost, and that equal-cost observation actions give a fixed conversion between the two. This subsection demonstrates end to end that the

denomination matters. A budget stated in test count and a budget stated in sensing cost are not interchangeable, because the agent’s mix of cheap and expensive tests can change with w . We introduce a Diagnosis variant with heterogeneous per-test costs. Its two test types are priced differently, the cheaper test at 0.5 and the more expensive test at 2.5 (the cost vector $[0.5, 2.5]$ in the implementation), while both tests keep the base environment’s accuracy. On this variant we estimate both the count-usage curve $U_{\text{count}}(w)$ and the cost-usage curve $U_{\text{cost}}(w)$ on the same agent family.

The mechanism is a shift in the agent’s test mix. Each Diagnosis test noisily reports which side of its fixed partition contains the true condition, and the two tests use different partitions (Section 5). Although the tests are equally accurate, the information each reveals depends on the current belief over its partition. At a given belief one test can therefore reveal more than the other. The swept curves show a step in mean cost per test, $U_{\text{cost}}(w)/U_{\text{count}}(w)$. It holds at 1.26 from $w=0$ through $w=10$ and rises to 1.37 and then 1.41 only at the two largest weights, where the number of tests also jumps (Figure 7). The spread is 12.0% relative to the grid-mean cost per test. Our reading of the step is an interpretation rather than a measurement. While information gain is weighted lightly, test prices keep the agent at its low-weight test mix. Once w is large enough to buy extra tests, test choice depends more on what a test reveals at the current belief than on its price. On this variant, that means using the expensive test more often. The direction is measured on the swept curves rather than implied by the cost vector, which by itself says nothing about which test reveals more. A count budget cannot see that shift, while a cost budget prices it. This confirms directly that the two denominations are not interchangeable.

Testing whether the two denominations select different weights requires a rule for pairing a count budget with a cost budget. A cost budget here is not a count budget converted through a cost per test. Instead each curve is measured on its own scale, and the two budgets of a pair are set at the same fractional height inside each curve’s own usage range. The usage range of a curve is the span from the smallest to the largest mean usage on the tested grid. A margin is held back at both ends of that span before budgets are placed, so that each budget has a crossing bracket on the grid. We call two budgets paired in this way comparable operating points. Since the two denominations are not interchangeable, one might expect them to select different brackets at such a pair. At both tested operating points, the paired budgets select the same crossing bracket at this grid’s resolution. A count budget $B=8.86$ has crossing bracket $(10, 31.6]$, and a cost budget $B_{\text{cost}}=11.50$ (the comparable operating point on the cost curve, a budget on total sensing cost rather than on test count) also has crossing bracket $(10, 31.6]$ (Figure 7). The same coincidence holds at the higher pair of budgets we test ($B=13.71$ count, $B_{\text{cost}}=19.20$ cost, both with crossing bracket $(31.6, 100]$).

The coincidence has a simple explanation. The cost curve is the count curve multiplied by the mean cost per test. If that multiplier were constant across weights, the pairing rule would express the same budget in two units. The two

curves would then cross their paired budgets at the same weight. Only the variation of the multiplier across the grid can pull the two crossings apart, by shifting where along the weight axis the cost curve crosses its budget. That multiplier moves by at most the 12% spread just reported, while adjacent positive grid weights are log-spaced a factor of $\sqrt{10} \approx 3.2$ apart. A bracket changes only if that shift carries the crossing past one of those grid weights. A rescaling that small need not carry a crossing into a neighboring grid cell, and at these two pairs it does not. The two denominations are therefore not guaranteed to disagree at every tested budget pair. They land in different brackets only when the variation in the cost per test is large enough, relative to the grid’s resolution, to move a crossing past a grid weight.

6.4 RockSample and Structural Inspection as Interleaved Settings

We now estimate usage curves and shadow prices in the interleaved observe-act settings of Sections 7.4 and 7.5, where sensing and acting alternate within an episode. RockSample also exercises the navigation between observations admitted by Proposition 3. The observe-then-commit environments above cannot exercise that case because they separate observation from commitment by construction. We use the interleaved tree-search agents that those two sections specify, counting check/test actions as sensing usage. The three instances are RockSample[5,3], RockSample[7,4], and Inspection- $N=16$. All three share one set of five seeds, and every episode count below is per seed at each grid weight. RockSample[5,3] uses a 10-point weight grid with 5 seeds $\times$ 50 episodes, and its usage range (Section 6.3) is [3.00, 8.83]. RockSample[7,4] uses an 8-point grid with 30 episodes and spans [2.00, 14.54]. Inspection- $N=16$ uses a 10-point grid with 50 episodes and spans [22.38, 58.56]. Budgets are placed inside each usage range by the rule of Section 6.3, at evenly spaced positions, with a narrower margin held back at each end than that section used. Inspection appears at $N=16$, the larger of the two instances evaluated in Section 7.5, while the staircases of Section 6.5 use the smaller $N=8$ instance. The two subsections together therefore cover both sizes. Figure 8 reports grid estimates of the shadow prices for each instance. Four budgets are evaluated per environment, each reported with a crossing bracket and the usage SE at its plotted grid weight.

The usage curves $U(w)$ from which these staircases are read off are nondecreasing between every pair of consecutive sampled grid points on RockSample[7,4] and Inspection- $N=16$. That is cleaner than the count-usage curves of Bandit and Tileworld, which the discussion after Proposition 5 noted are only roughly monotone and whose shadow-price staircases follow in Section 6.5. RockSample[5,3] is the exception to this clean monotonicity. Its instrumental floor, the usage the agent spends at $w=0$ because reward alone already motivates that much sensing, is $U(0)=3.70 \pm 0.11$ (SE), and usage dips from there down to 3.00 ± 0.00 at $w=0.316$ before rising, a gap of roughly six SEs. This is the same kind of non-monotone dip the observe-then-commit environments show, so the interleaved class does not uniformly enjoy the cleaner monotonicity of the other two instances.

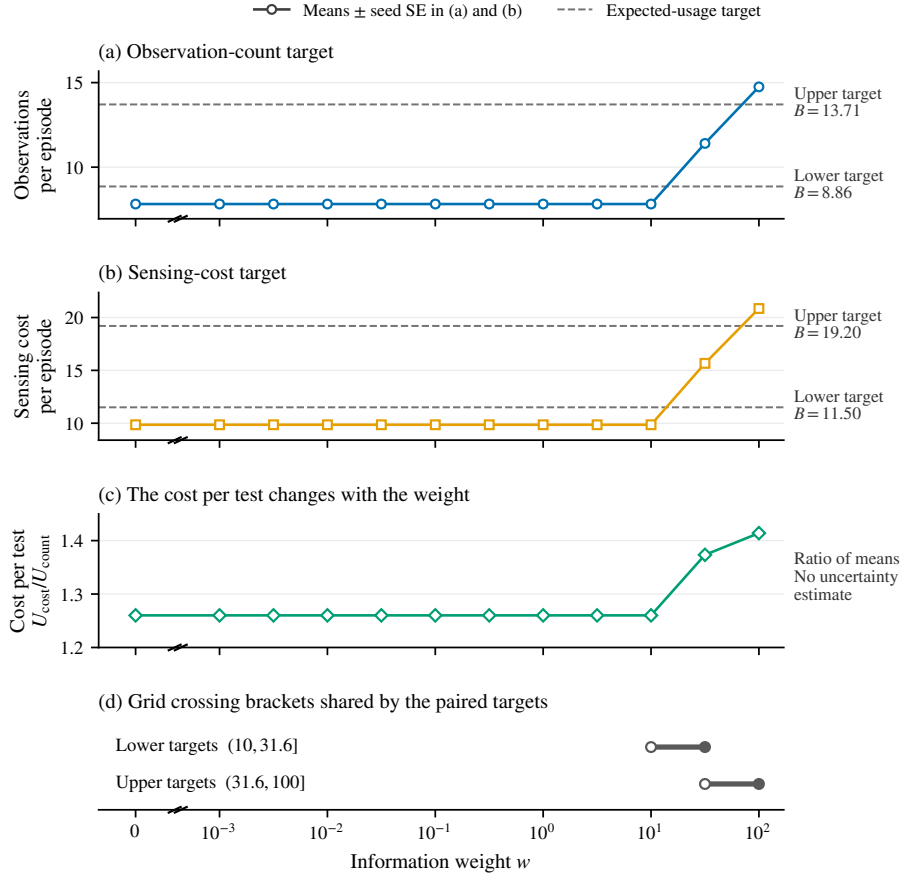


Fig. 7. Observation-count and sensing-cost targets on heterogeneous-cost Diagnosis. Panels (a) and (b) separate the two usage units, showing means ± 1 seed-level SE and the tested targets as dashed lines. The lower count/cost targets are 8.86/11.50, and the upper targets are 13.71/19.20. Panel (c) shows the ratio of mean sensing cost to mean observation count, which rises as the test mix changes. No uncertainty estimate is shown for this ratio. Panel (d) displays the paired targets' shared crossing brackets, (10, 31.6] and (31.6, 100]. These intervals measure grid resolution, not statistical uncertainty. Lines joining sampled means guide the eye.

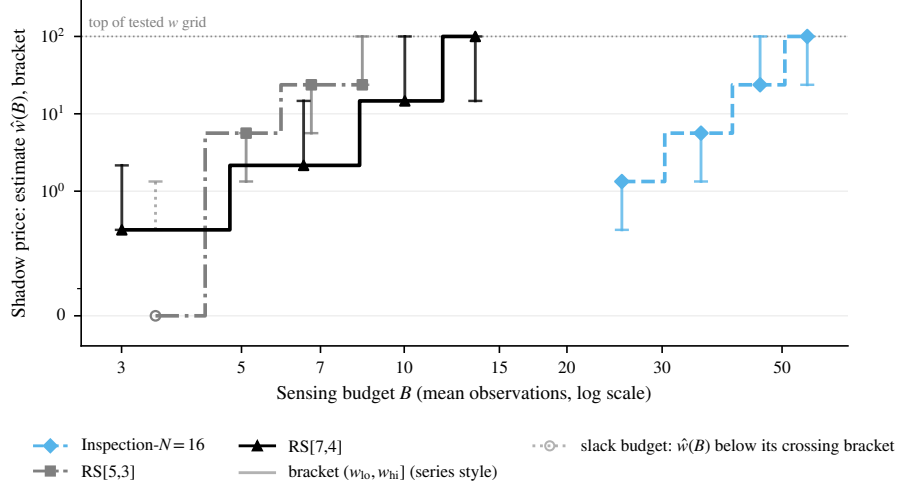


Fig. 8. Grid estimates of the shadow price on three interleaved (non-observe-then-commit) settings, RockSample[5,3], RockSample[7,4], and Inspection- $N=16$, with crossing brackets marked as vertical bars, capped where they reach the dotted rule marking the top of the tested w grid, and an open-circle marker where a budget is slack (the grid weight whose usage comes nearest to the budget sits strictly below the bracket’s lower edge, here $w=0$ on the lowest RockSample[5,3] budget). Every marker plots the nearest-point solution $\hat{w}(B)$, which can sit on a bracket’s lower edge. The budget axis is logarithmic and the weight axis uses a symmetric-log scale, linear near zero. The underlying usage curves $U(w)$ are monotone at every sampled grid point on RockSample[7,4] and Inspection- $N=16$, while RockSample[5,3] dips once near $w=0.316$ before rising (see text).

On the sampled RockSample[7,4] and Inspection- $N=16$ grids, the smallest measured usage occurs at $w = 0$, so no tested weight meets a lower target. The usages at zero weight are $U(0) = 3.70$ on RockSample[5,3], 2.00 on RockSample[7,4], and 22.38 on Inspection- $N=16$. These measurements identify lower limits within the sampled grids, not bounds for every untested weight. They show why attainable usage must be checked in interleaved settings as well as in the observe-then-commit class. RockSample[5,3] is again the exception, because there a positive weight buys less sensing than $w=0$ does, so budgets between 3.00 and 3.70 are achievable even though they lie below $U(0)$. The lowest RockSample[5,3] budget in Figure 8 lies in this band. The figure draws it as slack because the grid weight whose usage comes nearest to that budget is $w=0$ itself, which sits below the budget’s crossing bracket, not because the budget is unachievable.

6.5 Shadow-Price Staircases across Five Domains

The observe-then-commit domains and the smaller Inspection instance now receive the same treatment as the interleaved instances. Figure 9 reports grid

estimates of $w^*(B)$ over identifiable budgets inside $[U_{\min}, U_{\max}]$ for Tiger, Diagnosis, Bandit, Tileworld-6 $\times$ 6, and Inspection- $N=8$, where $U_{\min}$ and $U_{\max}$ are the endpoints of each curve’s usage range (Section 6.3). An identifiable budget is one placed strictly inside that range by the rule of that section, evenly spaced with the same margin held back at each end as in Section 6.4, so that the grid brackets it. The monotonicity at issue is that of the underlying usage curve $U(w)$ from which each staircase is read off, taken between consecutive sampled grid weights. On Tiger, Diagnosis, and Inspection that curve is nondecreasing between every pair of consecutive grid weights. On Bandit and Tileworld it is not, showing local non-monotonicity from discrete policy switches, the first of the two obstacles noted after Proposition 5. We report crossing brackets and usage standard errors for every domain (Definition 2). The standard error attached to a bracket is that of usage at a fixed weight, and the bracket’s width is set by the grid, so neither says how often a different sample of seeds would select a different bracket. We measured bracket-selection stability by resampling the five seeds with replacement 2000 times and re-solving each of the 25 budgets of Figure 9 on each resampled curve (`run_bracket_stability.py`, `results_price_bracket_stability.csv`, with the count of every bracket outcome in `results_price_bracket_stability_counts.csv`). The reported bracket is recovered in every resample at 19 budgets and in at least 99 percent of resamples at 21. The lowest agreement is 0.789, at Inspection’s lowest budget. In all but one disagreeing resample, the alternative is a neighbouring step of the staircase or, at Tiger’s lowest and highest budgets, no bracket at all. Those two Tiger budgets lie about one standard error of usage from the corresponding ends of its usage range. On Bandit and Tileworld a budget can be crossed more than once, and the reported bracket estimates the last crossing by the convention that definition adopts, so it names one threshold at grid resolution rather than displaying the reversals. Bandit’s budget $B=5.87$ is the one tested case. Usage first reaches it at $w=0.139$ (6.19 observations), falls back to 5.11 at $w=0.72$, and reaches it for the last time inside the reported bracket $(1.64, 3.73]$ (`results_price_shadow_curves.csv`), which does not show the earlier crossing. Tiger’s price is weakly identified, consistent with Table 33, in the sense that the budget pins the price down only loosely. Instrumental listening already saturates most of the usage range at $w=0$, so the curve is flat at its floor across the low-weight range and rises in a single step at high weight. Every tested budget lands in that one step’s bracket.

6.6 Unconstrained Reward Reference

To determine whether $w=1$ is reward-competitive at its natural operating point, we compare EFE with SARSOP [32] on Tiger, Diagnosis, and Bandit. The original APPL solver is built from source, each environment is exported through a common model path, and the solve uses precision 10^{-3} at $\gamma=0.999$. The returned alpha-vector policy is evaluated in the same simulator as EFE at five seeds and 500 episodes per seed (Table 2). SARSOP is a near-optimal reference, not a certified exact solution to the reported Monte Carlo comparison. Build and export details appear in Appendix T.1.

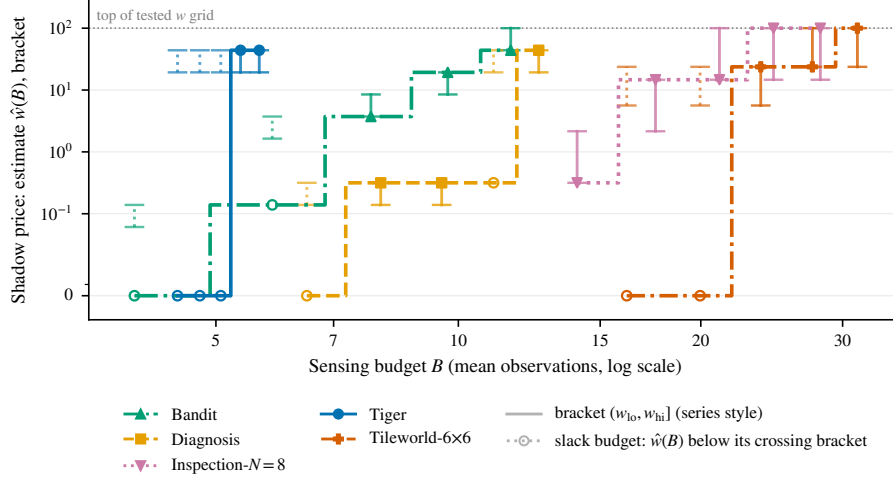


Fig. 9. Operational shadow-price staircases $w^*(B)$ across five observe-then-commit and inspection domains (Tiger, Bandit, Diagnosis, Inspection- $N=8$, Tileworld- 6×6), each in its own color, with a logarithmic budget axis and a symmetric-log weight axis that is linear near zero. Each staircase steps up as the sensing budget B rises, and the vertical bar at each step marks that budget’s crossing bracket $(w_{lo}, w_{hi}]$ (Definition 2), capped where the bracket’s upper edge coincides with the top of the tested w grid (dotted horizontal rule at 100). An open circle with a dotted bar marks a slack budget, one for which the grid weight whose usage comes nearest to the budget lies strictly below the lower edge of the crossing bracket ($w=0$ at seven of the nine marked budgets, $w=0.316$ at Diagnosis $B=11.06$, and $w=0.139$ at Bandit $B=5.87$, the twice-crossed budget discussed in the text), so the nearest-point solution returns a weight below the bracket while the dotted bar still marks the crossing. Every marker plots that nearest-point solution $\hat{w}(B)$, which can also sit on a bracket’s lower edge. The usage curves behind Tiger, Diagnosis, and Inspection never step downward between consecutive sampled weights, while Bandit and Tileworld show local non-monotonicity from discrete policy switches, so their brackets mark the last crossing of each budget by the convention of Definition 2.

Table 2. SARSOP and EFE ($w=1$) on the three core environments, five seeds and 500 episodes per seed, $\gamma=0.999$. Values are means $\pm$ seed-level SE. Usage counts observations. Fixed-margin TOST and its retrospective/prospective scope are described in the text.

Environment	SARSOP reward	EFE reward	SARSOP usage	EFE usage
Tiger	5.061 ± 0.158	5.061 ± 0.158	4.323 ± 0.045	4.323 ± 0.045
Diagnosis	-1.452 ± 0.336	-1.217 ± 0.184	9.68 ± 0.12	9.73 ± 0.13
Bandit	6.280 ± 0.134	6.261 ± 0.112	5.16 ± 0.09	5.09 ± 0.07

Tiger’s policies take identical actions on every evaluated episode. Diagnosis and Bandit show small reward differences, but non-significance alone would not establish equivalence. We therefore use two one-sided tests (TOST) [53] with a reward margin equal to one observation’s cost, ± 1.0 on Tiger and Diagnosis and ± 0.5 on Bandit. These operational margins were chosen after an earlier canonical-seed comparison had been inspected, so the five-seed TOST is retrospective. The additional fifteen seeds in the $n=20$ robustness study were fixed after the margin and before that study ran.

The primary TOST uses unpaired Welch tests on per-seed means. All three environments reject both one-sided nulls, including after Holm correction across environments, with $p_{\text{TOST}} = 0.0010$, 0.0458 , and 0.0130 for Tiger, Diagnosis, and Bandit. Their 90% intervals lie inside the specified margins. The $n=20$ study retains equivalence, with respective p -values 4.5×10^{-10} , 6.8×10^{-7} , and 6.0×10^{-11} . The nominal seed lists are shared, but differing policies consume random draws differently, so seed-paired differences are not episode-matched. A predeclared twenty-seed check reseeds every episode so both policies face the same hidden states and, while their actions agree, the same observation draws. It retains equivalence, with paired differences of $+0.11 \pm 0.09$ on Diagnosis and $+0.04 \pm 0.03$ on Bandit and paired p_{TOST} of 3.2×10^{-9} and 4.6×10^{-12} , while Tiger’s two policies act identically. Appendix W.4 gives intervals, mechanics, and archive filenames. The conclusion is equivalence within these fixed margins on these three environments, rather than universal near-optimality of $w=1$.

6.7 A Usage-Constrained Reference

A literal usage penalty supplies a different reference from an information bonus. We replace observation reward $-\text{cost}_j$ by $-(\text{cost}_j + \lambda)$ and solve with SARSOP over a 12-point penalty grid in $[0, 100]$. This estimates policies maximizing $\mathbb{E}[R] - \lambda \mathbb{E}[U]$ over the POMDP policy space, with $\gamma=0.999$ and five seeds of 300 episodes per penalty. The penalty is the Lagrangian price of the usage cap in Equation 6. The information weight w is not that multiplier.

At budget B , the reference is the feasible envelope of the sampled policy points, obtained by maximizing their mixture reward subject to mixture usage at most B . A solution uses at most two sampled policies. Policies below the target remain eligible, and the reference is flat at the largest sampled reward once the

cap is slack. This is a linear program over measured points, not equal-usage interpolation. The solver is approximate, the penalty grid is finite, and reward and usage are Monte Carlo estimates. Consequently the envelope is neither a certified population bound nor the full constrained frontier. We do not assume that finite-state strong-duality results [1] automatically extend to the continuous belief MDP. Appendix W.5 discusses these limitations.

Table 3. Estimated cap-constrained reference at EFE usage B_{EFE} . R_{ref} maximizes reward over mixtures of sampled SARSOP-Lagrangian policies with usage at most that budget. Gap is $R_{\text{ref}} - R_{\text{EFE}}$, and Gap % is $100(R_{\text{ref}} - R_{\text{EFE}})/|R_{\text{ref}}|$. Neither is a certified optimality gap. On each environment the best sampled reference is already feasible, so the cap does not bind. Values use five seeds and 300 episodes per seed. The actually evaluated usage-matched Planning+IG grid weights produce the same rewards as EFE in this run.

Environment	B_{EFE}	R_{ref}	R_{EFE}	Gap	Gap %
Tiger	4.32	5.016	5.016	0.000	0.00%
Diagnosis	9.82	-1.265	-1.224	-0.041	-3.27%
Bandit	5.10	6.314	6.320	-0.005	-0.08%

Table 3 evaluates the reference at EFE’s endogenous usage B_{EFE} . The highest-reward sampled reference policy is already feasible on each environment, so the cap does not bind. Its estimated gap from EFE on the canonical stream is zero on Tiger, -0.041 on Diagnosis, and -0.005 on Bandit, small relative to the estimates’ standard errors. Negative estimated gaps can arise from sampling or approximation error and do not establish reward above the population optimum. The usage-matched Planning+IG grid weights actually run, 0.00, 0.45, and 1.07, give rewards bit-identical to EFE here. This is an empirical plateau result rather than an extension of the equivalence theorem from $w=1$ to arbitrary weights. The reference remains limited to these small environments, with no corresponding optimality-gap estimate for the larger RockSample and Inspection instances.

6.8 Calibration-Derived Targets on Held-Out Seeds

The preceding comparison uses whatever sensing the weight-one policy happens to buy. We now test targets independent of that weight-one usage, chosen by a predeclared rule from each calibration curve. These targets stand in for application requirements but are not themselves application-specified. The study separates whether the procedure meets expected usage from whether doing so maximizes reward under a cap. Its original design was recorded in the producer’s docstring before results were inspected (`experiments/run_budget_frontier.py`).

Each core environment is calibrated on the canonical five seeds, 100 episodes per seed, using the 16-point weight grid and horizons of Section 6.5. We select

budgets at 0.25, 0.5, and 0.75 of the measured range $[U_{\min}, U_{\max}]$, at the midpoint of its largest jump, and at half of $U_{\min}$. The last is deliberately unattainable and must be declared so. Definition 2 supplies the crossing bracket and endpoint-mixture probability q . All selection uses calibration data alone, and evaluation uses held-out seeds $\{7, 8, 9, 10, 11\}$ with 100 episodes each. Tiger’s gap budget duplicates its middle-range budget. Bandit’s curve is nonmonotone, so the stipulated last-crossing convention determines its bracket.

The original within-family comparator is the highest-reward single grid member whose calibration usage is at most B . Two additional LP comparators were introduced after inspecting the first results, and that amendment is disclosed in the producer. One maximizes calibration reward over mixtures with usage equal to B , and one does so with usage at most B . Each uses at most two calibration-grid weights and is evaluated on held-out seeds. Their scope is the sampled grid. Selecting calibration usage below a cap does not guarantee that the population or held-out usage meets it.

Table 4 gives the principal held-out results. All three below-range targets are declared unattainable. At all twelve attainable targets, the selected endpoint mixture meets usage to within 0.13 observations. Errors are within one held-out SE except Bandit’s gap target, where an error of 0.09 is about 1.3 SE. The evaluated bracket endpoints and best feasible single member miss the targets by more. Endpoints miss by 0.3–1.1 observations on Tiger and 0.2–3.6 elsewhere. The selected feasible single member leaves 0.2–4.7 observations unspent on held-out data. Its usage is below the target in all twelve rows, an observed result in addition to the calibration selection fact. Conversely, the endpoint mixture’s held-out mean exceeds B on three rows, by at most 0.13. An expected-use target provides neither a hard per-episode cap nor finite-sample cap feasibility.

Meeting a target can buy useful information or impose an avoidable reward cost. On Diagnosis at $B=7.72$, 9.53, and its gap target 7.84, the mixture earns -2.14 , -2.11 , and -2.06 against the feasible member’s -2.86 . On Tiger, reward-only planning uses 4.32 observations and earns 5.68 on this held-out stream. Spending more to meet the three increasing targets reduces reward to 5.36, 4.96, and 4.65. Bandit likewise loses reward at its two larger interior targets. The endpoint procedure also need not select the best mixture meeting a target. At Bandit’s gap budget, the calibration LP mixes $w=0$ and $w=0.72$ and earns 5.94 ± 0.19 on held-out seeds, against the endpoint mixture’s 5.62 ± 0.21 . Its usage error is larger, 0.18 rather than 0.09. The chosen endpoint mixture therefore cannot stand for every mixture in the family, and its deficit at this budget cannot be attributed to planning depth alone.

A valid constrained-reward comparison must also control the evaluation stream. The original reference points were evaluated on canonical seeds with once-per-seed seeding, whereas the family uses held-out per-episode seeds. Their direct reward comparisons, retained in Appendix W.6, are descriptive cross-stream comparisons. The difference matters on Tiger. Its unpenalized reference earns 5.02 on the canonical stream but 5.68 on the held-out stream, removing the mixture’s apparent excess over the reference at the smallest budget.

Table 4. Endpoint-mixture calibration on held-out seeds, 100 episodes per seed over five seeds. B is the calibration-derived expected-use target. U and R are held-out mixture usage and reward. All $\pm$ values are seed-level SE. The cap gap is mixture reward minus the same-stream cap-reference envelope, and the target gap is mixture reward minus the best sampled reference mixture whose usage equals B , each with seed-level SE. Reference supports and weights are fitted on those same seed means and held fixed for the SE, so intervals derived from these columns are nominal pointwise plug-in summaries, as qualified in the text. Tiger’s gap row duplicates its middle target. All three below-range targets are declared unattainable. Complete within-family and reference comparisons appear in Table 34.

Environment	Target	B	U	R	Cap gap	Target gap
Tiger	0.25	4.72	4.64 ± 0.10	5.36 ± 0.10	-0.33 ± 0.04	0.08 ± 0.04
Tiger	0.50	5.07	5.04 ± 0.14	4.96 ± 0.14	-0.72 ± 0.06	0.03 ± 0.06
Tiger	0.75	5.42	5.35 ± 0.14	4.65 ± 0.14	-1.03 ± 0.06	0.07 ± 0.06
Tiger	gap	5.07	5.04 ± 0.14	4.96 ± 0.14	-0.72 ± 0.06	0.03 ± 0.06
Diagnosis	0.25	7.72	7.70 ± 0.25	-2.14 ± 0.48	-0.03 ± 0.28	-0.03 ± 0.28
Diagnosis	0.50	9.53	9.47 ± 0.20	-2.11 ± 0.67	-0.76 ± 0.15	-0.76 ± 0.15
Diagnosis	0.75	11.35	11.47 ± 0.37	-2.31 ± 0.45	-1.10 ± 0.22	-0.05 ± 0.16
Diagnosis	gap	7.84	7.86 ± 0.25	-2.06 ± 0.47	0.00 ± 0.22	0.00 ± 0.22
Bandit	0.25	5.51	5.62 ± 0.21	6.06 ± 0.26	-0.21 ± 0.12	-0.16 ± 0.12
Bandit	0.50	7.79	7.67 ± 0.37	5.70 ± 0.25	-0.57 ± 0.09	-0.08 ± 0.06
Bandit	0.75	10.07	9.95 ± 0.36	4.95 ± 0.22	-1.32 ± 0.08	0.05 ± 0.04
Bandit	gap	4.71	4.62 ± 0.07	5.62 ± 0.21	-0.59 ± 0.20	-0.59 ± 0.20

We therefore added a same-stream comparison after inspecting the original results. Every saved reference policy was first checked against its committed canonical estimate to 10^{-9} , then re-evaluated on the held-out seeds with the family’s per-episode seeding, and the feasible envelope was recomputed. The Cap gap column of Table 4 pairs each seed’s mixture reward with its reference reward on the same episode seeds. Reference support and mixture weights are fitted on those same held-out seed means and held fixed for the SE. The resulting Student- t intervals are nominal pointwise plug-in summaries. They omit support reselection, uncertainty in mixture weights and cap feasibility, and multiplicity across budgets. They do not establish coverage for the population constrained frontier.

Under that fitted comparison, negative 95% intervals exclude zero at eight of eleven distinct budgets, counting Tiger’s duplicate once. The cap is slack at seven budgets: every Tiger target, Bandit’s three interior targets, and Diagnosis’s largest. At these budgets, the reference is the unpenalized SARSOP policy, the estimated unconstrained reward optimum. A gap there combines the cost of spending B with any inefficiency of the family. A predeclared target-matched reference separates the two. Subsidized SARSOP policies, with negative usage penalties, extend the sampled points above the unpenalized usage, and the reference becomes the best sampled mixture whose usage equals B (`run_frontier_`

`target_reference.py`). Against it, intervals exclude zero at two of eleven distinct budgets on the held-out seeds. Replaying the held-out-fitted mixtures on ten fresh seeds again gives two shortfalls, all at budgets whose held-out cap reference binds. Fresh realized usage need not equal B . Only Bandit’s gap-budget shortfall, -0.59 held out and -0.50 fresh, appears on both streams, and its held-out interval just reaches zero when the reference is matched to the mixture’s realized usage rather than to B (Appendix W.6). Diagnosis’s held-out shortfall at $B=9.53$ mostly pairs a policy on the usage plateau containing $w=1$ with the unpenalized SARSOP policy at nearly equal usage. The twenty-seed episode-matched check of Section 6.6 finds that pair equivalent within the margin, although on these five held-out seeds it differs by 1.01 ± 0.18 . Both references are maxima over noisy sampled points, so selection bias remains.

Tiger’s paired gaps measure only extra listening, since neither the family nor any policy in either reference support makes a wrong commit in the 500 held-out episodes. A wrong commit costs 110 relative to a correct one, and the canonical reference error rate of 0.6% is worth about 0.66 expected reward, a rare-event uncertainty the nominal intervals do not capture. If the reference erred at that rate while the family never did, the smallest-budget cap shortfall would reverse and the middle one would nearly vanish.

This study supports expected-usage calibration within the tested family, including gap targets, while quantifying its limits as a reward optimizer. If the engineering requirement is to spend a specified expected sensing amount for coverage or verification that reward does not represent, the endpoint mixture directly addresses it. If the requirement is only an upper limit and reward is the objective, the feasible mixture or constrained reference is the relevant comparator. The complete three-panel table, selection details, reference construction, and paired archives are retained in Appendix W.6.

6.9 Reward-Irrelevant Sensing

An expected sensing target controls the amount of sensing, but does not ensure that it concerns the task. We test this limit with Distractor Diagnosis, whose hidden state combines the usual four-way condition with an independent binary nuisance bit. Commit rewards depend only on the condition. An added test observes only the nuisance bit, while the original tests observe only the condition. Under the independent generative process the belief stays factored, so the distractor carries exactly zero information about the condition. Appendix T.2 states and tests that property, whose scope excludes correlated priors introduced through misspecification.

We sweep ordinary Planning+IG over sixteen weights with five seeds and 100 episodes per weight and seed, refining the grid where distractor use begins. The distractor fraction is zero through $w=3.5$, 0.103 ± 0.001 at $w=4$, 0.233 ± 0.003 at $w=5$, and approximately one third at the largest weights (Figure 10). Thus the observed onset bracket is $(3.5, 4.0]$. The canonical $w=1$ agent avoids the distractor on this instance because it is below onset, not because its objective

is structurally immune. At sufficiently high weights, genuine information about the nuisance bit earns more epistemic value than the test costs.

A reward-relevance variant scores information gain on the marginal belief over reward-equivalence classes. Two states belong to the same class when every commit action pays the same in them, so the partition is determined by the commit reward matrix already supplied to the agent. On this benchmark, the two states differing only in the nuisance bit share a class and the four classes are the four conditions. The planner still propagates the full joint posterior and changes only the belief on which information gain is scored. If no states are reward-equivalent, the variant reduces to ordinary information gain. This construction uses the known reward model and the benchmark’s factorization, rather than learning task relevance from data.

The variant buys zero distractor tests at every swept weight. At $w=100$, ordinary information gain buys 6.46 distractor tests, 33.2% of its usage, and earns -10.48 ± 0.11 reward against the variant’s -3.64 ± 0.36 . Seed-level Welch tests with Holm correction over the weight-by-metric family detect usage and distractor-count differences at every tested weight past onset. Reward differences survive that correction at $w=31.6$ and 100, with $p = 1.2 \times 10^{-5}$ and 1.5×10^{-5} , but not at the smaller post-onset weights. Success rates are identical through 31.6, and the variant’s nominal advantage at 100 is not significant.

Relevance weighting does not solve over-observation. At the two largest weights the variant still buys 13.16 task-relevant tests against 9.77 at $w=10$. Nor is the IDS adaptation used here immune to nuisance sensing. Its information about the optimal commit is correctly zero for the distractor, but its zero-denominator fallback uses raw state-entropy reduction and yields a distractor fraction of 0.305 ± 0.005 . This is a failure of that implementation choice, not evidence against reward-aware information measures in general. A sensing target and relevance weighting therefore address different design requirements. Appendix W.7 preserves the full curves, implementation analysis, comparison family, and remaining numerical results.

6.10 Additional Onset and Operating-Point Checks

Positive-threshold two-state instances place the observed onset within one refined grid step, 3%, above the closed-form threshold, while standard negative-threshold instances observe already at $w=0$ (Appendix W.2). The atlas collects the implicit $w=1$ budgets and two grid brackets per instance from the measured curves (Appendix T.3). These are measured operating points, not a universal predictive formula for the price.

7 Results

The second battery locates the canonical EFE weight within the policy family and compares it with other planners. It provides empirical context for the equivalence result, whose validity itself rests on the stated assumptions and proof.

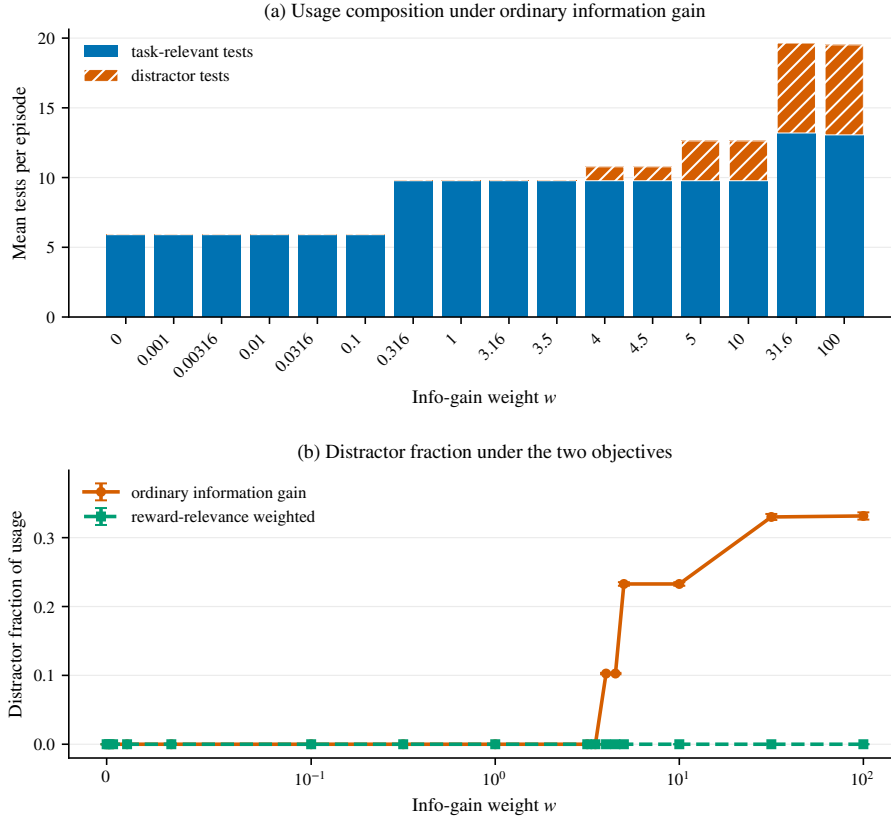


Fig. 10. Reward-irrelevant sensing on `DistractorDiagnosisEnv`. The upper panel decomposes ordinary Planning+IG’s usage into task-relevant and distractor tests as w grows. Distractor spend is zero below the onset bracket $(3.5, 4.0]$ and saturates at roughly a third of total usage. The lower panel plots the distractor fraction for ordinary information gain against the reward-relevance-weighted variant on the same grid. The weighted variant is flat at exactly zero everywhere. Error bars are seed-level SE over 5 seeds.

We retain the main comparisons here and place detailed per-environment mechanisms, robustness studies, and additional tables in Appendix X.

7.1 Core Environments

Table 5 contrasts $w=1$ EFE with myopic reward maximization, same-horizon reward-only Planning, and Planning+IG tuned for success within its own class. Reward-maximizing weights are considered separately below. On Tiger, EFE and Planning produce identical trajectories. With one observation action, the information term never changes the reward-based action choice at these stakes. This is a measured property of this instance rather than a general consequence of having a single sensor.

Table 5. Core comparison, 1,000 episodes per seed over five seeds. Reward is mean $\pm$ seed-level SE. Here $w^* = w^*_{\text{succ}}$ is tuned for success, with reward tuning reported separately. Bold identifies EFE and its exact Tiger tie with Planning, not a per-column winner. Success-tuned Planning+IG buys greater accuracy at a reward cost. Full agents, pooled uncertainties, and statistical comparisons are in Appendices B and I.

Environment	Agent	Obs.	Success	Reward
Tiger $H=6, w^*=50$	Myopic	1.00	85.3%	-7.15 ± 0.41
	Planning	4.20	99.4%	$+5.19 \pm 0.21$
	Planning+IG	5.66	99.9%	$+4.21 \pm 0.08$
	EFE	4.20	99.4%	$+5.19 \pm 0.21$
Diagnosis $H=3, w^*=50$	Myopic	2.00	64.7%	-13.17 ± 0.58
	Planning	5.86	88.8%	-2.57 ± 0.20
	Planning+IG	13.16	99.2%	-3.66 ± 0.12
	EFE	9.66	97.2%	-1.37 ± 0.18
Bandit $H=2, w^*=20$	Myopic	2.06	62.2%	$+5.56 \pm 0.09$
	Planning	3.29	71.0%	$+5.75 \pm 0.08$
	Planning+IG	9.57	98.9%	$+5.11 \pm 0.04$
	EFE	5.11	86.9%	$+6.27 \pm 0.04$

On Diagnosis, EFE raises success from 88.8% to 97.2% relative to Planning and improves reward from -2.57 to -1.37 . Seed-level Welch tests give $p = 4.0 \times 10^{-8}$ for success and $p = 0.002$ for reward. On Bandit it raises success from 71.0% to 86.9% and reward from 5.75 to 6.27, with $p = 2.1 \times 10^{-5}$ and $p = 0.001$. These differences survive the stated Holm correction. Thus EFE Pareto-dominates same-horizon reward-only Planning on these two environments. Success-tuned Planning+IG reaches higher accuracy, 99.2% and 98.9%, but pays for more sensing and attains lower reward. EFE supplies an untuned trade-off, not the best value of every metric.

Two baselines limit any attribution of these outcomes solely to explicit information gain. Posterior-vote, which draws states from the belief and votes over

their optimal actions without an information term, is not significantly different from EFE on any Bandit metric. Its reward is nominally higher, 6.32 against 6.27 ($p = 0.58$). It also beats EFE on Testbed, but performs worse on Diagnosis and Tiger. The Epistemic-only ablation commits immediately at chance-level success on all three core environments and Tileworld. The units force this outcome, since in bits no observation there yields as much information as it costs. Full baselines, pooled effect sizes, and seed-level tests appear in Appendices B, I, and X.1. Reward effects against Planning are small at the episode level on Diagnosis and Bandit despite statistically detectable mean differences. We do not interpret the mechanically inflated effect sizes computed from only five seed means.

POMCP provides a search-based comparison, reported in Appendix S. Its simulation count per decision is a computational budget distinct from the sensing budget B . The MCTS-EFE comparison matches simulation counts, while a separate Tiger control matches wall-clock computation. The POMCP comparisons outside the exploration sweep use untuned exploration constants. A separate sweep selects constants on tuning seeds. The selection rule was written after inspection of an exploratory sweep on the canonical seeds and was recorded before the runs reported here. Neither the POMCP comparison nor the component ablation isolates directed observation selection as the cause of every performance difference. These qualifications remain necessary when comparing search architectures, even when the information-weighted family performs well.

7.2 Where the Information-Unit Weight Sits

We sweep $w \in \{0.01, 0.1, 0.5, 1, 2, 5, 10, 20, 50, 100, 200\}$ on Tiger, Diagnosis, Bandit, Testbed, and Tileworld. Figure 11 shows reward against success. The $w=1$ Planning+IG and EFE markers coincide under Proposition 1 and shared tie-breaking. The empirical question is where that common policy lies relative to other weights, not whether equivalent implementations differ.

The weight-one policy is within the reward-maximizing run of sampled weights on Tiger (0.01 through 20), Diagnosis (0.5 through 20), and Bandit (1 and 2). These are sampled plateaus, not claims about every untested weight between grid points. Testbed instead maximizes reward across sampled weights from 0.01 through 0.5. Its weight-one policy buys 6.4 percentage points of success at a reward cost of 0.11. On Tileworld, $w=20$ Pareto-dominates $w=1$, with reward -18.95 against -21.53 and success 91.8% against 72.0%. Thus the canonical weight attains the sampled reward maximum on three of five environments and is suboptimal on two.

Success reaches its maximum at the upper grid endpoint 200 on all five environments, although it dips from $w=0.5$ to $w=1$ on Bandit and Tileworld. The reward cost need not grow smoothly. On Tiger and Diagnosis, the tested weights from 1 up to the point below 50 leave reward unchanged, and the step to 50 incurs the measured cost. The grid does not exclude a cheaper untested intermediate policy. The advantage of $w=1$ here is that it selects a useful default

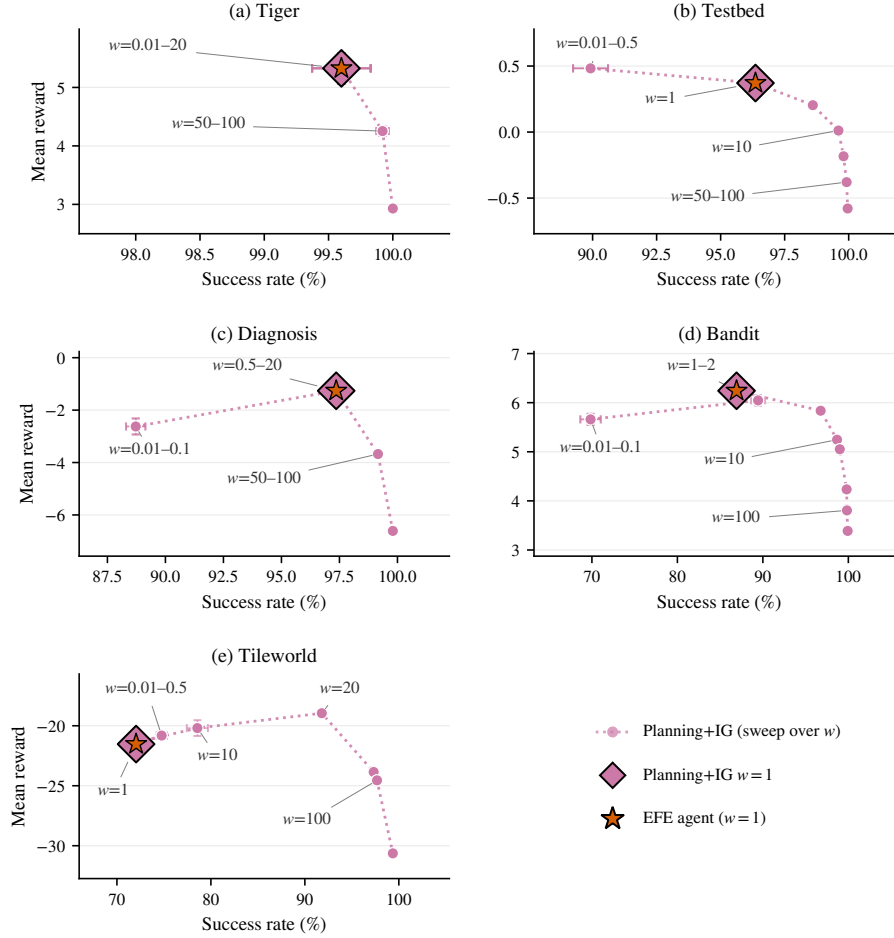


Fig. 11. Reward and success across eleven information weights from 0.01 to 200. The $w=1$ Planning+IG diamond and EFE star coincide under their equivalence assumptions and shared tie-breaking. They attain the largest sampled mean reward on Tiger, Diagnosis, and Bandit. On Testbed they trade reward for success, and on Tileworld $w=20$ dominates them on both axes. Labels identify sampled weights sharing a point. Error bars are seed-level SE, visible where larger than markers.

without environment-specific search under the stated reward convention. It does not remove dependence on units, promise best accuracy, or replace calibration when a particular sensing target is required. Appendix X.2 retains the detailed trade-offs, and Appendix R compares the implementation’s bit-unit default with the exactly nat-canonical weight.

7.3 Tileworld

Tileworld shows why horizon and observation structure must accompany a scaling claim. At 6×6 and $H=2$, EFE and Planning are not significantly different in reward (-21.53 against -21.39 , $p = 0.82$) or success (72.0% against 73.8%, $p = 0.17$). EFE uses fewer scans, 14.75 against 15.64 ($p = 8.8 \times 10^{-5}$). These are failures to reject reward and success differences, not TOST equivalence. The success-tuned $w=50$ agent reaches 97.3% accuracy but uses 32.25 scans and earns lower reward. Table 35 and Figure 25 retain the full comparison and an illustrative trajectory in Appendix X.3.

In a separate fixed- $H=2$ sweep (Figure 12), Planning takes no scans at 8×8 and achieves $1.4\% \pm 0.3\text{pp}$ success, while EFE achieves $69.4\% \pm 1.0\text{pp}$. The short horizon cannot value enough scans to repay their costs before committing on a large grid. The contrast is therefore with a non-scanning finite-horizon agent, not with reward-only planning at arbitrary depth. At 4×4 and 6×6 the two are within one SE, with no formal test for that sweep. The advantage appears at a horizon-and-scale threshold rather than widening uniformly with size. Its weighted comparators use fixed $w=100$, not class-specific tuning. Random and overlapping scan partitions sharply reduce all agents’ absolute performance and do not separate EFE from Planning on reward or success, although EFE uses fewer scans in all three modes. Appendix X.3 preserves these robustness results and their scope.

7.4 RockSample

RockSample tests the information weight within a shared factored belief-space search and a shared greedy leaf rule. On RS[5,3], EFE (16.47) and the $w=5$ comparator (16.41) are not significantly different. On RS[7,4], EFE earns 15.96 against 14.56 and leads the weighted family ($p < 10^{-5}$). On RS[7,8], EFE (21.85) and $w=5$ (23.76) are not significantly different after correction, and both outperform reward-only Planning (12.31). Raising the weight from 5 to 10 significantly hurts on two of these instances and does not significantly help on the third. Only weights 1, 5, and 10 were compared here, so this is not a dense reward-tuning claim.

The important counterexample is RS[11,11]. EFE and Planning both earn 13.23, while a standalone approach-then-check heuristic earns 28.66 ± 0.83 . That heuristic also beats both weighted agents on RS[7,8]. With this shared leaf rule, a larger state space therefore does not by itself increase the value of information weighting. At feasible search depths, distant useful checks lie beyond the tree, and the shared leaf rule values untouched rocks at zero under the uniform prior.

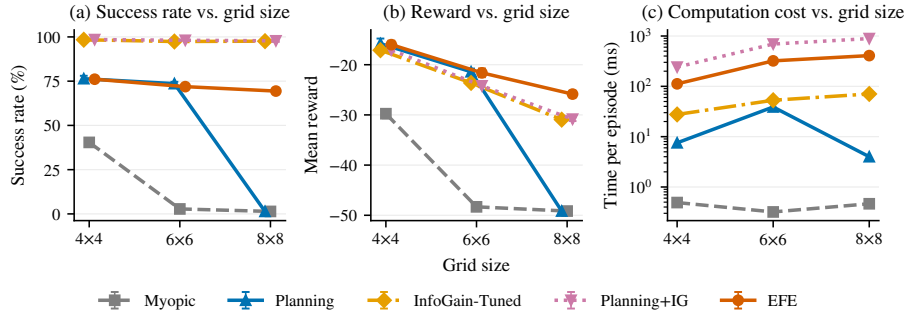


Fig. 12. Tileworld scaling ($H=2$, 200 episodes per seed $\times$ 5 seeds) across Myopic, Planning, InfoGain-Tuned, Planning+IG, and EFE. Planning collapses at 8×8 ($|S|=64$), while EFE maintains 69.4%. Planning+IG at a fixed $w=100$, not tuned in class, achieves 97.8% but at higher reward cost. Error bars in (a) and (b) are seed-level SE, tied series are offset slightly along the categorical axis for legibility, and per-episode time in (c) is a pooled mean over all episodes and seeds, shown without error bars on a log scale.

It also charges a phantom trip to the east edge before exit, although this variant allows exit anywhere, biasing depth-capped continuations toward early exit. The cost is shared across the weighted family but limits absolute performance. Deeper POMCP does not repair this instance in the tested configurations. Behavioral traces show many distant checks, and reward declines from 12.94 at horizon 10 to -54.78 at horizon 40. Table 6 gives the full comparison. Appendix X.4 retains the leaf-rule disclosure, depth check, and heuristic comparison, alongside the extended solver analysis in Appendix M.

7.5 Structural Inspection

Inspection makes the reward-accuracy trade-off explicit. At $N=8$, EFE improves accuracy from Planning’s 73.0% to 91.1% while reward falls from -17.85 to -20.95 . At $N=16$, it improves accuracy from 78.2% to 90.8%, with reward -45.80 against -45.42 , a non-significant difference ($p = 0.57$). The $w=5$ agent attains higher accuracy, 97.3%, but uses 38.3 tests against EFE’s 28.1 and earns -54.74 . These results demonstrate feasible factored search on a synthetic known-model problem with a hand-coded leaf rule. They do not validate industrial deployment or show that EFE uniquely optimizes either reward or accuracy. Table 7 reports both instances, with the qualified two-state interpretation in Appendix X.5.

Across the battery, the canonical weight is useful when the reward-only policy leaves valuable, representable sensing opportunities unused. Its benefit is bounded by the search horizon, sensor structure, reward convention, and objective of the comparison. It matches the sampled reward maximum on three of five swept environments, loses to other weights on two, and gains nothing over Planning on the largest RockSample instance. These limits matter alongside its

positive results on Diagnosis, Bandit, and the largest fixed-horizon Tileworld grid.

8 Discussion

What a sensing target determines. The method is useful when expected sensing usage is the intended design requirement. A target in observation count or cost gives an engineer an interpretable input, and scale equivariance makes the corresponding transformation of w predictable when reward units change. The target does not determine which uncertainty should be reduced or maximize reward under a cap. Ordinary information gain may reward irrelevant sensing. In the distractor study, scoring information on reward-equivalence classes removes that spending at every swept weight, while the budget controls how much sensing is bought overall (Section 6.9).

Even within the chosen family, calibration is not a general policy optimizer. A plateau can contain many weights with the same usage, and the selected endpoint mixture need not maximize reward among all mixtures meeting the target. The held-out budget study quantifies both distinctions, against a sampled constrained reference and against alternative mixtures over the calibration grid. Its targets come from a predeclared rule applied to that grid, not from an application-specified inspection requirement. Deployment would additionally require checking the supplied model and the stability of calibration under the intended operating conditions.

How to interpret the information-unit default. The recursive EFE identity fixes $w = 1$ once reward and information units have been declared. It does not eliminate that convention. The weight belongs to the level set of its implicit budget $U(1)$, and inversion need not return the same weight. For the specified recursion, the formal equivalence is exact under its assumptions. The claim that $w = 1$ is a useful default is empirical: it lies in the reward-maximizing tied run of sampled weights on three of five swept environments and is beaten on the other two. Its near-optimality in a random two-state family also depends on horizon and the chosen reward-gap criterion. Even at $H = 3$, approximately 21% of that family fails the study’s criterion (Appendix J).

The comparison between reward-tuned and success-tuned weights reflects different objectives. A larger weight can improve success while spending more on tests and reducing reward. Moderate transferred weights perform well on parts of the suite under shared reward units, but transfer at the top of the grid can be costly (Table 36). These observations support checking reward, success, and usage separately. They do not establish a universal operating point.

Which problem structures benefit. In the tested regimes, EFE helps when the available search can exploit informative observations that a reward-only planner undervalues. Multiple observation actions create opportunities to select useful tests, but neither their number nor state-space size alone predicts an advantage. On the largest Tileworld grid, the fixed-horizon reward-only planner takes

no scans, making the separation a joint horizon-and-scale effect. On RockSample[11,11], the advantage disappears, and on the two larger RockSample instances an unbounded-horizon heuristic beats the tree-search agents. Navigation supplies a further negative case: no epistemic agent significantly beats the reward-only greedy mover at any tested grid size (Appendix F).

Planning depth and discounting also change the value of information. At one-step depth the recursive objective reduces to myopic reward plus information gain at $w = 1$. The observed discount threshold in the benchmark sweep is not a general regime boundary. Appendix J retains the horizon study, Appendix K the discount comparison, and Appendix Y the weight-transfer results. Model accuracy matters independently. EFE and reward-only Planning respond identically to every tested Tiger sensor mismatch, so that study supplies no EFE-specific robustness benefit. Overestimating sensor accuracy can trigger premature commits (Appendix L).

What the tree-search controls establish. Exact belief-tree enumeration limits practical planning depth, motivating MCTS-EFE. The principal comparison with POMCP matches simulation counts, with different costs per simulation. Tuning POMCP’s exploration constant narrows the Tiger gap without closing it. With both solvers’ constants selected on disjoint tuning seeds, MCTS-EFE remains significantly ahead on Tiger success and reward. On Diagnosis and Tileworld, the gap survives the tested constants and information-gain rollouts. The ablation finds no detectable effect of either in-tree information gain or max-backup on success or reward at five seeds after multiplicity correction.

The remaining components, EFE leaf evaluation, exact beliefs, and exact commit-child values, were held fixed together. The experiments therefore evaluate their joint configuration and do not isolate directed observation selection as the cause of the advantage. Nor do they establish superiority over POMDP solvers in general. Full solver configurations, timings, controls, and outcomes appear in Appendices Y and S.

Structural and practical limits. The equivalence covers observation and navigation actions that preserve the hidden state. When sensing changes that state, the epistemic term must use transition-aware predictive and posterior distributions. The transition–observation coupling diagnostic Δ_T identifies a failure of the reduction’s hypothesis, but has not been proved to be an additive correction or a decision-error bound. Small transition probabilities do not guarantee a small diagnostic: with perfect sensing, a posterior support change can make the divergence infinite. Example 1 gives a concrete failure, and Appendix U develops the full scope discussion.

All experiments assume a known generative model. The large Structural Inspection state space demonstrates feasibility on a synthetic benchmark with a hand-coded leaf rule, not on a deployed inspection system. Model learning, more efficient search with many sensing actions, transition-aware epistemic objectives, and validation against an actual operational sensing requirement remain open

work. These limits distinguish the exact mathematical results from the empirical guidance drawn from this suite.

9 Conclusion

An expected sensing-usage requirement can calibrate the information-gain weight of a ρ -POMDP policy family. The operational shadow price is a crossing threshold, estimated by a grid bracket. Under stated conditions, mixing the bracket’s endpoint policies meets a usage-gap target in expectation. The receding-horizon planner is scale equivariant, and a projected decaying-step controller has a stationary-convergence guarantee under explicit hypotheses. Re-adaptation after reward-scale shifts is an empirical result, including a useful reset heuristic that the theorem does not cover.

The held-out study shows accurate usage calibration but also its reward cost. Meeting an equality target can earn less than spending below it, and the selected mixture need not be optimal within the family or against the constrained reference. Calibration is appropriate when expected usage itself is the requirement.

The recursive EFE objective supplies an information-unit anchor at $w = 1$, with exact reward–information identification under log scoring and an explicit convention for ordinary rewards. This weight is competitive in several benchmarks and suboptimal in others. Its practical value depends on the observation structure, planning depth, and model assumptions. The resulting design procedure is to state the intended usage, calibrate within a specified policy family, and assess usage, reward, and success together.

Competing interests. The authors have no competing interests to declare that are relevant to the content of this article.

Acknowledgments

The authors used Anthropic’s Claude and OpenAI’s Codex as assistants for drafting and revising text, implementing and testing code, running and summarizing experiments, checking citations, preparing figures, and generating simulated referee reviews. The authors are responsible for the originality, accuracy, and integrity of the manuscript and its associated artifacts. No AI system is an author of this work.

Table 6. RockSample results. RS[5,3] and RS[7,4] run 500 episodes $\times$ 10 seeds (5,000 episodes), and RS[7,8] and RS[11,11] run 100 episodes $\times$ 5 seeds (500 episodes, reduced protocol). Greedy samples without checking. POMCP is the search-tree solver at its tuned, frozen configuration (2,048 simulations, Appendix M). Rollout only runs POMCP’s approach-then-check rollout policy standalone, with no search tree. It is not significantly different from EFE on RS[5,3] and RS[7,4] and earns more on RS[7,8] and RS[11,11]. Bold marks the best mean reward per instance plus any agent not significantly different from it by a seed-level Welch t -test with Holm–Bonferroni correction, corrected within metric (complete pairwise statistics in the committed `results_rocksample*_stats.csv` files). Reward reported as mean $\pm$ seed-level SE (SE of the per-seed means over 10 seeds for RS[5,3] and RS[7,4] and over 5 seeds for RS[7,8] and RS[11,11]). Steps and Checks omitted here (per-instance step and check counts are in Appendix M).

Instance	Agent	Good	Bad	Reward
RS[5,3]	Greedy	1.49	1.51	$+3.71 \pm 0.26$
	Planning ($w=0$, $d=3$)	0.99	0.04	$+14.28 \pm 0.13$
	Plan+IG ($w=5$, $d=3$)	1.44	0.01	$+16.41 \pm 0.13$
	EFE ($w=1$, $d=3$)	1.44	0.03	$+16.47 \pm 0.12$
	POMCP (2048 sims)	0.68	0.03	$+12.74 \pm 0.11$
	Rollout only (approach)	1.41	0.07	$+16.67 \pm 0.13$
RS[7,4]	Greedy	1.98	2.02	-1.38 ± 0.36
	Planning ($w=0$, $d=4$)	0.95	0.05	$+13.97 \pm 0.11$
	Plan+IG ($w=5$, $d=4$)	1.69	0.01	$+14.56 \pm 0.14$
	EFE ($w=1$, $d=4$)	1.44	0.01	$+15.96 \pm 0.15$
	POMCP (2048 sims)	0.66	0.03	$+12.01 \pm 0.06$
	Rollout only (approach)	1.88	0.10	$+15.85 \pm 0.17$
RS[7,8]	Greedy	4.02	3.98	-5.60 ± 1.03
	Planning ($w=0$, $d=3$)	0.48	0.03	$+12.31 \pm 0.35$
	Plan+IG ($w=5$, $d=3$)	3.40	0.01	$+23.76 \pm 0.58$
	EFE ($w=1$, $d=3$)	2.61	0.06	$+21.85 \pm 0.79$
	POMCP (2048 sims)	1.71	0.10	$+15.07 \pm 0.45$
	Rollout only (approach)	3.83	0.19	$+28.33 \pm 0.82$
RS[11,11]	Greedy	5.48	5.52	-18.90 ± 1.04
	Planning ($w=0$, $d=2$)	0.48	0.03	$+13.23 \pm 0.18$
	Plan+IG ($w=5$, $d=2$)	0.50	0.03	$+13.18 \pm 0.15$
	EFE ($w=1$, $d=2$)	0.48	0.03	$+13.23 \pm 0.18$
	POMCP (2048 sims)	0.49	0.03	$+12.94 \pm 0.24$
	Rollout only (approach)	5.22	0.23	$+28.66 \pm 0.83$

Table 7. Structural Inspection results. $N=8$ and $N=16$ run 2,500 episodes each (500 per seed $\times$ 5 seeds). Reward and accuracy are reported as mean $\pm$ seed-level SE. Greedy diagnoses without testing. Bold marks the best accuracy and the best reward per instance, plus any value within sampling error of that best (seed-level Welch $p > 0.05$, uncorrected). The two are not always the same agent. EFE is a competitive untuned trade-off, not uniquely best on all metrics. Accuracy is the fraction of components correctly diagnosed.

Instance	Agent	Accuracy	Missed	Tests	Reward
$N=8$ $ S =256$	Greedy	$70.7 \pm 0.5\%$	2.34	0.0	-114.33 ± 1.96
	Planning ($w=0$)	$73.0 \pm 0.3\%$	0.07	12.7	-17.85 ± 0.23
	Planning+IG ($w=5$)	$96.3 \pm 0.1\%$	0.10	19.7	-27.98 ± 0.44
	EFE ($w=1$)	$91.1 \pm 0.2\%$	0.09	15.5	-20.95 ± 0.30
$N=16$ $ S =65,536$	Greedy	$70.5 \pm 0.4\%$	4.72	0.0	-233.09 ± 3.13
	Planning ($w=0$)	$78.2 \pm 0.2\%$	0.24	22.4	-45.42 ± 0.23
	Planning+IG ($w=5$)	$97.3 \pm 0.0\%$	0.14	38.3	-54.74 ± 0.40
	EFE ($w=1$)	$90.8 \pm 0.1\%$	0.24	28.1	-45.80 ± 0.60

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A Proof of Proposition 1

Proof. Let $V_\rho(a, b, d)$ be the ρ -POMDP action value at belief b and depth d , with ρ_{EFE} defined in the proposition. Define $V(a, b, d) := -\mathcal{G}(a, b, d)$. Negation reverses ordering, so the minimizing actions for $\mathcal{G}$ are the maximizing actions for V .

For any terminal commit action i and $d \leq H$, Equation 5 gives

$$V(\text{commit}_i, b, d) = \mathbb{E}_b[R_i] = V_\rho(\text{commit}_i, b, d),$$

because $\rho_{\text{EFE}}(b, \text{commit}_i) = 0$ and there is no continuation. At depth H , only these commit actions are available, establishing the induction boundary.

For observation action k at $d < H$, negating Equation 3, with its depth arguments explicit, gives

$$V(\text{observe}_k, b, d) = -c_k + I_k(b) + \mathbb{E}_o \left[\max_{a'} V(a', b'_o, d + 1) \right].$$

The ρ -POMDP backup has the same immediate reward $-c_k$, belief utility $I_k(b)$, posterior distribution, and continuation depth. Under $\gamma = 1$, it is therefore the same expression with V_ρ replacing V . Induction on the remaining horizon $H - d$ gives $V(a, b, d) = V_\rho(a, b, d)$ for every available action and belief node. Their maximizing action sets coincide, and the shared tie-breaking rule selects the same action. Hence the policies agree throughout the finite-horizon tree and, when replanned from the same beliefs, at each real decision.

B Full Per-Environment Results

This appendix reports the per-environment results for Tiger, Diagnosis, Bandit, and the two Tileworld grids, omitting the duplicate rows identified in Appendix X.1.

Table 8. Tiger problem (5,000 episodes at 1,000 per seed $\times$ 5 seeds, $H=6$). Reward asymmetry $+10/-100$. Bold marks the best-reward row and any row bit-identical with it or within sampling error of it (seed-level Welch $p > 0.05$, uncorrected, the conservative direction for a tie claim), here and in the Diagnosis and Bandit tables of this appendix. Printed uncertainty in these tables is pooled episode-level SE, while the bolding rule uses the seed-level test.

Agent	Listens	Success	Reward
Myopic	1.00	85.3%	-7.15 ± 0.55
Planning ($H=6$)	4.20	99.4%	$+5.19 \pm 0.12$
Info Gain ($w=20$)	4.20	99.4%	$+5.19 \pm 0.12$
Planning+IG ($w=50$)	5.66	99.9%	$+4.21 \pm 0.06$
Epistemic-only	0.00	49.7%	-45.31 ± 0.78
Posterior-vote	2.64	96.8%	$+3.79 \pm 0.28$
EFE ($H=6$)	4.20	99.4%	$+5.19 \pm 0.12$

Table 9. Diagnosis ($N=4$, $K=2$ tests, $H=3$, 5,000 episodes at 1,000 per seed $\times$ 5 seeds).

Agent	Tests	Success	Reward
Myopic	2.00	64.7%	-13.17 ± 0.41
Planning ($H=3$)	5.86	88.8%	-2.57 ± 0.27
Info Gain ($w=100$)	13.19	99.4%	-3.58 ± 0.09
Planning+IG ($w=50$)	13.16	99.2%	-3.66 ± 0.10
Epistemic-only	0.00	24.7%	-35.16 ± 0.37
Posterior-vote	5.83	88.4%	-2.78 ± 0.27
EFE ($H=3$)	9.66	97.2%	-1.37 ± 0.15

Table 10. Structured bandit ($K=4$ arms, $H=2$, 5,000 episodes at 1,000 per seed $\times$ 5 seeds).

Agent	Inspections	Success	Reward
Myopic	2.06	62.2%	$+5.56 \pm 0.06$
Planning ($H=2$)	3.29	71.0%	$+5.75 \pm 0.06$
Info Gain ($w=20$)	9.54	98.8%	$+5.12 \pm 0.04$
Planning+IG ($w=20$)	9.57	98.9%	$+5.11 \pm 0.04$
Epistemic-only	0.00	24.7%	$+3.23 \pm 0.05$
Posterior-vote	5.01	87.0%	$+6.32 \pm 0.05$
EFE ($H=2$)	5.11	86.9%	$+6.27 \pm 0.05$

Table 11. Tileworld 6×6 (2,500 episodes at 500 per seed × 5 seeds, $H=2$). This covers every agent in the Tileworld battery (`experiments/run_tileworld.py`). POMCP and MCTS-EFE results on this grid, run under their own protocols, are in Table 27 and Appendix S. Posterior-vote and PyMDP-AIF were not run on the spatial environments. Bold marks EFE and any agent not significantly different from it at the seed level (Welch $p > 0.05$, uncorrected), not the single best value per column. Epistemic-only commits immediately (0 scans) rather than exhausting the step cap, matching Myopic exactly (Section 7). Reward uncertainty is pooled episode-level SE, matching the other tables of this appendix.

Agent	Scans	Success	Reward
Myopic	0.00	2.9%	-48.25 ± 0.20
Planning ($H=2$)	15.64	73.8%	-21.39 ± 0.54
Info Gain ($w=1$)	5.20	31.7%	-36.16 ± 0.56
Info Gain ($w=100$)	32.30	98.0%	-23.50 ± 0.24
Planning+IG ($w=50$)	32.25	97.3%	-23.85 ± 0.26
Epistemic-only	0.00	2.9%	-48.25 ± 0.20
EFE ($H=2$)	14.75	72.0%	-21.53 ± 0.54

Table 12. Tileworld 8×8 (2,500 episodes at 500 per seed × 5 seeds, $H=2$). The agent set and bolding rule are those of the 6×6 table, so the higher success rates of the two heavily scanning tuned agents are unbolded by design. The separate spatial-scaling sweep (Figure 12), with a different per-seed episode count, gives 69.4% for EFE and 1.4% for Planning, gaps consistent with seed-to-seed variation. Planning, untuned Info Gain, and Epistemic-only take no scans and tie Myopic exactly, because a single noisy scan over 64 cells barely narrows the belief (Planning’s depth-2 case in Appendix X.3). Reward uncertainty is pooled episode-level SE, matching the other tables of this appendix.

Agent	Scans	Success	Reward
Myopic	0.00	1.8%	-48.94 ± 0.16
Planning ($H=2$)	0.00	1.8%	-48.94 ± 0.16
Info Gain ($w=1$)	0.00	1.8%	-48.94 ± 0.16
Info Gain ($w=100$)	39.45	97.2%	-31.15 ± 0.25
Planning+IG ($w=50$)	39.49	97.5%	-31.00 ± 0.25
Epistemic-only	0.00	1.8%	-48.94 ± 0.16
EFE ($H=2$)	17.60	69.8%	-25.69 ± 0.56

C Two-State Testbed

Table 13. Information-seeking testbed (5,000 episodes at 1,000 per seed $\times$ 5 seeds, $H=4$). Symmetric rewards $+1/-1$. Each tuned agent is success-tuned in its own agent class, as `tune_info_gain_weight` requires, at $w_{\text{succ}}^*=20$ for myopic Info Gain and $w_{\text{succ}}^*=10$ for Planning+IG (both recorded in `results_supplementary_tuned_weights.csv`). At their respective tuned weights the two converge to the same policy and are bit-identical on all three metrics. Bold marks the best value in the success and reward columns, with exact ties bolded together. Every row here is produced by `experiments/run_supplementary.py`.

Agent	Obs.	Success	Reward
Myopic	1.00	76.3%	$+0.43 \pm 0.01$
Planning ($H=4$)	3.15	89.8%	$+0.48 \pm 0.01$
Info Gain ($w_{\text{succ}}^*=20$)	9.79	99.5%	$+0.01 \pm 0.01$
Planning+IG ($w_{\text{succ}}^*=10$)	9.79	99.5%	$+0.01 \pm 0.01$
Posterior-vote	3.15	89.8%	$+0.48 \pm 0.01$
EFE ($H=4$)	5.50	96.3%	$+0.38 \pm 0.01$

Under symmetric rewards ($+1/-1$), EFE gathers more information than is instrumentally optimal ($+0.38$ vs. Planning’s $+0.48$, seed-level Welch $p=1.1 \times 10^{-5}$) because uncertainty reduction has intrinsic value under EFE. The success-tuned agents show the same effect more starkly, driving reward to $+0.01$ to reach 99.5% success. Posterior-vote matches reward-only Planning bit-for-bit ($+0.48$, 89.8%, 3.15 observations), and both beat EFE on reward. The Epistemic-only ablation of Table 1, which sets $\mathcal{G}(\text{commit}) = 0$, also matches reward-only Planning on observations and success here (`results_summary.csv`), because Testbed’s observation is cheap enough that information gain alone clears its cost. On Tiger, Diagnosis, Bandit, and Tileworld it instead commits immediately at the chance rate (Appendix B). With symmetric ± 1 rewards a second observation at the post-first-observation belief has zero instrumental value, since the disconfirming outcome returns the belief exactly to the tie point, so its 0.1 cost buys no reward. Proposition 2’s over-observation ceiling is correspondingly tight, $w_{\text{over}}^* \approx 1.01$ reward units per nat (Table 31), below the implemented bit-denominated $w=1$ at 1.44 reward units per nat, so in this regime EFE assigns too much weight to information gain. The high-asymmetry regime in which the interval analysis favors $w=1$ is discussed in Appendix Y. Within that regime, $w=1$ lies on the reward-maximizing plateau on Tiger and Diagnosis but is Pareto-dominated by $w=20$ on Tileworld (Section 7.2).

D Supplementary Figures

These figures support the main text but are not needed to follow it.

D.1 EFE Trajectory Decomposition

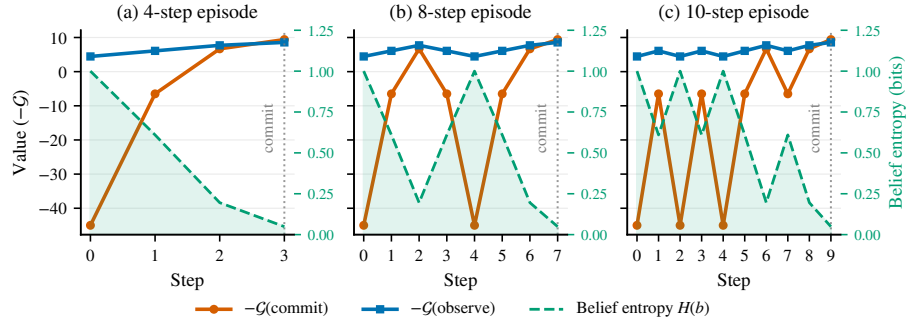


Fig. 13. EFE decomposition within Tiger episodes ($H=6$). The agent commits at the crossover where commit value (circles) exceeds observe value (squares). As belief entropy (dashed line with shading) decreases, the transition from exploration to exploitation occurs automatically.

Figure 13 decomposes EFE within Tiger episodes. The transition from exploration to commitment needs no separate stopping rule.

D.2 Observation Action Scaling and Adaptive Test Selection

Holding $N=8$ fixed on Diagnosis and varying K from 1 to 3 isolates the “which information” hypothesis (Figure 14). EFE and Planning are bit-identical at $K=1$ and within seed-level sampling error at $K=2$ (43.2% against 43.8% success). Evidence of a gap with two tests comes from the main Diagnosis battery of Section 7. They separate at $K=3$ (96.0% against 82.8%), a threshold rather than steady growth. Figure 15 shows the mechanism in one extended episode. Tests become differentially informative as belief concentrates, because some partitions stop mattering once the condition is narrowed down while others remain decisive, and the agent shifts its test selection accordingly. It optimizes a recursive pragmatic-epistemic score, so its chosen test need not maximize the one-step information gain plotted in panel (b).

D.3 Reward Asymmetry on Tiger

To test whether EFE’s Tiger parity holds across reward scales, we sweep the penalty magnitude from $|R^-|=1$ to 500 (Figure 16). EFE adapts its listening on its own, from 2.7 observations at the smallest penalty to 5.8 at the largest, tracking the stakes without reconfiguration. The sweep re-tunes the myopic InfoGain-Tuned agent’s weight at every penalty, and the tuned weight comes back as $w=20$ at all of them. At that fixed weight the myopic agent listens 4.37 times at every

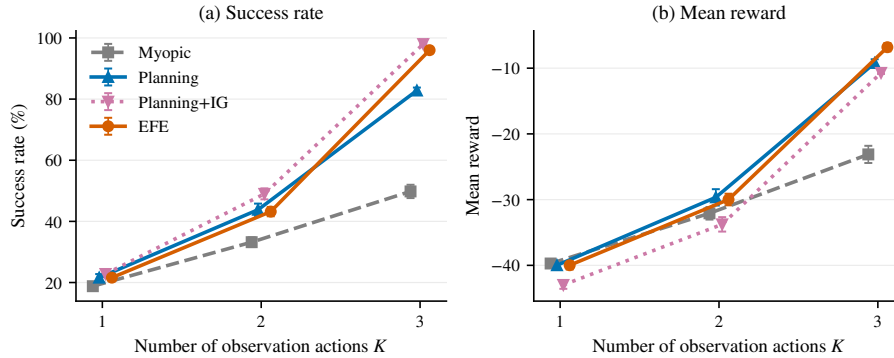


Fig. 14. Observation action scaling on Diagnosis ($N=8$, $H=3$, 500 episodes). EFE separates from Planning only once several tests compete. The gap is zero at $K=1$ (bit-identical policies), -0.6pp and within seed-level sampling error at $K=2$, and $+13.2\text{pp}$ at $K=3$, where choosing among differentially informative tests is exactly what the information-gain term contributes. InfoGain-Tuned is not part of this battery, so the legend carries four agents rather than the five of Figure 16.

penalty, while the deeper Planning+IG rises to 5.77 observations at penalties 200 and 500. At low stakes that over-listening costs about 1.6 reward against reward-only Planning. At the largest penalty the fixed-weight myopic agent sits marginally ahead of EFE (4.61 against 4.23), so the case for adaptation is robustness across the sweep rather than uniform dominance.

D.4 Belief Evolution

Figures 17 and 18 follow belief within single episodes on the 6×6 Tileworld and on extended Diagnosis instances. EFE narrows by partition. Early tests eliminate large groups of states and later tests discriminate among what is left. Reward-only Planning commits earlier and, over the battery, less accurately (Section 7 and Figure 14), though in the plotted episode its fourteenth observation already leaves the belief concentrated on the true state. InfoGain-Tuned keeps testing past the point where further tests pay for themselves in reward.

D.5 Exploration Efficiency

Figure 19 averages the temporal dynamics of epistemic foraging over 300 episodes. EFE reduces entropy at a rate comparable to tuned Planning+IG but commits earlier, staying out of the diminishing-returns regime. Its survival curve falls off through steps 10 to 25 while Planning+IG’s extends further right.

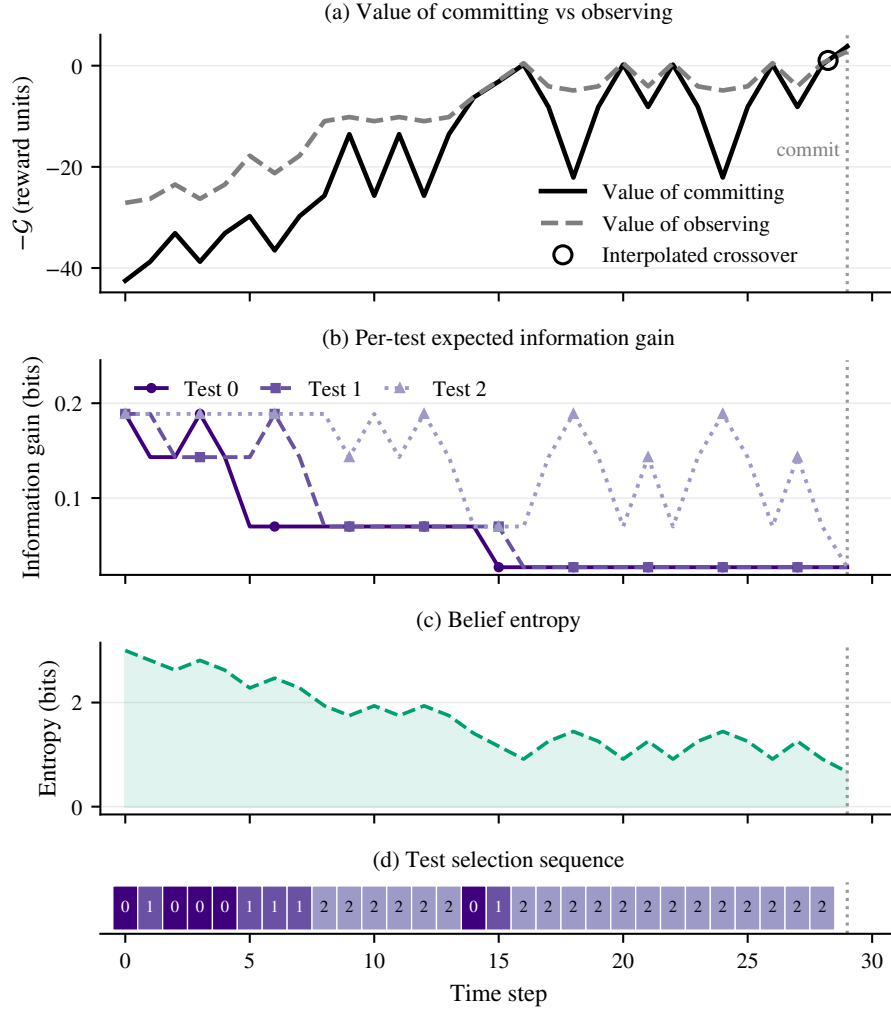


Fig. 15. EFE decomposition over an extended Diagnosis episode ($N=8$, $K=3$). (a) Value of committing vs. observing. The open circle marks the value crossover interpolated between sampled steps. The dotted vertical line marks the actual discrete commit step in all four panels. (b) Per-test expected information gain. Tests become differentially informative as belief concentrates, and the agent shifts its test selection as they do, though the recursive score it optimizes is not the one-step information gain plotted here. (c) Belief entropy decay. (d) Test selection sequence showing the agent’s adaptive strategy.

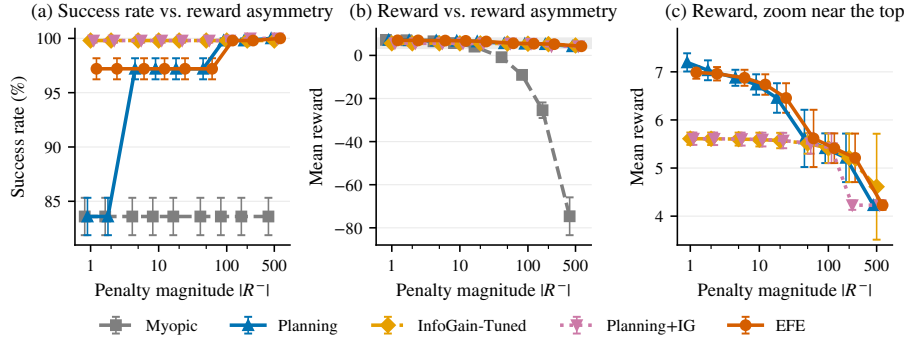


Fig. 16. Reward asymmetry sweep on Tiger ($H=6$, 500 episodes per point, 5 canonical seeds). EFE and Planning adapt their listening as the penalty grows, while the myopic InfoGain-Tuned agent at its fixed tuned weight listens 4.37 times at every penalty. The accompanying text gives the per-agent observation counts and reward gaps.

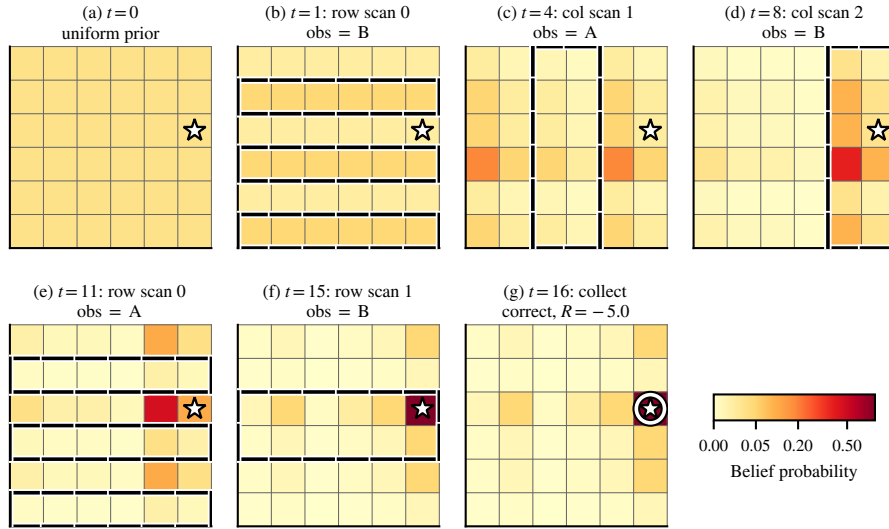


Fig. 17. Belief evolution within an EFE episode on the 6x6 Tileworld. Panels (a)–(g) show belief probability over the 36 grid cells on a shared square-root color scale (colorbar, darker = higher probability). Black outlines mark the cell group queried by each scan, each panel’s title reports that scan’s noisy binary readout as obs=A or obs=B naming which half of the queried group the signal favors, and the black-edged white star marks the target tile in every panel. Belief concentrates toward the target, and in the final panel a black-stroked white ring marks the committed cell.

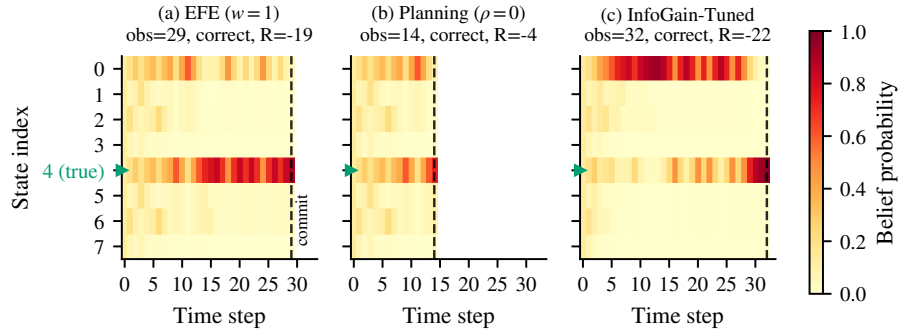


Fig. 18. Belief evolution within one illustrative Diagnosis episode per agent, not chosen to be typical in length ($N=8$, $K=3$, accuracy 0.75). Darker = higher probability (shared colorbar). The green arrow in the left margin marks the true state row, the dashed line marks the commit point, and the three panels share the time axis so episode lengths compare directly. In the plotted episodes all three agents commit correctly, Planning after 14 observations on a belief already concentrated on the true state, EFE after 29, and InfoGain-Tuned after 32 with its darkest band on a wrong row for most of the episode.

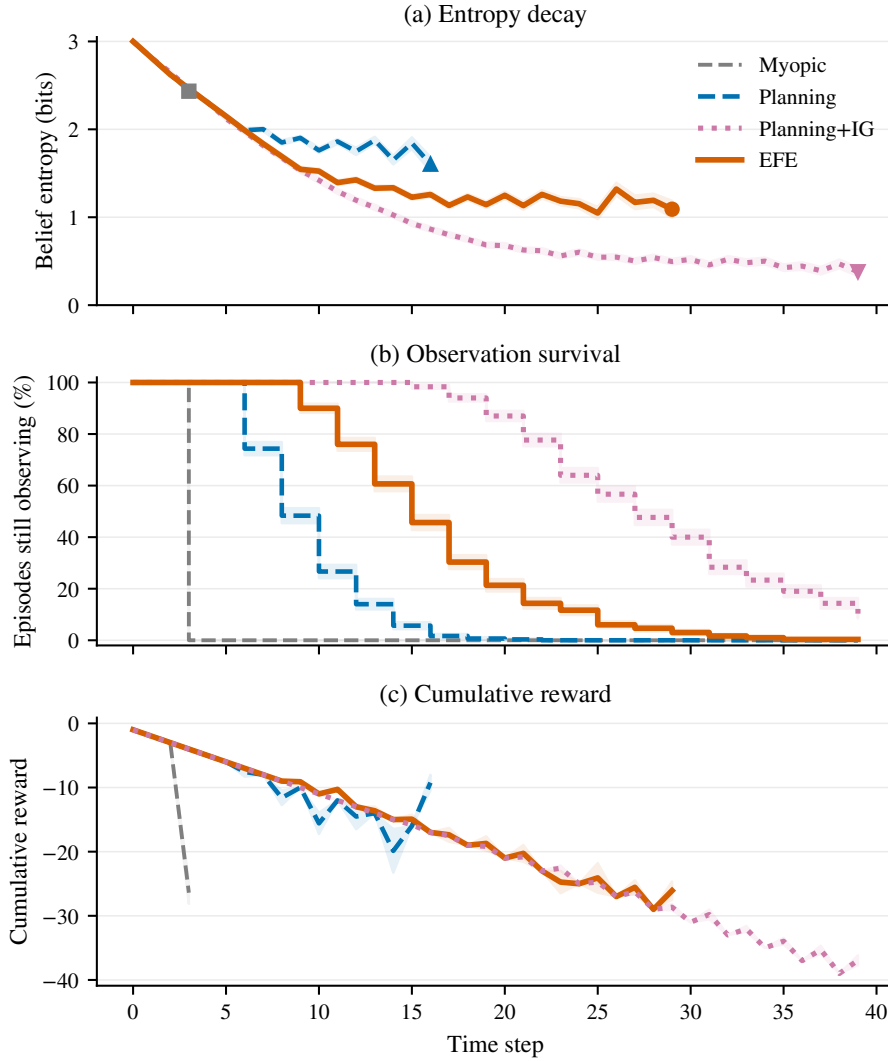


Fig. 19. Exploration efficiency on Diagnosis ($N=8$, $K=3$, 300 episodes). (a) Belief entropy decay. (b) Percentage of episodes still observing. (c) Cumulative reward. Panels are stacked on a shared time axis, shaded bands show ± 1 SE across episodes, and in panel (a) a marker terminates each series at its final plotted step. Panels (a) and (c) average, at each step, only the episodes still running at that step. Planning+IG uses the myopic Info Gain agent’s success-tuned weight, selected on 100 tuning episodes and transferred to Planning+IG for this visualization, not the own-class tuning protocol of the main tables (Section 3). EFE stops observing later than reward-only Planning and earlier than Planning+IG, whose survival curve lies at or above EFE’s at every step (panel b).

E Environment Specifications

RockSample check accuracy decays with distance from the rock (0.95 at range zero, halving every 4 cells, floored at 0.5). The tabulated value is the maximum. Each RockSample instance runs under an episode step cap of $N^2 + 10K$ steps, giving 55, 89, 129, and 231 for RS[5,3], RS[7,4], RS[7,8], and RS[11,11], and the fixed rock maps are those in `ROCKSAMPLE_CONFIGS` (`experiments/run_rocksample.py`). On RS[11,11] the start cell (5,0) coincides with a rock while every other rock lies at Euclidean distance 3.16 or more, at or beyond the limit of the sensor’s useful range (accuracy roughly 0.55 at distance 3.16, and floored at chance from 3.7 cells). This geometry is behind the single-informative-check pattern of Section 7.4. Its per-instance tree depth was fixed a priori before the reported runs, not selected on measured agent performance, and the depth-3 check on RS[11,11] (Section 7.4) shows the reported pattern is insensitive to one extra ply. Inspection also gives +2 for a correct nominal call and −5 for a false alarm, with fault prior 0.3.

The Diagnosis environment uses $K = \lceil \log_2 N \rceil$ binary tests, each providing noisy information about a different partition of the state space. The Navigation environment uses proximity-based warm/cold signals with accuracy depending on Manhattan distance to the hidden goal. Tileworld projects the Diagnosis partition structure onto a 2D grid. $K = 2\lceil \log_2 n \rceil$ scans, for grid side n , partition the grid by bit-level splits of the row and column indices, enabling complete spatial disambiguation.

Table 14. Full environment specifications for every environment reported in this paper’s results tables and appendices. Variant instances used outside the main results tables (Navigation 5×5 and 7×7 , Tileworld 4×4 , Diagnosis $N \in \{2, 8, 16\}$) differ from their tabulated siblings only in the stated size parameter. H is the planning horizon for observe-then-commit environments and the tree depth for the interleaved ones. γ is the discount used by the reporting agents. All main-text batteries are undiscounted ($\gamma=1.0$, the agent default), and γ is set to 0.999 inside the SARSOP/CPOMDP exports of Section 6.6, which require a discounted infinite-horizon model, and is otherwise varied only in the discount sweep of Appendix K and in two rows of the RockSample POMCP sensitivity battery. $|S|$ counts hidden-state configurations only, excluding fully observable components such as the RockSample agent position. Move actions are listed under observation actions only for Navigation, where they are the sole information source. Inspection additionally has four move (navigation) actions, omitted from its observation-action count.

Environment	$ S $	Obs. actions	Commit actions	Accuracy
Tiger	2	1	2	0.85
Testbed	2	1	2	0.75
Diagnosis				
($N=4$)	4	2	4	0.80
Bandit				
($K=4$)	4	4	4	0.80
Navigation	9	4 (move)	0 (implicit)	var.
Tileworld				
(6×6)	36	6 (scan)	36 (collect)	0.80
Tileworld				
(8×8)	64	6 (scan)	64 (collect)	0.80
Interleaved observe-act (Proposition 3). Undiscounted.				
RockSample				0.95
RS[5,3]	$2^3=8$	3 (check)	sample/exit	decaying
RockSample				0.95
RS[7,4]	$2^4=16$	4 (check)	sample/exit	decaying
RockSample				0.95
RS[7,8]	$2^8=256$	8 (check)	sample/exit	decaying
RockSample				0.95
RS[11,11]	$2^{11}=2,048$	11 (check)	sample/exit	decaying
Inspection				
($N=8$)	$2^8=256$	2 types $\times$ 8	diagnose	0.70/0.90
Inspection				
($N=16$)	$2^{16}=65,536$	2 types $\times$ 16	diagnose	0.70/0.90

Table 14. Full environment specifications (continued). Sensing costs, commit rewards, and planning horizon or tree depth H with discount γ . Every RockSample action costs -0.5 . Inspection additionally pays -0.5 per move, and its other outcomes are specified below.

Environment	Obs. cost	Correct	Incorrect	H / γ
Tiger	-1.0	$+10$	-100	$6 / 1.0$
Testbed	-0.1	$+1$	-1	$4 / 1.0$
Diagnosis				
($N=4$)	-1.0	$+10$	-50	$3 / 1.0$
Bandit				
($K=4$)	-0.5	$+10$	$+1$	$2 / 1.0$
Navigation	-0.5	$+20$	$-$	$2 / 1.0$
Tileworld				
(6×6)	-1.0	$+10$	-50	$2 / 1.0$
Tileworld				
(8×8)	-1.0	$+10$	-50	$2 / 1.0$
Interleaved observe-act (Proposition 3). Undiscounted.				
RockSample				
RS[5,3]	-0.5	$+10$	-10	$3 / 1.0$
RockSample				
RS[7,4]	-0.5	$+10$	-10	$4 / 1.0$
RockSample				
RS[7,8]	-0.5	$+10$	-10	$3 / 1.0$
RockSample				
RS[11,11]	-0.5	$+10$	-10	$2 / 1.0$
Inspection				
($N=8$)	$-0.5 / -2.0$	$+5$ fault	-50 missed	$3 / 1.0$
Inspection				
($N=16$)	$-0.5 / -2.0$	$+5$ fault	-50 missed	$2 / 1.0$

F Navigation Results

Navigation uses only move actions. The goal cell is hidden, and each move yields a proximity-based warm/cold observation (Appendix E). Unlike Diagnosis or Tileworld, there is no separate menu of observation actions. Information arrives as a side effect of movement, and greedy progress toward the belief mode already provides correlated evidence.

Table 15. Navigation scaling (150 episodes per seed $\times$ 5 seeds = 750 episodes per agent per grid). Step budget $3n^2$. NavEFE uses depth-limited planning $H=2$. Uncertainty is seed-level SE. See `results_navigation_scaling.csv` and the companion seed-level Welch tests in `results_navigation_scaling_seed_stats.csv`.

Grid	$ S $	Agent	Success	Mean reward
3×3	9	NavMyopic	$90.4 \pm 1.3\%$	$+2.04 \pm 1.51$
		NavInfoGain	$86.4 \pm 1.6\%$	-1.37 ± 1.69
		NavEFE ($H=2$)	$87.2 \pm 1.7\%$	-1.29 ± 1.58
5×5	25	NavMyopic	$80.7 \pm 1.4\%$	-16.64 ± 1.54
		NavInfoGain	$76.0 \pm 1.5\%$	-23.24 ± 1.75
		NavEFE ($H=2$)	$74.4 \pm 1.2\%$	-25.76 ± 1.14
7×7	49	NavMyopic	$65.7 \pm 2.8\%$	-38.42 ± 2.77
		NavInfoGain	$60.3 \pm 1.2\%$	-44.28 ± 1.12
		NavEFE ($H=2$)	$62.8 \pm 0.7\%$	-40.86 ± 0.77

NavMyopic (move toward the highest-probability cell) is the strongest baseline at every scale, nominally at 3×3 and 7×7 and significantly at 5×5 , where its lead over NavEFE holds on reward ($p=0.002$), success ($p=0.010$), and observation count ($p=0.001$, all seed-level Welch, Holm-corrected within metric at each grid size, `results_navigation_scaling_seed_stats.csv`). At the other two sizes no pairwise difference survives correction at 5 seeds, so at no size does an epistemic agent significantly beat the myopic rule. Exploratory detours pay the per-move cost without the sharp information returns available when tests can be chosen explicitly. At 7×7 , NavEFE’s mean reward is nominally above one-step NavInfoGain’s (-40.86 against -44.28 , seed-level $p=0.040$ uncorrected, not significant after Holm), with comparable success. The contrast with Tileworld (Figure 12) supports the multiple-observation-actions condition of Section 8. EFE’s clearest wins appear when the agent must select among differentially informative observation actions, not only where to translate next under a smooth proximity field.

G Scaling Analysis

At $H=2$, EFE and Planning show no consistent gap at any N tested here (bit-identical at $N=2$ and $N=4$, at $N=8$, $84.08\% \pm 0.66\text{pp}$ vs. $83.92\% \pm 0.82\text{pp}$, and at

Table 16. Scaling analysis. Diagnosis $N = 2$ to 16 (2,500 episodes at 500 per seed $\times$ 5 seeds, $H=2$). Bold marks the best success and reward per N and the tests of those rows, plus any row within sampling error of it (seed-level Welch $p > 0.05$, uncorrected, `results_scaling_stats.csv`). At $N=8$ and $N=16$ the EFE and Planning rows are statistical ties.

N	Agent	Tests	Success	Reward
2	Myopic	1.00	80.3%	-2.83
	Planning	2.91	94.1%	+3.56
	Info Gain	1.00	80.3%	-2.83
	EFE	2.91	94.1%	+3.56
4	Myopic	2.00	64.5%	-13.29
	Planning	5.83	88.8%	-2.58
	Info Gain	2.00	64.5%	-13.29
	EFE	5.83	88.8%	-2.58
8	Myopic	3.00	51.1%	-22.33
	Planning	8.79	83.9%	-8.44
	Info Gain	3.00	51.1%	-22.33
	EFE	8.83	84.1%	-8.38
16	Myopic	4.00	40.8%	-29.50
	Planning	11.65	77.4%	-15.23
	Info Gain	4.00	40.8%	-29.50
	EFE	11.64	77.1%	-15.37

$N=16$, $77.12\% \pm 0.78\text{pp}$ vs. $77.36\% \pm 0.98\text{pp}$, seed-level SE, with no consistent direction), both substantially outperforming Myopic and untuned Info Gain. This contrasts with the $H=3$ Diagnosis result (Table 5), where EFE outperformed Planning by +8.3 pp, consistent with EFE’s advantage depending on recursive depth, not merely on a larger state space. EFE-over-Myopic advantage grows from +13.8 pp at $N=2$ ($94.12\% \pm 0.50\text{pp}$ vs. $80.28\% \pm 0.91\text{pp}$) to +36.3 pp at $N=16$ ($77.12\% \pm 0.78\text{pp}$ vs. $40.84\% \pm 1.69\text{pp}$, seed-level SE), consistent with deeper planning becoming more valuable as state spaces grow, independent of whether EFE or reward-only Planning supplies that depth.

H PyMDP Consistency Check

We check our EFE computation against the `pymdp` library [26] by wrapping `pymdp`’s standard Agent class. The check tests consistency between implementations, not numerical equivalence. Run as an observe-then-commit agent on the two-state testbed under the canonical protocol, `pymdp-AIF` returns 3.15 observations, 89.8% success, and +0.480 reward. Our myopic Info Gain agent at $w=1$, which by Proposition 1 is the myopic EFE agent, returns the same three values (`results_summary.csv`), so the two single-step implementations agree. Our $H=4$ recursive EFE agent differs on this environment (5.50 observations) through planning depth, so the check says nothing about recursive EFE. It is also weak evidence, because on this environment the myopic $w=1$ agent and `pymdp-AIF` are both bit-identical to reward-only Planning, so the comparison cannot separate a correctly implemented epistemic term from its absence. Both implementations agree on observe-vs-commit decisions across all tested belief states (uniform, intermediate, and confident beliefs on the two-state environments). Information gain values computed by our recursive scheme and `pymdp`’s `control` module agree qualitatively, with differences attributable to our recursive multi-step evaluation vs. `pymdp`’s single-step policy enumeration.

I Supplementary Statistics

Pairwise comparisons in the main results tables use independent two-sample Welch t -tests on per-seed means with Holm–Bonferroni correction for family-wise error control, and comparisons reported without that correction are marked uncorrected where they appear. The companion pooled episode-level column in the six stats files behind the main observe-then-commit and Tileworld tables is an equal-variance two-sample test rather than a Welch test, which changes no non-degenerate comparison’s outcome at $\alpha=0.05$ across the 510 pooled comparisons in those files. The RockSample and Structural Inspection producers use Welch for both levels. The correction family is the full set of pairwise agent comparisons and metrics (reward, success/accuracy, and, where reported, observation count) reported together within a single table. For the main results tables (Table 5 and the RockSample and Structural Inspection tables), the family is scoped to one environment or instance at a time, since each table’s rows are computed

and corrected together, independently of every other table. Concretely, with A agents and M corrected metrics the family size is $M \cdot \binom{A}{2}$, giving 108 for Tiger ($A=9$, $M=3$ over reward, success, and observations), 84 for Diagnosis and Bandit ($A=8$, $M=3$), and 18 for each Structural Inspection instance ($A=4$, $M=3$ over reward, accuracy, and tests). These are nominal sizes. A comparison whose seed-level p -value is undefined, because both agents' per-seed values have zero variance, is excluded from its family, which leaves 107 defined tests for Tiger and 83 for Diagnosis. RockSample corrects within metric rather than pooling, giving 21 tests per metric per instance over its seven agents, because the RockSample table's bolding rule treats a non-rejection as a tie with the best row and pooling metrics into one larger family would loosen exactly that rule. The exceptions are the discount-sweep appendix table (Table 19), where the family spans every environment and γ value in that sweep jointly (24 tests, from 3 environments $\times$ 4 discount factors $\times$ 2 metrics), and the two MCTS batteries of Appendix S, where the ablation corrects within metric across all three environments jointly and the exploration sweep corrects within metric within each of its three comparison families. Bootstrap 95% confidence intervals use 10,000 resamples with fixed seed for reproducibility. Effect sizes use Cohen's d with pooled standard deviation. Because episodes within a seed are not independent, those primary tests aggregate each seed's episodes to a single mean before testing ($n=5$ seeds), and the pooled episode-level columns are reported alongside them. The main-text table (Table 5) reports the seed-level SE as the primary uncertainty measure. Pooled episode-level and seed-level SE describe different sources of variation, episode-level precision and between-seed replicability, so neither replaces the other. The Tiger, Diagnosis and Bandit tables of Appendix B report the pooled episode-level SE only and omit the untuned Info Gain rows that duplicate Myopic exactly, together with Tiger's pymdp-AIF row. The two Tileworld tables report point estimates for every agent run, duplicates included. The companion `*_stats.csv` files committed alongside the main battery carry the seed-level Welch t -tests and Cohen's d this section describes. The Tileworld scaling battery reports seed-level standard errors for reward and success without a companion Welch file.

Table 17. Cohen’s d for EFE vs. each baseline on mean episode reward (top) and vs. Planning on success rate (bottom). Positive d favors EFE. Interpretation. $|d| < 0.2$ negligible, 0.2–0.5 small, 0.5–0.8 medium, $|d| > 0.8$ large. Success-rate d uses the same pooled- s convention as reward (proportions treated as per-episode Bernoulli outcomes).

Environment	Comparison	d (reward)	Interpretation
Tiger	EFE vs. Myopic	+0.44	small
Tiger	EFE vs. Planning	0.00	negligible (exact tie)
Tiger	EFE vs. InfoGain-Tuned	0.00	negligible (exact tie)
Tiger	EFE vs. Planning+IG	+0.14	negligible
Tiger	EFE vs. Posterior-vote	+0.09	negligible
Testbed	EFE vs. Myopic	−0.07	negligible
Testbed	EFE vs. Planning	−0.18	negligible
Testbed	EFE vs. InfoGain-Tuned	+0.69	medium
Testbed	EFE vs. Planning+IG	+0.69	medium
Testbed	EFE vs. Posterior-vote	−0.18	negligible
Diagnosis	EFE vs. Myopic	+0.55	medium
Diagnosis	EFE vs. Planning	+0.08	negligible
Diagnosis	EFE vs. InfoGain-Tuned	+0.25	small
Diagnosis	EFE vs. Planning+IG	+0.25	small
Diagnosis	EFE vs. Posterior-vote	+0.09	negligible
Bandit	EFE vs. Myopic	+0.18	negligible
Bandit	EFE vs. Planning	+0.14	negligible
Bandit	EFE vs. InfoGain-Tuned	+0.36	small
Bandit	EFE vs. Planning+IG	+0.37	small
Bandit	EFE vs. Posterior-vote	−0.01	negligible
<i>Success rate (d favors EFE)</i>			
Tiger	EFE vs. Planning	0.00	negligible (exact tie)
Testbed	EFE vs. Planning	+0.26	small
Diagnosis	EFE vs. Planning	+0.33	small
Bandit	EFE vs. Planning	+0.40	small

J Near-Optimality across Planning Horizons

Proposition 2 derives w_{thresh}^* at $H=1$ and w_{over}^* at $H=2$. To check whether the resulting near-optimality claim survives recursive multi-step planning, we generated 100 random two-state environments (reward asymmetry $\alpha \in [1, 50]$, observation accuracy $p \in [0.55, 0.95]$, cost $\in [0.1, 5]$). For each environment and each $H \in \{1, 2, 3\}$, we selected the weight w_{ret}^* with the largest estimated mean reward on the grid and recorded whether its estimated reward advantage over $w=1$ is at most $\max(0.05|\hat{R}_{\text{best}}|, 0.5)$, where $\hat{R}_{\text{best}}$ is that largest estimated mean and 0.5 is measured in reward units. Each (environment, weight, horizon) cell runs 50 episodes at a single seed, and w_{ret}^* is selected over the six-point grid $\{0.01, 0.5, 1, 2, 5, 50\}$ (`experiments/run_nearopt_horizon.py`). This is a deliberate deviation from the canonical five-seed protocol, because the study is a descriptive sweep over 100 synthetic environments rather than a headline comparison, and its conclusions are read as coarse rates rather than per-environment estimates. Figure 20 reports the fraction of environments where $w=1$ passes this test at each horizon, stratified by α . The near-optimality rate rises from 30% at $H=1$ to 58% at $H=2$ and 79% at $H=3$ (26% $\rightarrow$ 53% $\rightarrow$ 79% restricted to $\alpha \geq 10$), so deeper planning widens the basin in which the untuned information-unit weight is reward-competitive rather than narrowing it. Stated as a failure rate, $w=1$ fails this test in about 21% of the random family even at $H=3$ and in 70% of it at $H=1$, and the criterion is lenient by construction, since the absolute floor of 0.5 reward units does most of the work on environments whose reward range is small. The trend is qualitatively consistent with Bandit, where a loose low-asymmetry reading would lead one to expect the least from $w=1$ yet multi-step EFE at $H=2$ reaches 86.9% success (Table 5), although the two-state threshold formula does not apply to Bandit and no threshold is computed for it.

Agreement across planning horizons. Table 18 breaks the same 100 environments down by all three sampled regime axes and compares the $H=1$ classification with $H \geq 2$ per environment. Eight of the nine regime strata show near-optimality rarer at $H=1$ than at $H=3$, most sharply for high-informativeness observations ($p \geq 0.85$, 17% $\rightarrow$ 90%). The exception is the low-informativeness stratum ($p < 0.65$, 25 environments), flat and mildly non-monotone at 80% $\rightarrow$ 84% $\rightarrow$ 80%, since weak observations leave little for a deeper horizon to exploit. Low reward asymmetry ($\alpha < 3$, 7 environments) rises less sharply, from 71% at $H=1$ to 86% at both $H=2$ and $H=3$. Following the producing script, we classify an environment as agreeing across horizons when $w=1$ is near-optimal at $H=1$ and at some deeper horizon, or at neither. This holds for 44 of the 100 environments. In 30 the $H=1$ check passes and near-optimality recurs deeper, at every deeper horizon in 26 of them and at exactly one of the two in the other four (environments 2, 42, 63, and 95 in the committed CSV). In 14 the check fails and never recovers. The remaining 56 fail at $H=1$ yet are near-optimal at some deeper horizon (the table’s myopic underclaims), a missed opportunity rather than a risk. No environment in this sample is a myopic overclaim, where $H=1$ predicts near-optimality and no deeper horizon delivers it. This is an empirical

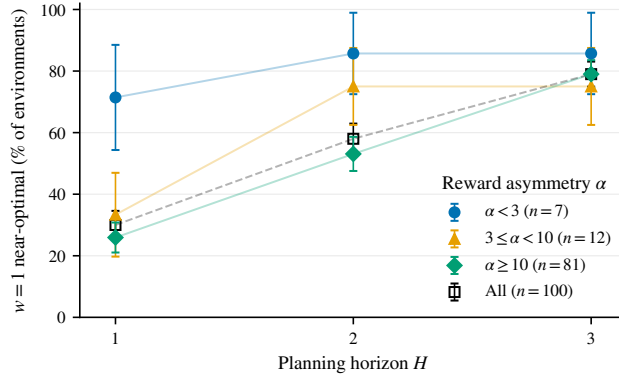


Fig. 20. Fraction of randomly generated two-state environments where $w=1$ is near-optimal, as a function of planning horizon H , stratified by reward asymmetry α . The near-optimality basin widens with horizon, consistent with multi-step planning amplifying the value of observation. 100 environments, $\alpha \in [1, 50]$, $p \in [0.55, 0.95]$, cost $\in [0.1, 5]$. Markers show the three measured horizons $H \in \{1, 2, 3\}$ with binomial standard-error bars, the legend reports the number of environments in each α stratum, and the light guide lines connect measured points only.

boundary map over one random family of two-state environments. It is not an adaptive-submodularity result, and it does not show that the $H=1$ check is monotone or that such overclaiming is impossible in general (per-environment data in `results/results_nearopt_horizon.csv`, aggregated by `experiments/build_horizon_map.py` into `results/results_horizon_map_agreement.csv`).

Table 18. Horizon map, the fraction of randomly generated two-state environments where $w=1$ is near-optimal: its estimated reward gap from the best grid weight is at most $\max(0.05|\hat{R}_{\text{best}}|, 0.5)$ reward units, where $\hat{R}_{\text{best}}$ is the largest estimated grid reward. Results are stratified by each sampled regime axis and planning horizon H . Same 100 environments as Figure 20 (Appendix J). This is an empirical boundary map, not an adaptive-submodularity guarantee.

Regime	$H=1$	$H=2$	$H=3$
<i>α (reward asymmetry)</i>			
$\alpha < 3$	71%	86%	86%
$3 \leq \alpha < 10$	33%	75%	75%
$\alpha \geq 10$	26%	53%	79%
<i>p (observation informativeness)</i>			
$p < 0.65$	80%	84%	80%
$0.65 \leq p < 0.85$	11%	40%	71%
$p \geq 0.85$	17%	63%	90%
<i>cost (observation cost)</i>			
$\text{cost} < 1$	16%	16%	58%
$1 \leq \text{cost} < 3$	26%	55%	79%
$\text{cost} \geq 3$	40%	79%	88%

$H=1$ vs. $H \geq 2$ agreement: 44% of 100 environments agree; myopic overclaims 0, underclaims 56.

K Discount Factor Sensitivity

We sweep $\gamma \in \{0.9, 0.95, 0.99, 1.0\}$ for EFE and same-horizon Planning on three environments under the canonical 5-seed protocol, so that the gap between the two agents can be read at each discount.

Sensitivity to discounting. On Tiger (single observation action), both agents achieve $\geq 96\%$ success across all γ values, with a significant but small +2.7 pp EFE advantage at the heaviest discounting tested ($\gamma=0.9$, seed-level Welch $p < 10^{-3}$). At $\gamma \geq 0.95$ the two agents are numerically identical ($p=1.0$). On Diagnosis and Bandit, heavy discounting ($\gamma=0.9-0.95$) renders both agents effectively myopic and erases EFE’s advantage, because discounting truncates the effective horizon below the number of observations needed for disambiguation. Both agents observe the same number of times on Bandit (~ 2) and Diagnosis (~ 6) at these discount levels, and the residual success gaps are not statistically significant (Diagnosis diff $\approx 0.04\text{pp}$, $p=0.96$, Bandit diff $\approx -0.16\text{pp}$, $p=0.95$, Table 19). The transition between $\gamma=0.95$ and $\gamma=0.99$ is sharp, and the Diagnosis and Bandit gaps at $\gamma \geq 0.99$ are significant after Holm correction. At $\gamma=0.99$, EFE’s success gap over Planning jumps to +7.9 pp on Diagnosis and +22.4 pp on Bandit (both $p < 10^{-4}$), matching the undiscounted pattern (+8.6 pp Diagnosis, +17.0 pp Bandit at $\gamma=1.0$, both $p < 10^{-4}$). On this grid, EFE’s advantage therefore requires a discount that preserves a sufficient effective horizon for recursive epistemic evaluation.

Table 19. EFE and Planning agent performance across discount factors γ (2,500 episodes at 500 per seed $\times$ 5 seeds, lighter than Table 5’s 5,000-episode battery, so small numeric differences from that table on the $\gamma=1.0$ row are expected). $\Delta\text{Succ.}$ shows the success rate gap (EFE – Planning) at each γ , with the seed-level Welch p -value in parentheses. * marks significance after Holm–Bonferroni correction.

Environment	Agent	γ	Obs.	Success	Reward	$\Delta\text{Succ. (}p\text{)}$
Tiger	EFE	0.90	4.23	99.6%	+5.33	+2.7 ($p < 10^{-3}$)*
	Planning	0.90	2.65	96.9%	+3.91	
	EFE	0.95	4.23	99.6%	+5.33	0.0 ($p=1.0$)
	Planning	0.95	4.23	99.6%	+5.33	
	EFE	0.99	4.23	99.6%	+5.33	0.0 ($p=1.0$)
	Planning	0.99	4.23	99.6%	+5.33	
	EFE	1.00	4.23	99.6%	+5.33	0.0 ($p=1.0$)
	Planning	1.00	4.23	99.6%	+5.33	
Diagnosis	EFE	0.90	5.83	88.8%	−2.58	+0.0 ($p=0.96$)
	Planning	0.90	5.86	88.7%	−2.62	
	EFE	0.95	5.83	88.8%	−2.58	+0.0 ($p=0.96$)
	Planning	0.95	5.86	88.7%	−2.62	
	EFE	0.99	9.67	97.5%	−1.19	+7.9 ($p < 10^{-4}$)*
	Planning	0.99	5.89	89.6%	−2.15	
	EFE	1.00	9.68	97.4%	−1.26	+8.6 ($p < 10^{-4}$)*
	Planning	1.00	5.85	88.8%	−2.59	
Bandit	EFE	0.90	2.04	61.9%	+5.55	−0.2 ($p=0.95$)
	Planning	0.90	2.04	62.0%	+5.56	
	EFE	0.95	2.04	61.9%	+5.55	−0.2 ($p=0.95$)
	Planning	0.95	2.04	62.0%	+5.56	
	EFE	0.99	5.16	86.9%	+6.24	+22.4 ($p < 10^{-4}$)*
	Planning	0.99	2.38	64.5%	+5.61	
	EFE	1.00	5.16	86.9%	+6.24	+17.0 ($p < 10^{-4}$)*
	Planning	1.00	3.24	69.9%	+5.67	

L Model Misspecification Sensitivity

Our main results assume the agent’s generative model matches the true environment dynamics. Here we investigate robustness when the agent’s believed observation accuracy p_{agent} differs from the true accuracy p_{true} , creating a systematic model mismatch.

Table 20. Model misspecification on Tiger ($p_{\text{true}}=0.85$, $H=4$, 2,500 episodes across 5 seeds, lighter than Table 5’s 5,000-episode battery, so small numeric differences from that table’s calibrated row are expected). The mismatch column shows $p_{\text{agent}} - p_{\text{true}}$. Reward reported as mean $\pm$ seed-level SE (`results_model_misspec.csv`). EFE and Planning rows are bit-identical at every level tested, not merely close, so no significance test applies to that comparison.

Agent	p_{agent}	Mismatch	Obs	Success	Reward
EFE	0.70	−0.15	4.23	99.6%	+5.33 $\pm$ 0.26
EFE	0.75	−0.10	4.23	99.6%	+5.33 $\pm$ 0.26
EFE	0.80	−0.05	4.23	99.6%	+5.33 $\pm$ 0.26
EFE	0.85	± 0.00	4.23	99.6%	+5.33 $\pm$ 0.26
EFE	0.90	+0.05	2.65	96.9%	+3.91 $\pm$ 0.38
EFE	0.95	+0.10	2.65	96.9%	+3.91 $\pm$ 0.38
Planning	0.70	−0.15	4.23	99.6%	+5.33 $\pm$ 0.26
Planning	0.85	± 0.00	4.23	99.6%	+5.33 $\pm$ 0.26
Planning	0.95	+0.10	2.65	96.9%	+3.91 $\pm$ 0.38

Asymmetric sensitivity to sensor error. On Tiger, success falls at most 2.7pp below the calibrated 99.6% across the tested range of -0.15 to $+0.10$, and the pattern is asymmetric. Underestimating accuracy or calibrating exactly leaves behavior unchanged (4.23 observations, 99.6% success at every level from -0.15 to 0.00), whereas overestimating it cuts observations to 2.65 and success to 96.9%. EFE and Planning give bit-identical rows at every tested level (Table 20), not merely statistically indistinguishable ones. With a single observation action, the believed accuracy alone determines when committing becomes attractive, and information gain has no separate observation action to redirect toward. The study therefore shows no EFE-specific buffering on Tiger, since reward-only Planning, with no information-gain term, degrades identically.

On Diagnosis, EFE’s behavior steps between exactly two levels rather than degrading smoothly with mismatch magnitude (Table 21). The high-testing level (9.68 observations, 97.4% success) occurs at mismatch $\in \{-0.10, -0.05, 0.00\}$. The low-testing level (5.86 observations, 88.7% success) occurs at mismatch $\in \{-0.15, +0.05, +0.10\}$. Calibration therefore ties with two levels of underestimation rather than uniquely maximizing testing. Overestimating accuracy makes each test look more decisive than it is, so the agent commits prematurely on insufficient evidence at every overestimate tested. Mild underestimation (-0.05 ,

Table 21. Model misspecification on Diagnosis ($p_{\text{true}}=0.80$, $H=3$, 2,500 episodes across 5 seeds). Reward reported as mean $\pm$ seed-level SE (`results_model_misspec.csv`).

Agent	p_{agent}	Mismatch	Obs	Success	Reward
EFE	0.65	-0.15	5.86	88.7%	-2.62 ± 0.31
EFE	0.70	-0.10	9.68	97.4%	-1.26 ± 0.34
EFE	0.75	-0.05	9.68	97.4%	-1.26 ± 0.34
EFE	0.80	± 0.00	9.68	97.4%	-1.26 ± 0.34
EFE	0.85	+0.05	5.86	88.7%	-2.62 ± 0.31
EFE	0.90	+0.10	5.86	88.7%	-2.62 ± 0.31
Planning	0.65	-0.15	5.84	89.2%	-2.35 ± 0.51
Planning	0.70	-0.10	9.71	97.6%	-1.12 ± 0.26
Planning	0.75	-0.05	9.68	97.3%	-1.29 ± 0.32
Planning	0.80	± 0.00	5.85	88.8%	-2.59 ± 0.37
Planning	0.85	+0.05	5.86	88.7%	-2.62 ± 0.31
Planning	0.90	+0.10	5.82	88.7%	-2.61 ± 0.40

-0.10) leaves behavior unchanged, while the most severe underestimate tested (-0.15) triggers the same reduced-testing response as overestimation. Reward-only Planning shows a comparable but not identical step pattern (Table 21), so this threshold structure is not unique to EFE’s epistemic term. We report the pattern without assigning a mechanism to the precise transition point.

Implications for learned models. Observation models learned from data will carry estimation error. On these two environments, mismatches up to 0.15 change behavior in discrete steps without catastrophic failure, overestimating sensor informativeness is the more harmful direction, and reward-only Planning shows comparable sensitivity.

M RockSample as an Interleaved Observe-Act Setting (Extended)

This appendix provides extended RockSample results beyond the main text (Section 7.4), including POMCP at its tuned frozen configuration, POMCP’s approach-then-check rollout policy run standalone, the two-point Planning+IG sweep ($w=5,10$), and per-instance step and check counts. The extended table (Table 22) is read directly from the same CSVs as Table 6 (`results_rocksample_{5x3,7x4,7x8,11x11}.csv`), regenerated by `experiments/build_rocksample_tables.py`. The tuning, budget-scaling, horizon, and sensitivity numbers quoted in the POMCP discussion come from their own producer CSVs (`results_rocksample_pomcp_{tuning,budget,horizon,sensitivity}.csv`).

The Greedy agent moves to rocks and samples them without checking quality first, incurring frequent bad-rock penalties ($1.51\text{--}5.52 \pm 0.01\text{--}0.05$, seed-level SE, across the four instances). EFE moves toward uncertain rocks, checks them to

resolve quality, then samples selectively, so bad samples stay low ($0.01\text{--}0.06 \pm 0.001\text{--}0.013$, seed-level Welch $p < 5 \times 10^{-8}$ vs. Greedy at every instance). On RS[11,11], all tree-search agents, including Planning at $w=0$, instead follow low-activity policies, discussed in Section 7.4.

The RockSample POMCP (`rho_aif/agents/rocksample_pomcp.py`) builds a search tree over action-observation histories, selects with UCB1, backs up means as UCT does, and evaluates newly expanded leaves by Monte Carlo rollout. The search procedure makes three departures, all keyed to fully observable quantities. The belief representation and depth-cap leaf value are separate implementation choices, disclosed and sensitivity-checked below. The depth bound shrinks as the episode step cap approaches. The action set is masked to exclude wall bumps, checks on already-sampled rocks, and samples where no rock sits. The subtree under the realized history is not reused, since under a fixed depth bound statistics accumulated at depth one would be credited with a full horizon after promotion. Its configuration was selected on three tuning seeds disjoint from every evaluation seed, over a two-stage coordinate grid covering the rollout policy, the exploration constant, the planning horizon, and the root criterion (`results_rocksample_pomcp_tuning.csv`), and frozen before any evaluation number was inspected. Belief is the exact factored posterior by default, which is available here because the RockSample posterior factorizes exactly. A literal unweighted particle filter with rejection sampling is also implemented and scores within sampling error of the exact version ($+12.93 \pm 0.25$ against $+12.81 \pm 0.43$, $p=0.81$), so the exact belief is not what carries the result. The depth-cap leaf value is a second implementation choice worth the same check. Replacing the frozen exit-valued leaf with a greedy-belief leaf scores $+14.52 \pm 0.34$ against the frozen configuration’s $+12.81 \pm 0.43$ ($p=0.015$ uncorrected), still below that battery’s own standalone rollout at $+17.16 \pm 0.44$ ($p=0.002$ uncorrected, and neither comparison survives that battery’s per-metric Holm family). The leaf choice moves the number without changing the ordering, so we report it rather than retune around it.

At the frozen configuration POMCP reaches $+12.74 \pm 0.11$ on RS[5,3] at the full protocol, against EFE’s $+16.47 \pm 0.12$, and it costs 46.6 ms per decision against EFE’s 1.76 ms. The gap holds at every instance though its ratio varies, 767.5 against 103.5 ms on RS[7,4], 75.5 against 7.2 on RS[7,8], and 87.6 against 0.46 on RS[11,11] (`mean_wallclock_ms_per_step` in the instance CSVs). Two further measurements qualify that comparison.

First, the search tree is not what produces the score. The rollout policy POMCP was tuned to use, run standalone with no tree at all, scores $+16.67 \pm 0.13$ on the same protocol at 0.005 ms per decision, which is 3.9 reward above the full search (seed-level Welch $p=1.9 \times 10^{-14}$ at the full protocol, significant after Holm–Bonferroni). On this domain the tree subtracts value from the policy it rolls out with. The mechanism is the position-independent exit acting as a zero-variance outside option worth +9.50 from any cell, against mean UCT backup. The forced exploration that makes UCT sound is what drags the alternatives

below it, so the tree commits to exiting after 7.5 steps where its own rollout plays a 13.5-step collection tour.

Second, the frozen planning horizon handicapped POMCP. Sweeping the horizon post hoc on RS[5,3] at two simulation budgets (`results_rocksample_pomcp_horizon.csv`) shows the frozen $H=10$ is the worst setting tested at both. At 16,384 simulations, $H=10$ gives $+12.52 \pm 0.26$ while $H=15$, $H=25$ and $H=40$ give $+15.60 \pm 0.37$, $+15.24 \pm 0.39$ and $+15.62 \pm 0.36$. The headline 2,048-simulation budget was fixed before evaluation as double the tuning stage’s 1,024, not itself a tuned choice. The budget-scaling direction reverses with the horizon. At $H=10$ the sweep is non-monotone (256 scores $+11.73$, the worst tested, and the peak sits at 1,024), and more simulations beyond 1,024 make POMCP worse, from $+13.19$ to $+12.52$ at 16,384, because a longer search only sharpens a conclusion that is wrong for being truncated. At $H=15$ more simulations help, from $+14.00$ to $+15.60$. This is a gap in our tuning protocol, not a property of POMCP. Stage B selected the horizon at a fixed budget of 1,024 simulations, so a budget-by-horizon interaction was invisible to it. The horizon sweep ran on the canonical evaluation seeds and is therefore reported as a disclosed post-hoc diagnostic, not used to re-select the frozen configuration.

The same diagnostic on RS[11,11] (`results_rocksample_pomcp_horizon_11x11.csv`, at the 2,048-simulation tier only, since the 16,384 tier is prohibitively slow at this instance’s per-step planning cost) shows the opposite pattern, with deeper horizons monotonically worse. $H=10$ scores $+12.94 \pm 0.24$, $H=15$ falls to -7.54 ± 1.20 , $H=25$ to -34.65 ± 1.12 , and $H=40$ to -54.78 ± 0.86 , with every step of the decline significant (seed-level Welch $p \leq 4.3 \times 10^{-5}$, all surviving Holm correction). The behavioral trace says why. At $H=10$ the tree exits after 3.3 steps with 1.6 checks. At $H=40$ it plays 196-step episodes with 144 checks, thirteen times its own rollout’s 11, yet still collects only 3.4 good rocks against the rollout’s 5.2. The deeper tree does see the collection tour, but what it assembles from noisy distant checks is a bad tour. The -0.5 cost on every one of those extra actions swamps the sampling profit. So the tuning-protocol gap does not transfer. The frozen $H=10$, the worst measured setting on RS[5,3], is the best by a wide margin on RS[11,11].

Taken together, the fair reading is that POMCP at its best measured configuration on RS[5,3] reaches roughly $+15.6$ at 530 ms per decision. EFE run under that diagnostic’s own protocol (5 seeds $\times$ 100 episodes, `results_rocksample_efe_diagnostic.csv`) scores $+16.90 \pm 0.45$ at 1.7 ms per decision, nominally 1.3 above POMCP’s best and not significantly so (seed-level Welch $p=0.058$ against $H=15$ and $H=40$, $p=0.024$ against $H=25$, none surviving Holm correction over the nine diagnostic rows). The tree therefore reaches statistical parity with the closed-form recursion on this instance at roughly three hundred times the per-decision cost. On RS[11,11] the frozen $H=10$ ’s $+12.94$ is matched but not clearly beaten by any measured alternative, the best being the 4,096-simulation budget at $+13.14 \pm 0.15$, a nominal gain inside sampling error. On RS[5,3], POMCP’s remaining deficit against its own rollout policy at 16,384 simulations and $H \geq 15$ is 1.5 to 1.9 reward, 9 to 11 percent of the rollout’s return (seed-level Welch

$p=0.012$ to 0.028 uncorrected, none surviving the seed-level Holm correction over the battery’s 36 reward comparisons, though the $H=25$ deficit does survive the pooled-level correction). There the tree therefore comes within roughly ten percent of the rollout’s return at five orders of magnitude more compute per decision. On RS[11,11] the gap does not close, since the 4,096-simulation budget still trails its own rollout’s $+28.66 \pm 0.83$ (seed-level SE) by 15.5 reward (seed-level Welch $p=3.2 \times 10^{-5}$, surviving Holm correction at both levels, **results_rocksample_pomcp_budget_stats.csv**). The main table reports the predeclared configuration, and this diagnostic shows what a more generous one achieves.

Planning+IG at $w=5$ and $w=10$ lets us check whether $w=1$ leaves performance on the table when a higher weight is allowed. Appendix X.4 gives the per-instance comparisons of EFE with $w=5$, $w=10$, and the standalone heuristic, with their significance tests. Two further details come from the extended table. On RS[7,4], EFE ($w=1$) beats $w=10$ as well as $w=5$ ($+15.96$ vs. $+14.27$ and $+14.56$, both $p < 10^{-5}$), and the extra checking of the tuned weights buys more good rocks but not enough to pay for the extra steps. On RS[7,8], $w=10$ ’s extra checking (57.47 steps, 18.26 checks per episode, versus $w=5$ ’s 40.27/14.19) buys 0.54 more good rocks per episode but spends 17.2 more steps at the -0.5 per-step move cost to get them. POMCP at its frozen configuration trails EFE and both tuned Plan+IG weights significantly on the three smaller instances, beats reward-only Planning on RS[7,8] ($+15.07$ vs. $+12.31$, $p=0.002$), and sits within sampling error of the entire family on RS[11,11] (seed-level Welch $p \geq 0.36$, none significant after correction). The weight sweep bounds what tuning w can buy, not what this instance admits.

Table 22. RockSample extended results with the full agent roster, including POMCP at its tuned frozen configuration, POMCP’s rollout policy run standalone (Rollout only), and Planning+IG at both $w=5$ and $w=10$. Steps and checks are episode means. RS[5,3]/RS[7,4]: 500 episodes $\times$ 10 seeds. RS[7,8]/RS[11,11]: 100 episodes $\times$ 5 seeds.

Instance	Agent	Steps	Checks	Good	Bad	Reward
RS[5,3]	Greedy	12.00	0.00	1.49	1.51	$+3.71 \pm 0.26$
	POMCP (2048 sims)	7.47	2.97	0.68	0.03	$+12.74 \pm 0.11$
	Planning (d=3)	10.38	3.71	0.99	0.04	$+14.28 \pm 0.13$
	Plan+IG w=5 (d=3)	15.81	5.36	1.44	0.01	$+16.41 \pm 0.13$
	Plan+IG w=10 (d=3)	20.35	6.63	1.48	0.00	$+14.60 \pm 0.11$
	EFE w=1 (d=3)	15.26	4.79	1.44	0.03	$+16.47 \pm 0.12$
	Rollout only (approach)	13.48	3.00	1.41	0.07	$+16.67 \pm 0.13$
RS[7,4]	Greedy	22.00	0.00	1.98	2.02	-1.38 ± 0.36
	POMCP (2048 sims)	8.67	3.46	0.66	0.03	$+12.01 \pm 0.06$
	Planning (d=4)	10.00	2.00	0.95	0.05	$+13.97 \pm 0.11$
	Plan+IG w=5 (d=4)	24.55	8.15	1.69	0.01	$+14.56 \pm 0.14$
	Plan+IG w=10 (d=4)	26.82	8.80	1.78	0.01	$+14.27 \pm 0.15$
	EFE w=1 (d=4)	16.81	5.36	1.44	0.01	$+15.96 \pm 0.15$
	Rollout only (approach)	23.98	4.00	1.88	0.10	$+15.85 \pm 0.17$
RS[7,8]	Greedy	32.00	0.00	4.02	3.98	-5.60 ± 1.03
	POMCP (2048 sims)	22.18	11.77	1.71	0.10	$+15.07 \pm 0.45$
	Planning (d=3)	4.51	1.00	0.48	0.03	$+12.31 \pm 0.35$
	Plan+IG w=5 (d=3)	40.27	14.19	3.40	0.01	$+23.76 \pm 0.58$
	Plan+IG w=10 (d=3)	57.47	18.26	3.94	0.01	$+20.55 \pm 0.29$
	EFE w=1 (d=3)	27.26	9.84	2.61	0.06	$+21.85 \pm 0.79$
	Rollout only (approach)	36.02	8.00	3.83	0.19	$+28.33 \pm 0.82$
RS[11,11]	Greedy	57.00	0.00	5.48	5.52	-18.90 ± 1.04
	POMCP (2048 sims)	3.28	1.60	0.49	0.03	$+12.94 \pm 0.24$
	Planning (d=2)	2.50	1.00	0.48	0.03	$+13.23 \pm 0.18$
	Plan+IG w=5 (d=2)	3.09	1.56	0.50	0.03	$+13.18 \pm 0.15$
	Plan+IG w=10 (d=2)	3.09	1.56	0.50	0.03	$+13.18 \pm 0.15$
	EFE w=1 (d=2)	2.50	1.00	0.48	0.03	$+13.23 \pm 0.18$
	Rollout only (approach)	62.44	11.00	5.22	0.23	$+28.66 \pm 0.83$

N IDS Baseline (Observe-then-Commit)

We adapt information-directed sampling [48] to the observe-then-commit setting by minimizing the information ratio Δ^2/I over observation actions, where Δ is the opportunity cost of observing versus committing now and I is expected information gain about the optimal commit. Two fidelity gaps against Russo and Van Roy must be stated. Their IDS minimizes the ratio over randomized action distributions $\pi \in D(A)$. The journal version of their work [49] gives an example in which every stationary deterministic policy incurs linear regret for some prior, which they read as randomization playing a fundamental role. Our baseline instead takes a deterministic argmin over single observation actions. And where their formulation assumes the information gain is nonzero, our implementation substitutes a raw state-entropy term when the gain about the optimal commit vanishes. Table 23 reports a matched comparison (2,500 episodes at 500 per seed $\times$ 5 seeds, the canonical seed set) against Myopic, Planning, success-tuned Planning+IG, and EFE.

Table 23. IDS vs. baselines (2,500 episodes at 500 per seed $\times$ 5 seeds, lighter than Table 5’s 5,000-episode battery, so small numeric differences from that table’s EFE/Planning rows are expected). Reward reported as mean $\pm$ seed-level SE. IDS is competitive on Tiger (exactly tying Planning and EFE there) but over-explores on Diagnosis and Bandit relative to EFE’s untuned $w=1$, matching success-tuned Planning+IG on Diagnosis reward and trailing even it on Bandit. Success-tuned Planning+IG runs at the Planning+IG-class weights recorded in `results_supplementary_tuned_weights.csv`.

Environment	Agent	Obs.	Success	Reward
Tiger	Myopic	1.00	84.6%	-7.90 ± 0.67
	Planning	4.23	99.6%	$+5.33 \pm 0.26$
	Planning+IG ($w=50$)	5.66	99.9%	$+4.25 \pm 0.08$
	EFE	4.23	99.6%	$+5.33 \pm 0.26$
	IDS	4.23	99.6%	$+5.33 \pm 0.26$
Diagnosis	Myopic	2.00	64.5%	-13.29 ± 0.88
	Planning	5.85	88.8%	-2.59 ± 0.37
	Planning+IG ($w=50$)	13.17	99.2%	-3.68 ± 0.14
	EFE	9.68	97.4%	-1.26 ± 0.34
	IDS	13.17	99.2%	-3.63 ± 0.16
Bandit	Myopic	2.05	61.6%	$+5.52 \pm 0.13$
	Planning	3.24	69.9%	$+5.67 \pm 0.13$
	Planning+IG ($w=20$)	9.72	99.0%	$+5.05 \pm 0.07$
	EFE	5.16	86.9%	$+6.24 \pm 0.09$
	IDS	10.62	99.5%	$+4.65 \pm 0.07$

On multi-observation environments, IDS reaches near-ceiling success but at a reward cost similar to success-tuned Planning+IG, whereas EFE ($w=1$) re-

tains higher mean reward. A seed-level Welch test on reward rejects the null in both cases (Diagnosis -1.26 vs. -3.63 , $p=0.0009$, Bandit $+6.24$ vs. $+4.65$, $p=1.5 \times 10^{-6}$, `results_ids.csv`). The adaptive information ratio does not eliminate the need for a stopping trade-off calibrated to the reward scale. Under our shared reward convention, the fixed information-unit weight remains the stronger untuned default for expected reward.

O Proper-Scoring Calibration

The reward and success-rate metrics used throughout this paper score decisions, not beliefs. To check that the epistemic term is doing more than inflating a task-performance proxy, we separately score the terminal belief itself against the true hidden state using two proper scoring rules [22], the log score ($\ln b(s^*)$, the same functional the epistemic value term is derived from) and the Brier score ($\sum_s (b(s) - \mathbf{1}[s = s^*])^2$), where s^* is the true state. For Structural Inspection, each rule is applied to each component’s binary belief and true fault status, then averaged over components. Its reported score is not a joint-state score. The implementation floors the probability used in the log score at 10^{-12} to keep numerical output finite. These scores assess overall terminal-belief quality, which depends on both calibration and how informative the belief is. A uniform posterior before any observation and an exact point-mass posterior after a perfect observation can both be calibrated, although the latter has better expected scores. Table 24 reports mean $\pm$ SE over 5 seeds for Planning (reward-only), EFE ($w=1$), and a reference near-optimal or IG-heavy policy on the three core OTC environments plus both Structural Inspection instances (500 episodes per seed for Tiger, Diagnosis, Bandit, and Inspection-N8, 200 per seed for Inspection-N16). The reference is the point-based near-optimal solver SARSOP [32], run as a self-built alpha-vector policy evaluated in-simulator, for the three OTC environments whose small state spaces allow near-optimal convergence in reasonable time. For Structural Inspection, whose state space exceeds what our SARSOP build solves in reasonable time, the reference is Planning+IG at the atlas’s higher canonical budget (Appendix T.3).

Planning (reward-only) is dominated on both terminal-belief scores by every epistemic-aware agent on every instance except Tiger, where its single observation action already suffices and all three agents converge to the same policy and hence the same terminal belief. On Diagnosis, Bandit, and both Inspection instances, EFE’s terminal belief scores substantially better under both rules than reward-only Planning’s, and on the three OTC environments EFE ($w=1$) lies within overlapping SE bands of SARSOP’s terminal-belief scores. This is evidence that the epistemic term produces better-informed terminal beliefs, not only better terminal actions. The two need not coincide, since an agent can commit correctly from a confidently wrong belief on environments with few reachable outcomes.

Table 24. Proper-scoring calibration of the terminal belief against the true hidden state (log score in nats and Brier score, mean $\pm$ SE over 5 seeds). Reference is SARSOP for the three OTC environments and Planning+IG at the atlas’s higher canonical budget (Appendix T.3) for Structural Inspection, where the near-optimal solver used here does not scale to the instance’s state space. For Inspection, scores are averaged over binary component beliefs. Log-score probabilities are floored at 10^{-12} .

Instance	Agent	Log score	Brier score
Tiger	Planning (reward-only)	-0.035 ± 0.006	0.011 ± 0.002
Tiger	EFE ($w=1$)	-0.035 ± 0.006	0.011 ± 0.002
Tiger	SARSOP	-0.035 ± 0.006	0.011 ± 0.002
Diagnosis	Planning (reward-only)	-0.433 ± 0.018	0.199 ± 0.011
Diagnosis	EFE ($w=1$)	-0.134 ± 0.004	0.049 ± 0.002
Diagnosis	SARSOP	-0.154 ± 0.021	0.058 ± 0.010
Bandit	Planning (reward-only)	-0.901 ± 0.033	0.463 ± 0.019
Bandit	EFE ($w=1$)	-0.525 ± 0.027	0.240 ± 0.016
Bandit	SARSOP	-0.510 ± 0.034	0.231 ± 0.018
Inspection-N8	Planning (reward-only)	-0.417 ± 0.004	0.286 ± 0.003
Inspection-N8	EFE ($w=1$)	-0.245 ± 0.004	0.144 ± 0.003
Inspection-N8	Planning+IG ($w=100$)	-0.008 ± 0.001	0.002 ± 0.000
Inspection-N16	Planning (reward-only)	-0.382 ± 0.005	0.251 ± 0.003
Inspection-N16	EFE ($w=1$)	-0.257 ± 0.003	0.150 ± 0.002
Inspection-N16	Planning+IG ($w=100$)	-0.013 ± 0.003	0.004 ± 0.001

P Per-Test Value-of-Information Audit Trail

At the root of each decision, the Planning+IG score of an observation action is its weighted immediate information gain plus an expected task value. That task value carries the sensing cost and the value of continuing, including any information bonuses deeper in the tree. An optional audit record at the shared Planning+IG evaluation boundary, used by `PlanningInfoGainAgent`, `EFEAgent` (at $w=1$), and `InspectionTreeSearchAgent`, exposes this decomposition. For every candidate action considered at the root of the decision, the record stores the sensing cost, the expected information gain in bits, the weighted information gain $w \cdot \text{IG}$, the expected task value, and the resulting total score, plus which action was selected. Enabling this record does not change any action selection. It is a read-only projection of quantities the agent already computes, and unit tests confirm that the recorded total score always ranks the chosen action first and that its components sum correctly.

Table 25 shows one representative decision point from a Structural Inspection-N8 episode ($w=5$) where the ranking by total score disagrees with the ranking by raw information gain. `test1` carries about $5\times$ the information gain of `test0`, yet the agent selects `test0` because `test1`’s higher sensing cost is not offset by a compensating gain in expected task value. The audit trail lets a practitioner

Table 25. Per-action value-of-information audit for one representative decision (Inspection-N8, info-gain weight $w = 5$, step 2 of the episode seeded at 42). Total score is the maximized planning objective (Section 3). Task value is net of sensing cost, so Total score = Task value + $w \cdot \text{IG}$, with Cost shown for reference. The chosen action need not have the largest raw information gain once sensing cost and downstream task value are accounted for.

Action	Cost	IG (bits)	$w \cdot \text{IG}$	Task value	Total score
test0	0.50	0.021	0.106	-26.29	-26.18
test1	2.00	0.106	0.528	-26.87	-26.34
diagnose(nominal)	0.00	0.000	0.000	-26.86	-26.86
diagnose(faulty)	0.00	0.000	0.000	-31.05	-31.05

check, for any single decision, whether an unexpected action was chosen for an information-theoretic reason or a task-value reason. It is an interpretability artifact and changes no reported result.

Q Reward-Rescaling Invariance

To test the scale objection directly, we multiply all rewards and costs by $k \in \{0.1, 1, 10\}$ (the factor written α in Proposition 4, renamed here because α also denotes reward asymmetry) on Diagnosis and Bandit and re-sweep Planning+IG weights. Maximizing $\mathbb{E}[kR] + wI$ induces the same policy as maximizing $\mathbb{E}[R] + (w/k)I$. Matching a given unscaled policy therefore requires scaling the weight up with k , and $w_{\text{ret}}^*(k) \approx k w_{\text{ret}}^*(1)$. Figure 21 and Table 26 report the sweep (200 episodes $\times$ 3 seeds per (k, w) pair). Results confirm covariance of w_{ret}^* with k . $w=1$ is not scale-invariant, and the paper’s near-optimality claims are conditional on the shared reward convention used throughout the suite.

Table 26. Reward-optimal Planning+IG weight under reward scale k . The direction of covariance is as predicted (larger scales favor larger weights). On Bandit the ratio w_{ret}^*/k is exactly constant across the three tested scales. On Diagnosis it varies twofold (flat high- w plateaus mean several grid weights tie for w_{ret}^* , and ties are broken toward the smallest weight reaching the maximum), so the table is qualitative evidence of covariance on Diagnosis and a tight quantitative match on Bandit, not a precise stability estimate on this finite grid.

Environment	k	w_{ret}^*	w_{ret}^*/k
Diagnosis	0.1	0.1	1.0
Diagnosis	1.0	0.5	0.5
Diagnosis	10.0	5.0	0.5
Bandit	0.1	0.1	1.0
Bandit	1.0	1.0	1.0
Bandit	10.0	10.0	1.0

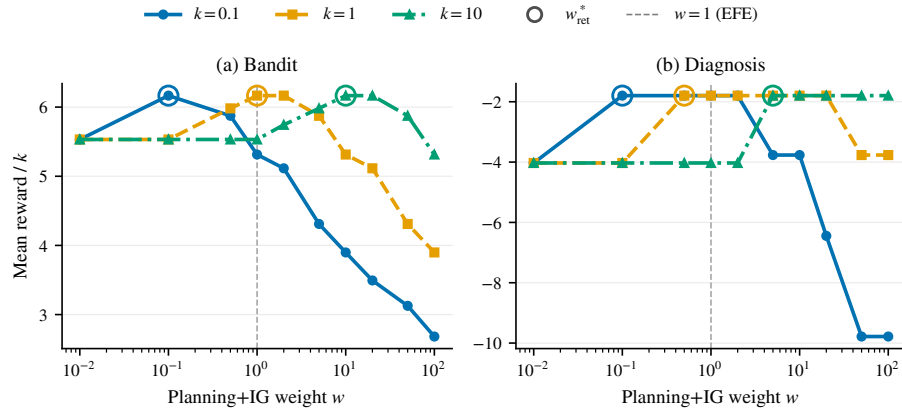


Fig. 21. Reward rescaling on Bandit (a) and Diagnosis (b). Scale-normalized mean reward (mean reward divided by k) vs. Planning+IG weight w under global scale factors k . Normalization places the three scales on one axis, so the equivariance is read directly. The three curves share one shape, and the reward-maximizing weight (open circles, w_{ret}^*) shifts toward larger w as k grows, consistent with $w_{\text{ret}}^*(k) \propto k$. The dashed rule marks $w=1$, the EFE weight.

R Information Base and the Nat-Canonical Agent

The implementation measures information in bits, so the implemented $w=1$ agent carries an effective nat-denominated weight of $1/\ln 2 \approx 1.44$, and the exactly nat-canonical identification of Proposition 1 corresponds to $w = \ln 2 \approx 0.69$ in the implementation’s bit units (Section 3.3). The weight sweep of Section 7.2 answers whether this gap matters only in part, because 0.69 is not a grid weight. On Bandit the reward-maximizing tied run (the closed range of adjacent grid weights whose sweep rewards are exactly tied at the maximum) has grid endpoints $[1, 2]$, and the nearest swept weight below 0.69, $w=0.5$, scores 6.05 against the tied 6.24 (`results_pareto_sweep.csv`). We therefore ran the exactly nat-canonical agent, $w = \ln 2$ in the implementation’s bit units, together with a fresh $w=1$ run on the same seeds, under the sweep’s own protocol (500 episodes $\times$ 5 seeds) on all five swept environments (`results_nat_canonical_check.csv`, `experiments/run_nat_canonical_check.py`). On Tiger, Diagnosis, Bandit, and Testbed the two agents are bit-identical in reward, success rate, and mean observation count (5.33 and 99.6% on Tiger, -1.26 and 97.4% on Diagnosis, 6.24 and 86.9% on Bandit, 0.37 and 96.4% on Testbed), so on Bandit the nat-canonical weight lands inside the tied run rather than with $w=0.5$. On Testbed both agents score 0.37 against the 0.48 that the weights of the tied reward-maximizing run $[0.01, 0.5]$ achieve, and the separation of the two conventions by the $H=2$ interval is discussed after Table 31.

Tileworld is the one swept environment where the base changes the policy. There the nat-canonical agent behaves like the sweep’s $w=0.5$ grid point rather

than like its $w=1$ grid point (-20.81 reward, 74.7% success, and 15.64 scans against -21.53 , 72.0% , and 14.75 , `results_nat_canonical_check.csv`). The difference is significant in observation count and within sampling error in reward and success rate. Seed-level Welch t -tests under the protocol of Appendix I give $p=9.5\times 10^{-5}$ for observation count, surviving Holm correction over the three metrics, against $p=0.34$ and 0.070 for reward and success rate. The conclusion Tileworld supports is unchanged by the check. Under either unit convention $w=1$ is Pareto-dominated there by the larger weight $w=20$, which beats it on both axes of the sweep, reward and success rate (-18.95 and 91.8% , `results_pareto_sweep.csv`).

S POMCP Baseline Comparison

We compare against POMCP [55], a standard online POMDP solver that handles exploration through Monte Carlo tree search with UCB1 action selection. This addresses whether EFE’s explicit information gain valuation provides benefit over the implicit exploration in POMCP’s tree search. Our implementation adapts the algorithm to the observe-then-commit action structure, and two fidelity gaps against Silver and Veness must be stated before the numbers. The depth-cap leaf returns the best commit reward for the sampled state, a clairvoyant quantity. And rollouts credit the belief-expected commit reward rather than the sampled-state reward. The first gives an optimistic terminal value, and the second replaces a sampled payoff by its conditional expectation. Their effects on the resulting policy are not isolated here. Each rollout starts from the belief updated along the simulated action-observation history that reached its leaf. Tests check that simulated observations are charged their cost and that rollouts start from the path belief (`tests/test_pomcp_cost_sign.py`, `tests/test_pomcp_rollout_belief.py`). Neither gap is present in the RockSample POMCP of Appendix M, whose own departures from the published algorithm, all keyed to fully observable quantities, are stated there. Our POMCP rollout policy is semi-informed. Observation actions are selected uniformly at random, but the rollout terminates (stochastically, with 30% probability per step) with a belief-optimal commit, i.e., the commit action maximizing expected reward under the Bayesian-updated rollout belief. This is strictly stronger than a fully random rollout, which would also randomize over commit actions. Each of the five outer seeds initializes both the environment stream and the planner’s internal random stream (particle sampling and rollouts) from that seed, so the reported variability includes planner randomness. The success-rate gap is smallest on Tiger, which has a single observation action, and larger on Diagnosis, Bandit, and Tileworld, where the agent must choose among several tests, and the exploration sweep below shows that neither the exploration constant nor the rollout policy closes it on the two swept multi-test environments. EFE chooses which test to run based on information gain, while POMCP samples tests uniformly, but we do not attribute the gap to that difference in isolation, since exact EFE also differs from this POMCP in its exact tree enumeration and its closed-form leaf

and commit values, and the ablation below does not separate those from one another. We sweep POMCP simulation budgets across $\{500, 1000, 2000, 5000\}$ and report wall-clock timings for a simulation-budget scaling analysis. This comparison fixes POMCP’s simulation budgets without matching computation to exact EFE or Planning. The MCTS-EFE comparison below separately matches simulation counts between the two Monte Carlo planners. Tree-node action selection uses UCB1 with exploration constant 10.0 (`rho_aif/agents/pomcp.py`), fixed across all environments. This was not tuned per environment, and because reward scales differ across our benchmarks (Section 5), the same exploration constant implies a different effective exploration–exploitation balance on each one, a caveat symmetric to the reward-scale sensitivity we report for w (Appendix Q). Tree and rollout depth are capped at the reporting agent’s planning horizon plus three steps (`experiments/run_pomcp.py`).

Table 27. POMCP comparison (5,000 total episodes across 5 per-run seeds $\{42, 123, 456, 789, 1024\}$ for Tiger/Diagnosis/Bandit and 1,000 for Tileworld). The protocol matches the main table on the three core environments. EFE and Planning are evaluated inside the POMCP harness and reproduce Table 5 exactly. POMCP uses 1,000 simulations per decision. Reward is mean $\pm$ seed-level SE. Bold marks EFE and any agent bit-identical to it, the convention of Table 5, so it flags EFE’s row rather than a per-column best. On Tileworld 6×6 Planning is nominally ahead of EFE on both displayed metrics (73.6% against 71.9%, -21.41 against -21.62), within seed-level sampling error.

Environment	Agent	Obs.	Success	Reward
Tiger	Planning	4.20	99.4%	$+5.19 \pm 0.21$
	POMCP (1000)	1.64	89.2%	-3.57 ± 0.18
	EFE	4.20	99.4%	$+5.19 \pm 0.21$
Diagnosis	Planning	5.86	88.8%	-2.57 ± 0.20
	POMCP (1000)	3.19	70.5%	-10.91 ± 0.63
	EFE	9.66	97.2%	-1.37 ± 0.18
Bandit	Planning	3.29	71.0%	$+5.75 \pm 0.08$
	POMCP (1000)	1.34	47.2%	$+4.58 \pm 0.09$
	EFE	5.11	86.9%	$+6.27 \pm 0.04$
Tileworld 6×6	Planning	15.57	73.6%	-21.41 ± 0.64
	POMCP (1000)	0.05	2.6%	-48.49 ± 0.28
	EFE	14.76	71.9%	-21.62 ± 0.87

POMCP comparison. Table 27 shows POMCP underperforming both EFE and Planning on reward and success rate on every environment. On Tiger, POMCP at this table’s untuned exploration constant commits prematurely (1.64 observations vs. 4.20 for EFE), yielding 89.2% success against EFE’s 99.4% (here numerically identical to Planning, per the exact tie documented in Section 7). On

Diagnosis, POMCP achieves 70.5% success (-10.91 reward) while EFE reaches 97.2% (-1.37 reward). On Bandit, POMCP inspects 1.34 times per episode against EFE’s 5.11 and reaches $47.2 \pm 1.0\%$ success against EFE’s $86.9 \pm 0.3\%$ (seed-level SE), at $+4.58 \pm 0.09$ reward against $+6.27 \pm 0.04$. On Tileworld (6×6 , $|S|=36$), POMCP collapses to 2.6% success with almost no scans (0.05 per episode), close to the $1/36$ rate of committing without information, plausibly because the 36 commit actions and 6 scan actions create a branching factor that 1,000 simulations do not resolve.

Simulation-budget scaling. Increasing POMCP’s simulation budget to 5,000 does not qualitatively change the picture. On Diagnosis, POMCP(5000) reaches $71.6 \pm 0.9\%$ success (seed-level SE), still far short of EFE’s $97.2 \pm 0.3\%$, while taking more wall-clock time per episode (323.6 ms against EFE’s 93.8 ms). On Tiger, POMCP(5000) reaches $89.5 \pm 0.3\%$, within sampling error of its $89.2 \pm 0.2\%$ at 1,000 simulations. On Bandit the sweep is not monotone. POMCP(1000) reaches $47.2 \pm 1.0\%$ success, below POMCP(500)’s $50.9 \pm 1.1\%$ and POMCP(2000)’s $51.6 \pm 1.1\%$, and POMCP(5000) rises to $62.1 \pm 1.2\%$ at $+5.56 \pm 0.10$ reward, still below EFE on both metrics, at roughly 5.7 times the wall-clock cost of the 1,000-simulation budget. On Tileworld no budget exceeds 3.8% success. As configured here, with this rollout policy and a single untuned exploration constant, the gap is therefore not a simulation-budget artifact. A tuned constant narrows it on Tiger (Table 29), so we do not claim the comparison bounds what a best-practice POMCP can achieve.

MCTS-EFE. To disentangle EFE’s information valuation from the planning mechanism, we use MCTS-EFE, the UCB1 tree search defined in Appendix Y, which uses EFE as the leaf heuristic rather than random rollouts, and whose simulation-matched Tiger comparison against POMCP is reported there. As a reference point, exact EFE reaches 99.9% success on Tiger at $H=6$ (64 ms per episode) under the MCTS battery’s own 200-episodes-per-seed protocol rather than Table 5’s battery. MCTS-EFE and POMCP differ in more than the leaf, so the two components that distinguish this search from a UCT tree are ablated below. A compute-matched check on the same machine (`experiments/run_pomcp_compute_matched.py`) times MCTS-EFE(500) end to end for the full protocol (5 seeds $\times$ 200 episodes) at 13.87s ($97.7\% \pm 0.5\text{pp}$ success, seed-level SE), with no other battery running. Calibrating POMCP’s wall-clock cost at 200/500/1,000 simulations on the same protocol and machine and solving for the simulation budget matching 13.87s gives 661 simulations, roughly 32% more than the simulation-matched comparison. POMCP(661) reaches $88.0\% \pm 0.6\text{pp}$ success (-4.80 ± 0.69 reward, 13.88s), and a seed-level Welch test against the simulation-matched POMCP(500) ($88.1\% \pm 0.2\text{pp}$ success, -4.67 ± 0.19 reward, 10.42s) finds no significant difference (success $p=0.89$, reward $p=0.85$). Giving POMCP the same wall-clock budget as MCTS-EFE rather than the same simulation count does not close the gap at this exploration constant. Both POMCP configurations remain significantly behind MCTS-EFE on both metrics (seed-level Welch $p < 0.0001$ in every comparison). On Tiger, MCTS-EFE’s success

Table 28. MCTS-EFE component ablation under the MCTS battery’s protocol (200 episodes per seed $\times$ 5 seeds, at the same horizons and simulation budgets as the MCTS-EFE runs of `results_mcts_efc.csv`). The first row of each block, max-backup with the in-tree information gain, is the manuscript’s MCTS-EFE. Uncertainty is seed-level SE. A dagger marks a variant whose success or reward differs from that row at seed-level Welch $p < 0.05$ after Holm–Bonferroni within metric over the whole battery. No variant carries one, so no component shows a detectable effect on success or reward on any environment tested. See `results_mcts_efc_ablation.csv` and its `_stats` companion.

Environment	Backup	In-tree IG	Success	Reward	Obs.
Tiger	max	yes	$97.7 \pm 0.5\%$	$+3.73 \pm 0.53$	3.73
	mean	yes	$97.9 \pm 0.3\%$	$+4.01 \pm 0.33$	3.68
	max	no	$97.4 \pm 0.7\%$	$+3.66 \pm 0.75$	3.48
	mean	no	$97.7 \pm 0.5\%$	$+3.95 \pm 0.60$	3.52
Diagnosis ($N=4$)	max	yes	$94.2 \pm 0.5\%$	-3.26 ± 0.30	9.78
	mean	yes	$95.8 \pm 0.5\%$	-2.55 ± 0.38	10.03
	max	no	$96.3 \pm 0.6\%$	-2.49 ± 0.53	10.27
	mean	no	$95.7 \pm 0.8\%$	-2.97 ± 0.40	10.39
Tileworld-6 $\times$ 6	max	yes	$95.4 \pm 0.6\%$	-25.02 ± 0.25	32.26
	mean	yes	$96.2 \pm 0.5\%$	-24.35 ± 0.39	32.07
	max	no	$94.7 \pm 1.0\%$	-26.34 ± 0.57	33.16
	mean	no	$95.7 \pm 0.3\%$	-25.29 ± 0.22	32.71

and observation count are insensitive to both horizon (6, 8, 10) and simulation budget (200, 500, 1,000) in this range, saturating at 97.7% success and 3.735 observations throughout. This reflects Tiger’s small reachable-belief set, which the tree already covers well within 200 simulations, rather than the horizon or budget parameters having no effect in general.

Component ablation. MCTS-EFE differs from a UCT tree in its leaf evaluation, in its exact belief representation and exact commit-child values, and in two switchable places, the in-tree information gain and the max-backup. Table 28 runs all four combinations of those two switches under this battery’s own protocol. Neither switchable component shows a detectable effect on success or reward. No variant differs from the reported configuration on either metric on any of the three environments after Holm correction within metric. The only difference surviving that correction anywhere is Tiger’s observation count, which falls from 3.73 to 3.48 when the in-tree information gain is removed (seed-level Welch $p=0.0014$). At five seeds this is a failure to detect an effect rather than evidence that there is none. What remains to carry the contribution is the EFE-greedy leaf evaluation together with the exact belief representation and the exact commit-child values, which the ablation holds fixed. That is a narrower attribution than crediting the EFE objective throughout the tree, and it is the one these data support.

Exploration constant and informed rollouts. Table 29 sweeps POMCP’s UCB1 exploration constant on the MCTS battery’s three environments, Tiger, Diagnosis, and Tileworld-6×6, over the seven values $\{1, 2, 5, 10, 20, 50, R\}$, where R is the commit reward range that MCTS-EFE takes as its own default. Bandit is not swept. Outside the sweep, the observe-then-commit POMCP runs with a single untuned constant per battery, 10.0 in Table 27 and 5.0 in the MCTS-EFE battery of `results_mcts_efe.csv`, and Table 29 writes the constant as c , a quantity distinct from the observation cost c of Proposition 2. The sweep also adds an information-gain rollout that selects the observation with the largest exact one-step information gain at every rollout step. Three findings follow. First, the constant matters on Tiger, and tuning it does not close the gap. POMCP at $c=50$ reaches $93.2\% \pm 0.5$ success at $+0.39 \pm 0.52$ reward, against 87.3 to 89.0% at constants up to 10, and it remains behind MCTS-EFE at its default constant on both metrics after Holm correction (seed-level Welch $p=0.0002$ and $p=0.002$). Second, MCTS-EFE’s own default constant is not its best either. At $c=5$ on Tileworld-6×6 it earns -22.26 ± 0.46 against -25.02 ± 0.25 at the default (seed-level Welch $p=0.0017$) at indistinguishable success, so comparing the two solvers at their respective defaults is not a tuned-against-tuned comparison in either direction. Taking each solver at a configuration selected on the disjoint tuning seeds $\{11, 22, 33\}$ at 100 episodes per seed instead (highest success on those seeds, ties broken by reward, `results_pomcp_exploration_tuning.csv` and `experiments/select_pomcp_constants.py`), and reading the frozen selection off the canonical five-seed rows of Table 29, MCTS-EFE remains ahead on Tiger success ($98.2\% \pm 0.3$ at $c=5$ against POMCP’s $93.2\% \pm 0.5$ at $c=50$, seed-level Welch $p=0.0001$) and reward ($+4.33 \pm 0.28$ against $+0.39 \pm 0.52$, $p=0.0005$), both surviving Holm correction within the stated six-test family of success and reward on the three environments. The rule was recorded before the selection was run, with success the primary metric, but the rule and its candidate grid were written after an exploratory sweep on the canonical seeds had been inspected. It picks $c=5$ for MCTS-EFE on Tiger and Diagnosis and its default on Tileworld, and for POMCP $c=50$ with uniform rollouts on Tiger, $c=5$ with uniform rollouts on Diagnosis, and $c=5$ with information-gain rollouts on Tileworld. On Tiger and Tileworld these are the configurations the evaluation seeds would also have selected, and on Diagnosis the evaluation seeds would have selected $c=2$ for POMCP. Third, no swept constant rescues POMCP where the agent must choose among tests. Its tuning-seed selection reaches $65.6\% \pm 1.4$ on Diagnosis and $8.2\% \pm 0.6$ on Tileworld-6×6 against MCTS-EFE’s $94.7\% \pm 0.4$ at $c=5$ and $95.4\% \pm 0.6$ at its default, every comparison significant after Holm correction. The evaluation seeds’ own best POMCP configuration on Diagnosis, $c=2$ at $69.1\% \pm 1.7$, is not the selected one and is reported here only as a descriptive row of the sweep. On Tileworld-6×6 POMCP with uniform rollouts barely scans at $c \geq 5$, at most 0.09 scans per episode, and succeeds on 1.8 to 3.1% of episodes, while $c=1$ reaches 7.1% with about one scan. That environment’s best uniform-rollout constant therefore sits at the grid’s lower edge, and a constant below 1 was not tested. Information-gain rollouts leave both gaps open. At $c=5$

they put Diagnosis at 66.8% against 65.6, not a significant difference ($p=0.57$), and Tileworld at 8.2% against 2.6, which survives Holm correction on success ($p=0.0002$) but not on reward ($p=0.012$). They leave Tiger bit-identical, since with one observation action selecting the most informative observation and selecting uniformly are the same policy. POMCP’s planner stream is seeded per run from the outer seed in this battery, as it is in Table 27’s, while the MCTS battery of `results_mcts_efe.csv` left it at a fixed constant, so these POMCP rows are not bit-identical to that battery’s.

Table 29. POMCP exploration-constant sweep and informed rollouts under the MCTS battery’s protocol (200 episodes per seed $\times$ 5 seeds, at the same horizons and simulation budgets as the MCTS-EFE runs of `results_mcts_efe.csv`). R is the commit reward range (110 on Tiger, 60 elsewhere), MCTS-EFE’s default constant. The info-gain rollout takes the observation with the largest exact one-step information gain at every rollout step. POMCP’s planner stream is seeded per run from the outer seed here, which the MCTS battery left at a fixed constant, so these POMCP rows are not bit-identical to that battery’s. Uncertainty is seed-level SE. A dagger marks a row whose success or reward differs from MCTS-EFE at its default constant at seed-level Welch $p < 0.05$ after Holm–Bonferroni within metric over the whole battery. See `results_pomcp_exploration_sweep.csv` and its `_stats` companion.

Environment	Solver	c	Rollout	Success	Reward
Tiger	POMCP	1	uniform [†]	$87.3 \pm 0.4\%$	-5.51 ± 0.39
	POMCP	2	uniform [†]	$88.1 \pm 0.7\%$	-4.67 ± 0.72
	POMCP	5	uniform [†]	$89.0 \pm 0.3\%$	-3.74 ± 0.29
	POMCP	10	uniform [†]	$88.4 \pm 0.8\%$	-4.40 ± 0.87
	POMCP	20	uniform [†]	$91.5 \pm 0.5\%$	-1.24 ± 0.48
	POMCP	50	uniform [†]	$93.2 \pm 0.5\%$	$+0.39 \pm 0.52$
	POMCP	R	uniform [†]	$87.6 \pm 0.4\%$	-5.01 ± 0.39
	POMCP	5	info-gain [†]	$89.0 \pm 0.3\%$	-3.74 ± 0.29
	POMCP	R	info-gain [†]	$87.6 \pm 0.4\%$	-5.01 ± 0.39
	MCTS-EFE	R	EFE	$97.7 \pm 0.5\%$	$+3.73 \pm 0.53$
	MCTS-EFE	5	EFE	$98.2 \pm 0.3\%$	$+4.33 \pm 0.28$
Diagnosis ($N=4$)	POMCP	1	uniform [†]	$64.0 \pm 1.3\%$	-14.57 ± 0.75
	POMCP	2	uniform [†]	$69.1 \pm 1.7\%$	-11.59 ± 0.99
	POMCP	5	uniform [†]	$65.6 \pm 1.4\%$	-13.56 ± 0.85
	POMCP	10	uniform [†]	$67.4 \pm 1.0\%$	-12.59 ± 0.62
	POMCP	20	uniform [†]	$59.9 \pm 1.1\%$	-16.31 ± 0.67
	POMCP	50	uniform [†]	$50.7 \pm 1.2\%$	-21.30 ± 0.68
	POMCP	R	uniform [†]	$50.3 \pm 0.7\%$	-21.40 ± 0.44
	POMCP	5	info-gain [†]	$66.8 \pm 1.5\%$	-13.09 ± 0.87
	POMCP	R	info-gain [†]	$53.3 \pm 1.6\%$	-19.76 ± 0.98
	MCTS-EFE	R	EFE	$94.2 \pm 0.5\%$	-3.26 ± 0.30
	MCTS-EFE	5	EFE	$94.7 \pm 0.4\%$	-2.75 ± 0.31
Tileworld-6 $\times$ 6	POMCP	1	uniform [†]	$7.1 \pm 1.0\%$	-46.78 ± 0.59
	POMCP	2	uniform [†]	$3.0 \pm 0.8\%$	-48.61 ± 0.48
	POMCP	5	uniform [†]	$2.6 \pm 0.5\%$	-48.53 ± 0.31
	POMCP	10	uniform [†]	$3.1 \pm 0.3\%$	-48.16 ± 0.18
	POMCP	20	uniform [†]	$2.7 \pm 0.6\%$	-48.40 ± 0.39
	POMCP	50	uniform [†]	$3.1 \pm 0.5\%$	-48.15 ± 0.29
	POMCP	R	uniform [†]	$1.8 \pm 0.4\%$	-48.93 ± 0.24
	POMCP	5	info-gain [†]	$8.2 \pm 0.6\%$	-46.92 ± 0.38
	POMCP	R	info-gain [†]	$2.1 \pm 0.3\%$	-48.85 ± 0.18
	MCTS-EFE	R	EFE	$95.4 \pm 0.6\%$	-25.02 ± 0.25
	MCTS-EFE	5	EFE [†]	$94.9 \pm 0.7\%$	-22.26 ± 0.46

T Budgeted ρ -POMDP Supplementary Details

This appendix gives the details behind the budgeted experiments that the main text summarizes. It records how the SARSOP models were built and exported, proves the factorization property the distractor-diagnosis environment relies on, and presents the atlas of measured shadow prices.

T.1 SARSOP Build and Export Details

Section 6.6’s reference solver is the original APPL C++ toolkit (`pomdpso1`), built from source by `tools/build_sarsop.sh`, which applies and documents four compatibility patches for a modern clang toolchain on Apple Silicon (compiler flags, iterator construction, assertion syntax, and the legacy parser). The binary is not committed. `experiments/run_sarsop_baseline.py` exports any discrete observe-then-commit benchmark generically from the same hidden-state, action-layout, and observation-model definitions the agents already use (an absorbing done state, discount $\gamma=0.999$), solves to precision 10^{-3} , parses the resulting alpha-vector policy from its XML output, and evaluates it through the same episode runner used for every other agent so that mechanics and metrics are shared exactly. Four unit tests (`tests/test_sarsop_export.py`) cover the exporter, the policy parser, and the alpha-vector argmax agent without requiring the compiled binary, so the export pipeline is checked even in environments where the solver is unavailable.

T.2 Distractor-Diagnosis Environment and Its Factorization Guarantee

Section 6.9’s `DistractorDiagnosisEnv` joint state space is the product of a 4-way condition and a binary nuisance factor. Each observation likelihood, for a condition test or the distractor test, depends on exactly one of the two factors. Terminal commit rewards depend only on the condition and are not observations. For a belief that itself factors as a product of independent condition and nuisance marginals, observing the distractor test therefore updates the nuisance marginal and leaves the condition marginal exactly unchanged. The same holds symmetrically for condition tests and the nuisance marginal. This is verified by 13 unit tests (`tests/test_distractor_diagnosis.py`). They include a direct check that the condition marginal is unchanged before and after a distractor observation (up to floating-point tolerance) when the prior belief is factored, which makes their KL divergence exactly zero, and a symmetric check for condition tests against the nuisance marginal. The guarantee is stated for factored beliefs specifically. A belief with residual condition–nuisance correlation (which does not arise under this environment’s independent generative process, but could under model misspecification) is outside the scope of the proof.

Table 30. Atlas of operational sensing budgets across benchmark instances, giving the usage range of the Planning+IG family, the implicit EFE budget $B_{\text{EFE}} = U(w=1)$ (mean $\pm$ SE over seeds), and crossing brackets estimating $w^*(B)$ at two canonical budgets per instance. Each bracket estimates a single crossing threshold from the finite weight grid (Definition 2). No closed-form meta-model is implied.

Instance	$[U_{\min}, U_{\max}]$	B_{EFE}	B_1 : w^* bracket	B_2 : w^* bracket
Bandit	[3.23, 12.34]	5.11 ± 0.11	4.6: (0.0611, 0.139]	11.0: (19.3, 43.9]
Diagnosis	[5.90, 13.16]	9.77 ± 0.14	7.0: (0.139, 0.316]	12.1: (19.3, 43.9]
Inspection-N16	[22.38, 58.56]	28.02 ± 0.13	27.8: (0.316, 1.33]	53.1: (23.7, 100]
Inspection-N8	[12.70, 29.49]	15.56 ± 0.09	15.2: (0.316, 2.15]	27.0: (14.7, 100]
RS[5,3]	[3.00, 8.83]	4.79 ± 0.05	3.9: (0.316, 1.33]	8.0: (5.62, 23.7]
RS[7,4]	[2.00, 14.54]	5.37 ± 0.07	3.9: (0.316, 2.15]	12.7: (14.7, 100]
Tiger	[4.37, 5.77]	4.37 ± 0.11	4.6: (19.3, 43.9]	5.6: (19.3, 43.9]
Tileworld-6x6	[14.74, 32.73]	14.85 ± 0.13	17.4: (5.62, 23.7]	30.0: (23.7, 100]

T.3 The w^* Atlas

Table 30 collects, for the eight benchmark instances whose usage curves we estimate, the usage range each curve spans from its floor to its ceiling, the implicit EFE budget $B_{\text{EFE}} = U(w=1)$, and two crossing brackets (Definition 2) at two canonical budgets. It is generated by `experiments/run_w_atlas.py` from the usage curves of Sections 6.4 and 6.5 (`results_price_usage_curves.csv`, `results_price_interleaved_curves.csv`). The two canonical budgets are placed inside each instance’s own usage range, one a fixed fraction of that range up from its floor and one the same fraction down from its ceiling, so that every instance is bracketed at comparable relative positions on its curve. Brackets are grid intervals by construction. Read across instances, the table shows the implicit EFE budget at or near the floor of the usage range on some instances and part of the way up it on others. It is an atlas of measured operating points and implies no closed-form relationship between B_{EFE} and environment parameters such as reward scale, state count, or horizon, and none should be inferred from it.

U Methodological and Budget-Theory Details

This appendix retains the derivations, boundary examples, and implementation qualifications supporting Sections 3 and 4.

U.1 Observe-then-Commit Structure and EFE Scope

Proposition 1 covers one specific formulation of EFE, and we name what it does not cover before interpreting it. Equation 3 instantiates the recursive, closed-loop form of EFE associated with sophisticated inference [12, 20]. Among the four EFE formulations that Champion et al. [9] relate, this is the information-gain and

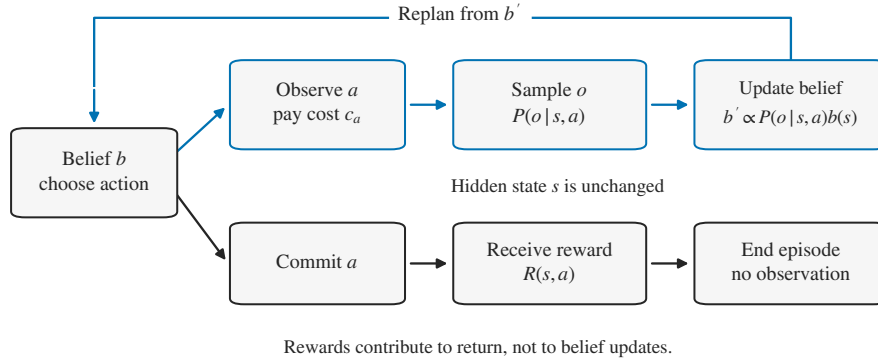


Fig. 22. The observe-then-commit decision loop, shown schematically. An observation incurs a cost and updates the belief about a hidden state that remains fixed. The agent then chooses again from the updated belief. A commit instead pays a state-dependent reward and ends the episode without emitting an observation. Rewards are not inputs to the belief update. The loop describes the episode, while the planner’s finite horizon limits how far it looks ahead at each decision.

pragmatic-value form, with information gain about the hidden state evaluated recursively over belief trajectories. We establish the $w=1$ coefficient for that formulation only. Whether the same coefficient survives under the Free Energy of the Expected Future [40], generalised free energy [42], the epistemic-prior variational treatment of de Vries et al. [13], or the Bethe-Lagrangian stationary point of Kouw [31] is not established here. A negative answer for any of them would narrow the scope of Proposition 1 without touching the budgeted results of Section 4.

Reward-to-nats calibration. The equivalence needs an exchange rate between reward and information. The pragmatic term of EFE is an expected log-probability, measured in nats, while task reward arrives in whatever units the environment specifies, so identifying the two means fixing how many nats one reward unit is worth. Identifying pragmatic value with task reward chooses $P(o|C) \propto \exp(\beta R)$ with $\beta=1$ per reward unit, which sets that rate at one nat per reward unit. This identification is a convention adopted in assumption (i) of Proposition 1, not a derivation, and the proposition is a statement about the recursion of Equation 3 as written, in which the pragmatic value of each scored outcome is its task reward and no normalizer appears. Deriving that recursion from a normalized preference distribution would add the accumulated term $-T \ln Z$ of Section 3.3, which is policy-dependent because observe-then-commit paths score different numbers of outcomes. It vanishes under log scoring ($Z=1$) and otherwise cancels only when every path scores the same number of outcomes against one preference distribution over a common outcome space, for example when a committed episode keeps emitting a zero-reward null outcome from an absorbing state at every re-

maining step. We do not adopt such a construction, and the proposition’s scope does not extend to a general normalized-preference derivation for variable-length episodes. Three things are therefore kept apart in this article. The exact case is that of Bernardo [4], where rewards are log scoring rules, so the reward is already a log probability measured in nats and no normalizer arises. The proved case is the reward-based recursion, conditional on assumption (i). The empirical convention is one reward unit per bit, stated next. Rescaling all rewards by a positive factor k (the factor written α in Proposition 4) at fixed w induces the same policy as weight w/k , so $w=1$ is not scale-invariant, and the reward-maximizing weight w_{ret}^* (the subscript abbreviates return) scales as $w_{\text{ret}}^*(k) \propto k$ (Appendix Q). Our claim is weaker and operational. Within a fixed reward convention, the information-unit weight $w=1$ is a robust untuned default.

The implementation measures information in bits (`scipy.stats.entropy` with `base=2`) and runs every reported EFE ($w=1$) agent at one reward unit per bit, which is $1/\ln 2 \approx 1.44$ reward units per nat. The exactly nat-canonical identification of Proposition 1 is therefore $w = \ln 2 \approx 0.69$ in the implementation’s bit units. Appendix R runs that agent on all five swept environments. It is bit-identical to the implemented $w=1$ agent in reward, success rate, and observation count on Tiger, Diagnosis, Bandit, and Testbed, and on Tileworld, the one environment where the base changes the policy, $w=1$ remains Pareto-dominated by $w=20$ under either convention.

U.2 Two-State Thresholds and Their Empirical Scope

The two parts of Proposition 2 concern distinct horizons. We retain the derivation and descriptive comparisons here.

Proof (Proof sketch). Part (a). At $H=1$, the agent observes once and commits. A Planning+IG agent with weight w observes iff $-c + w \cdot I(b) + \mathbb{E}_o[\max_i \mathbb{E}_{b'_o}[R_i]] > \max_i \mathbb{E}_b[R_i]$, where the left side is the observe-then-commit value and the right side is the immediate commit value. For the uniform prior, the immediate commit value is $(R^+ + R^-)/2$. After one observation, the posterior concentrates. With probability p the agent is correct, yielding expected commit value $p \cdot R^+ + (1-p) \cdot R^-$. The net gain from observing is $(p - \frac{1}{2})(R^+ - R^-) - c + w \cdot I_{\max}$. Setting this to zero gives $w_{\text{thresh}}^* = [c - (p - \frac{1}{2})(R^+ - R^-)]/I_{\max}$. Substituting $R^- = -\alpha R^+$ yields the second form. When $\alpha \gg 1$, the marginal reward improvement $(p - \frac{1}{2})(1 + \alpha)R^+$ dominates the cost c , making w_{thresh}^* negative, so the agent should observe at any $w \geq 0$, including $w=1$.

Part (b). Fix the post-first-observation belief $b=(p, 1-p)$. The agent prefers a second observation over committing when $-c + w \cdot I(b) + \mathbb{E}_{o'}[\max_i \mathbb{E}_{b'_{o'}}[R_i]] > \max_i \mathbb{E}_b[R_i]$, i.e., when $w > (c - \text{VoI}_2(b))/I(b)$ whenever $\text{VoI}_2(b) < c$. That cutoff is w_{over}^* . Combining both inequalities yields the $H=2$ near-optimality interval.

Under the paper’s reward convention, $w=1$ lies in this interval for Tiger, the one tabulated instance that is genuinely two-state with $|R^-| > R^+$ and so falls under the proposition as stated. For Diagnosis and Tileworld, which are multi-state, multi-action environments outside the proposition’s stated hypothesis, the

same thresholds are applied heuristically to a two-state reduction, for which no formal correspondence is proved. The reduction takes each environment’s observation accuracy as p , its per-observation cost as c (a test on Diagnosis, a scan on Tileworld), and its correct and incorrect commit rewards as R^+ and R^- , and it keeps nothing of the state and observation-action counts. Diagnosis and Tileworld share all four of those values (Appendix E), which is why their two rows in the table coincide. Table 31 provides a descriptive comparison against Proposition 2 on this basis. It reports α , η , the lower and upper thresholds w_{thresh}^* and w_{over}^* , and the observed reward-maximizing weight from the Pareto sweep (Section 7.2).

Table 31. Reward asymmetry (α), informativeness (η in nats per reward unit), lower observation threshold w_{thresh}^* , upper over-observation threshold w_{over}^* , and the observed reward-maximizing weight (or tied run) w_{ret}^* from a fresh Pareto sweep (Section 7.2, the eleven-point grid $w \in \{0.01, 0.1, 0.5, 1, 2, 5, 10, 20, 50, 100, 200\}$, 5 seeds $\times$ 500 episodes, `results_pareto_sweep.csv`). Where the sweep’s reward is exactly tied across several adjacent grid weights, w_{ret}^* is reported as the closed range of tied grid weights rather than an arbitrary point within it. That range is a flat set of maximizers, the analogue of the level set of Definition 2, not a crossing bracket. Only Tileworld has a genuinely unique argmax in this grid. The thresholds are in reward units per nat. w_{ret}^* is in the implementation’s reward units per bit. The $w=1$ column asks about the canonical nat-denominated weight, while the implemented bit-denominated agent corresponds to 1.44 reward units per nat (see text). Bandit omitted as multi-arm, so Prop. 2 does not apply directly.

Environment	α	η	w_{thresh}^*	w_{over}^*	$w=1$ in $(w_{\text{thresh}}^*, w_{\text{over}}^*)$?	w_{ret}^*
Testbed	1.0	1.31	−3.06	1.01	Nat units only (see text)	[0.01, 0.5] (tied)
Tiger	10.0	0.27	−138.66	6.89	Yes	[0.01, 20] (tied)
Diagnosis	5.0	0.19	−88.20	7.91	Yes	[0.5, 20] (tied)
Tileworld	5.0	0.19	−88.20	7.91	Yes	20 (unique)

Table 31 shows two patterns. On the three high-asymmetry environments (Tiger, Diagnosis, Tileworld) $w=1$ sits comfortably inside the interval in either unit convention, whereas Testbed sits on the interval’s upper edge and on the boundary of the proposition’s hypothesis, and we take Testbed first. On Testbed, the lower threshold is negative and the upper bound is tight ($w_{\text{over}}^* \approx 1.01$). Nat-denominated $w=1$ sits just inside, while the implemented agent’s bit-denominated $w=1$ corresponds to 1.44 reward units per nat and falls outside the interval. This is consistent with the over-exploration observed relative to the reward-maximizing run $w_{\text{ret}}^* \in [0.01, 0.5]$ (Appendix C). Under the sweep protocol at $H=4$, however, the nat-canonical agent of Appendix R over-explores exactly as the bit-denominated agent does, matching it in reward, success rate, and observation count (`results_nat_canonical_check.csv`), so the interval’s tight upper endpoint separates the two conventions on paper but not in measured behavior at that horizon. Testbed’s $\alpha=1.0$ sits at the boundary of Proposition 2’s

strict assumption $|R^-| > R^+$. Its row is included descriptively rather than as an instance of the proposition.

The second pattern is the high-asymmetry one. Tiger, Diagnosis, and Tileworld have $w_{\text{thresh}}^* \ll 0$ and $w_{\text{over}}^* \approx 7-8$. The near-optimality interval between those thresholds is derived at $H=2$, and the multi-step case beyond $H=2$ is harder to analyze because thresholds shift with belief. Read as a prediction about the sweep, the interval says that $w=1$ lands among the reward-maximizing weights wherever it lies inside the interval. On Tiger and Diagnosis the prediction is borne out, since $w=1$ sits inside the reward-maximizing tied run. On Tileworld it is not. Tileworld’s reward-maximizing weight is uniquely $w=20$ in the implementation’s bit units (28.9 reward units per nat), which lies above the interval’s nat-denominated upper endpoint of 7.91, and $w=1$ is Pareto-dominated there.

Extension to discounting. Discounting preserves the ρ -POMDP equivalence with $\rho(b, a) = I_a(b)$, but it changes the balance between the two terms along a branch. The measured net effect runs toward fewer observations rather than more. With $\gamma < 1$, the recursive EFE becomes $\mathcal{G}(\text{obs}_k) = c_k - I_k(b) + \gamma \mathbb{E}_o[\min_a \mathcal{G}(a, b'_o)]$, giving $V(\text{obs}_k) = -c_k + I_k(b) + \gamma \mathbb{E}_o[\max_a V(a, b'_o)]$. The information gain term $I_k(b)$ appears undiscounted at the current step, while future values are discounted. The intuition is that the information gain at a node is credited immediately, while the reward that information eventually buys arrives only at the commit action and is discounted by γ raised to the number of steps remaining until then. With $\gamma < 1$ the epistemic share of an observe branch’s value therefore rises with the length of the branch. This does not make the agent more exploratory overall, because discounting also shrinks the whole observe branch relative to committing now, and the measured effect runs the other way, with EFE’s observation count on Diagnosis falling from 9.68 at $\gamma=1$ to 5.83 at $\gamma=0.95$ (Table 19). Appendix K compares EFE with Planning, the reward-only planner of the same horizon, across discount factors.

U.3 State-Preservation Proof and Destructive Sensing

Proof (Proof sketch). When action a preserves s_{hid} , the transition on the hidden component is $T_{\text{hid}}(s'_{\text{hid}} | s_{\text{hid}}, a) = \delta(s'_{\text{hid}} = s_{\text{hid}})$. The belief update over s_{hid} is then $b'(s_{\text{hid}}) \propto O(o | s_{\text{hid}}, a) b(s_{\text{hid}})$, depending only on the observation likelihood, identical to the observe-then-commit case. The posterior that would incorporate transitions, $b'_{o,T}$, coincides with the observation-only posterior b'_o , so $D_{\text{KL}}[b'_{o,T} || b'_o] = 0$. For observation actions, EFE reduces to cost minus information gain plus expected continuation, matching the ρ -POMDP Bellman equation with $\rho = I_a(b)$. For navigation actions, the observation is uninformative ($I_a(b) = 0$), giving $\rho = 0$.

Proposition 3 extends the formal bridge from observe-then-commit to any POMDP where the hidden state is preserved by information-gathering and navigation actions. Where the observe-then-commit argument had a single run of observations followed by one commit, the agent here may interleave observation,

navigation, and exploitation in any order. The planner enumerates action sequences as a tree over belief states, using the recursive tree search of Equation 3 extended with navigation and exploitation branches. At each decision point, the EFE-as- ρ equivalence holds on every branch of that tree made up of observation and navigation actions. Because navigation actions never touch the hidden state, interleaving them with observation actions does not disturb the equivalence, and ρ_{EFE} is well defined along any sequence of actions drawn from $\mathcal{A}_{\text{obs}} \cup \mathcal{A}_{\text{nav}}$, regardless of how many such actions precede a given decision.

Exploitation actions are not covered by Proposition 3, and they carry no belief-dependent utility term for two separate reasons, depending on whether they end the episode. An exploitation action that ends the episode, such as exit, carries $\rho_{\text{EFE}} = 0$ for the reason Proposition 1 gives for commit actions, that termination leaves no successor belief to score. An exploitation action that does not end the episode, such as sampling a rock, yields a reward that depends on the hidden state. That its reward already enters the objective is not by itself a reason for it to carry no epistemic term, because a payoff that depends on s_{hid} can reveal s_{hid} . The reason is instead an assumption of the formal model that the benchmark shares. Rewards are not observations. The belief is conditioned on the observation symbol alone, a non-terminal exploitation action emits an uninformative observation, and its payoff is never fed back into the belief, so its expected information gain under the model’s observation channel is zero and $\rho_{\text{EFE}} = 0$ follows from the definition of $I_a(b)$ rather than from a separate stipulation. In RockSample the implemented agents mark a sampled rock as sampled, which removes its quality from every later decision because re-sampling earns nothing, and they do not infer the quality from the sampling payoff (`rho_aif/agents/rocksample_agents.py`). A model in which the payoff of an exploitation action is itself observed and conditioned on would give that action a positive information gain, and the equivalence would then have to include an epistemic term for it. We do not treat that case. Whatever a non-terminal exploitation action does to the visible state is handled at the next decision point, where replanning starts from the resulting belief and recomputes ρ_{EFE} fresh. The coupling term Δ_T only reenters if an action nominally classified as observation or navigation in fact changes s_{hid} , which Definition 1 rules out by construction rather than by approximation.

The extension therefore covers RockSample’s check and navigation actions. Beyond that benchmark, it covers mobile sensor placement and sequential testing with spatial access costs, domains where the agent must navigate to observation locations before gathering information. Section 7.4 exercises this extension empirically across four RockSample instances, and empirical agreement exercises the algebraic result but does not prove it.

The factored observation structure models settings where the measured quantity is static over the decision horizon (Table 32). The key requirement, that $T_{\text{hid}}(s'_{\text{hid}}|s_{\text{hid}}, a) = \delta(s'_{\text{hid}} = s_{\text{hid}})$ for observation and navigation actions, breaks when information-gathering itself alters the hidden state. In such settings the coupling term may be nonzero, $\Delta_T \neq 0$ at some beliefs, and the identity $\rho_{\text{EFE}}(b, a) =$

$I_a(b)$ behind the information-unit-weight equivalence (the EFE-as- ρ equivalence at $w=1$) then fails in general. We discuss this further in the limitations paragraph of Section 8. Example 1 makes precise what fails and how badly.

Remark 1 (The scope restriction is on the reduction, not on what EFE can price). Example 1 shows a failure of the factored ρ -POMDP reduction, not a failure of Expected Free Energy. For an action a that violates Definition 1’s state-preserving hypothesis, $\Delta_T(b, a)$ may be nonzero. For such an action, replacing the naive epistemic term $I_a(b) = \mathbb{E}_{o|a} D_{\text{KL}}[Q(s_{\text{hid}} | o, a) \| Q(s_{\text{hid}} | a)]$, computed from the pre-transition posterior b'_o , with the transition-aware term computed from the correct joint posterior $b'_{o,T}(s') \propto \sum_s T(s' | s, a) O(o | s, a) b(s)$, under this article’s observe-then-transition timing convention where the observation is emitted from the pre-transition hidden state and the dynamics then act, removes the illusory information credit. In Example 1 the post-transition state is degenerate, the transition-aware epistemic term vanishes, and the substitution recovers the reward-only Bellman value $V_{\text{true}}(a_D)$ exactly. In general it restores the correct belief dynamics while still crediting whatever genuine post-transition information the action carries, so it prices destructive sensing correctly rather than reducing to reward-only planning. Propositions 1–3 therefore delimit which ρ -POMDPs the factored reduction targets (those whose information-gathering and navigation actions preserve the hidden state), not which decision problems Expected Free Energy can correctly evaluate. EFE’s epistemic term is defined over post-transition beliefs by construction, so it prices destructive sensing correctly once the epistemic term is recomputed from transition-aware predictive and posterior distributions as above. A nonzero Δ_T witnesses failure of the state-preserving hypothesis, but a zero value alone does not establish state preservation. We have not shown this diagnostic to be an additive correction to the EFE value or a bound on decision error. We leave a transition-aware treatment of the budgeted machinery of Section 4, and any perturbation bound in terms of Δ_T , as future work (Section 8).

U.4 Agent Specifications and Weight Selection

Planning+IG is the critical baseline. It uses the same tree search as the EFE agent with an additive information-gain (IG) bonus at the same horizon, meaning that at every node of the tree the value of an observation action is its reward-only value plus w times the expected information gain $I_a(b)$ of that observation. Planning+IG represents the explicit-information-reward approach associated with POMDPs with information rewards (POMDP-IR) and ρ -POMDPs [2, 59]. Satsangi et al. [51] establish equivalence between the two formulations for piecewise-linear convex belief rewards. We do not invoke that equivalence for Planning+IG’s concave expected-IG reward. By Proposition 1, the EFE agent is exactly Planning+IG with $w=1$, so any difference between EFE and Planning+IG at a tuned weight, in either direction, is attributable to the weight choice rather than a different mechanism. That weight-only reading presumes the

Table 32. Taxonomy of real-world POMDPs by factored observation structure. Factored settings preserve the hidden state under observation and navigation actions, while non-factored settings do not.

Preserves hidden state	May change hidden state
Non-destructive testing	Destructive testing (drilling)
Medical imaging (CT, MRI)	Biopsy / tissue sampling
Structural inspection	Active interventions
Environmental monitoring	Predator-prey (target moves)
Security screening	Chemical testing (consumes sample)
Mineral exploration	Quantum measurement
Mobile sensor networks	Adversarial surveillance

EFE implementation computes the epistemic term it claims to, so the EFE computation itself is checked for consistency against `pymdp` [26], a reference Python library for active inference (Appendix H). This is a qualitative agreement check on the epistemic term, not a certified numerical equivalence.

The Epistemic-only agent is the ablation that removes the pragmatic term. It sets $\mathcal{G}(\text{commit}) = 0$, removing reward awareness from the decision of when to commit, so that the comparison between observing and committing sees only observation costs and information gain. Once that comparison favors committing, the agent takes the commit action for the state its belief rates most probable, the belief-argmax commit, which is how it can still succeed once its belief is sharp. Success, for an observe-then-commit environment, is the fraction of episodes in which the agent’s commit is correct. On Tiger, Diagnosis, Bandit, and Tileworld no observation’s information gain alone clears its cost, so the agent commits immediately from its prior and succeeds at the chance rate, the accuracy the prior alone affords (Section 7, per-agent numbers in Appendix B). Testbed is the exception, since its observation is cheap enough that information gain alone clears the cost, and there the ablation matches reward-only Planning exactly (Appendix C). The pragmatic term is what makes commit timing reward-aware, and the four chance-level environments show what its removal costs.

Four of the remaining baselines need a brief description here. Myopic ($\rho=0$, $H=1$) has no epistemic term at all, yet it observes on many environments. It does so because its one-step lookahead compares committing now against observing once and then committing, so it can see when a commit made after an observation is worth more than one made now. That difference is the instrumental value of information (VoI) of the observation, and Myopic observes for that reason rather than because of an epistemic bonus. The Greedy baseline provides a no-information lower bound in RockSample and Structural Inspection, both specified in Section 5. In RockSample it moves to rocks and samples

them without first checking whether they are good. In Structural Inspection it declares a diagnosis for each component without running any test, so its declarations rest on the prior alone. We also include an observe-then-commit baseline built on information-directed sampling (IDS) [48], which picks the observation action that minimizes its information ratio, the squared expected regret of taking that action divided by the information the action yields about the optimal commit (results in Appendix N). The fourth baseline warrants a naming note. The posterior-vote baseline draws 100 states from the belief, finds for each sampled state the action that would be optimal if that state were the true state, and takes the majority vote over those actions. As the sample count grows, this vote approaches the most-likely-state rule, which acts as if the belief’s most probable state were the true state. It is not Thompson sampling, and we do not claim it as such. Thompson sampling draws a single state and acts optimally for that one draw, so the frequency with which it takes each action matches the belief’s probability that the action is optimal, whereas the hundred-sample majority vote collapses toward a single action. The committed CSVs record it under the label `ThompsonSamplingAgent`.

Two of the six core agents carry a weight w that must be set. They are Info Gain and Planning+IG (Table 1). We report two tuning criteria for that weight, a success-maximizing criterion used in the main tables and a reward-maximizing criterion used in the Pareto analysis. In the main tables, the tuned weight for Info Gain and Planning+IG is the success-maximizing weight chosen by grid search over $\{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100\}$. It is written w_{succ}^* in the results tables and w^* where the criterion is clear from context. Each grid point is evaluated on 200 episodes, except on *Tileworld*, where 100 episodes are used because its computational cost per episode is an order of magnitude higher than that of the other observe-then-commit environments. Every grid point is evaluated on one common tuning stream, seed 7, disjoint from the five evaluation seeds and reset for each candidate so that candidates face the same episodes, and ties go to the smallest weight reaching the maximum (`tune_info_gain_weight` in `experiments/run_experiment.py`). Each tuned weight is therefore a selection on one stream of 200 or 100 episodes and carries that stream’s sampling noise. Seed 7 is also the first held-out seed of Section 6.8, which evaluates only the calibrated family and never a tuned baseline. This tuned baseline weight is a different quantity from the shadow price $w^*(B)$ of Section 4 and from the thresholds of Proposition 2, which share the star notation. The Pareto analysis (Section 7.2), which traces mean reward against success as w varies, sweeps a wider eleven-weight grid, listed in that section, and reports the reward-maximizing weight w_{ret}^* . The two criteria can select different weights, and both are reported so that a reader can see which one a comparison uses. Section 4 then offers a third way to fix the weight, reading it off a sensing budget as the crossing threshold $w^*(B)$, estimated by the bracket of adjacent grid weights between which the agent’s induced sensing usage crosses that budget.

U.5 Proof and Numerical Check of the Usage-Onset Corollary

The scale-equivariance and comparative-statics proofs appear with Propositions 4 and 5. We give the onset argument and its implementation check here.

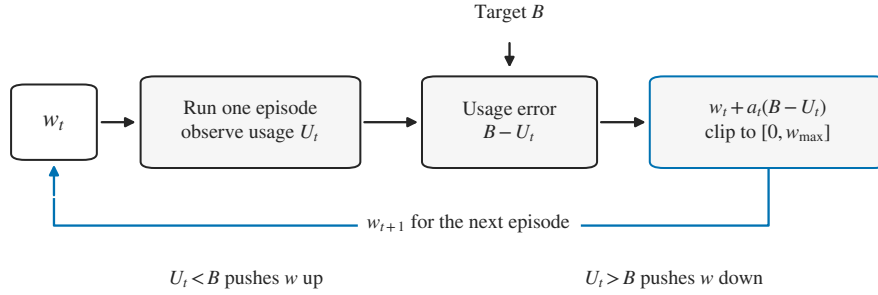
Proof. At $H=1$ the only two plans are commit now and observe once then commit, and Proposition 2 establishes the observe plan is preferred iff $w > w_{\text{thresh}}$. Consequently $U(w) = \mathbf{1}[w > w_{\text{thresh}}]$ exactly for this single observe-or-commit decision, so U is 0 below w_{thresh} and 1 above it. For any $B \in (0, 1)$ the supremum of the weights with $U(w) < B$ that precede a weight with $U(w) \geq B$ is therefore w_{thresh} , which is the stated crossing threshold, and a grid whose sampled usages straddle B returns a bracket whose closed enclosure contains w_{thresh} . The claim is about this $H=1$ decision, not about the receding-horizon usage curves of Section 6, which take many values and need not be monotone.

The corollary is verified numerically (`TestProp2OnsetExact` in `tests/test_budget.py`). The check runs the implemented $H=1$ planner over whole episodes rather than over the single decision the corollary analyzes. Within an episode that planner re-plans after each observation and is free to observe again before committing, so the test asserts the location of the onset rather than a usage of exactly 1. Usage is exactly 0 at $0.9 w_{\text{thresh}}$ and at least 1 at $1.2 w_{\text{thresh}}$ on a positive-threshold Testbed with observation accuracy $p=0.6$, observation cost $c=0.3$, and correct and incorrect commit rewards $R=\pm 1$, whose threshold is $w_{\text{thresh}} \approx 3.44$ in the implementation's bit units. Those bit units are the convention of the swept weight grid, and the calibration paragraph following Proposition 1 gives the conversion to the nat units of the proposition. Appendix W.2 extends the check to a refined weight grid on two such environments.

U.6 Proof and Implementation Scope of the Online Controller

Three versions of this controller appear in the article, and it matters which one each statement is about. The first is the projected decaying-step controller, the recursion (9) with a finite w_{max} and $\delta > 0$. It is the algorithm Proposition 6 covers, and it is the algorithm every reported run executes, with $w_{\text{max}}=100$, the top of the swept weight grid, and $\delta=0.02$ (`DualWeightAgent` in `rho_aif/agents/dual_descent.py` with `max_weight` set, wrapping `dual_update` in `rho_aif/budget.py`). No reported iterate exceeded 3.1, more than an order of magnitude below w_{max} , so the upper bound was never reached, although the projection remained in force. The second is the released library default, which sets no upper projection and uses a constant step ($\delta=0$). The proposition covers neither of those two departures, and no result in this article is reported for that default. The third is the reset-on-shift heuristic described below, which the proposition also does not cover.

The recursion originates with Robbins and Monro [47], with classical almost-sure extensions due to Blum [6]. Kushner and Clark [33] treats constrained stochastic approximation. We prove the stated bounded scalar case directly,



Stationary guarantee requires a single sign-consistent crossing and the stated assumptions.

Decaying step $a_t = \eta_0 / (1 + \delta t)$, $\delta > 0$. Reset-on-shift is a heuristic.

Fig. 23. One episode of the projected weight controller, shown schematically. The current weight produces an observed usage U_t . The controller adds a step proportional to $B - U_t$ and clips the result to $[0, w_{\max}]$. Underspending raises the weight and overspending lowers it, subject to clipping. This direction corrects usage under the sign-consistent crossing assumptions of Proposition 6. The proposition covers the projected decaying-step update in a stationary setting under its four stated conditions, not arbitrary usage curves or the reset-on-shift heuristic.

including discontinuous usage curves. Let $\mathcal{F}_t$ be the history before episode t , $e_t = w_t - w^*$, and $M(w) = U(w) - B$. By (ii), $\mathbb{E}[U_t | \mathcal{F}_t] = U(w_t)$ and $(B - U_t)^2$ is bounded by a constant C . Projection onto an interval containing w^* cannot increase distance from w^* , so

$$\mathbb{E}[e_{t+1}^2 | \mathcal{F}_t] \leq e_t^2 - 2a_t e_t M(w_t) + C a_t^2.$$

Condition (iii) gives $e_t M(w_t) \geq 0$, also when $w_t = w^*$ without requiring $M(w^*) = 0$. Thus $Y_t = e_t^2 + C \sum_{k \geq t} a_k^2$ is a nonnegative supermartingale and converges almost surely. Summing the expected drift inequality gives $2 \mathbb{E}[\sum_t a_t e_t M(w_t)] \leq \mathbb{E}[Y_0] < \infty$, so that nonnegative series is finite almost surely. Since the tail $\sum_{k \geq t} a_k^2$ vanishes, e_t^2 has an almost-sure limit. If it were positive, the bounded interval and condition (iv) would keep $e_t M(w_t)$ bounded away from zero eventually, contradicting $\sum_t a_t = \infty$. Hence $w_t \rightarrow w^*$ almost surely.

The running-average claim follows from that convergence. Because w^* is interior and the increments $a_t(B - U_t)$ vanish, usage being bounded by (ii), the projection is inactive from some episode on, so $\sum_t a_t(B - U_t)$ converges, its partial sums being differences of the convergent iterates. Since $1/a_t = (1 + \delta t)/\eta_0$ increases to infinity, Kronecker's lemma gives $a_n \sum_{t < n} (B - U_t) \rightarrow 0$, and because $n a_n \rightarrow \eta_0/\delta > 0$ under that schedule, the one property of it the running-average claim uses, this is $\frac{1}{n} \sum_{t < n} (B - U_t) \rightarrow 0$ almost surely. Under a constant step, which no reported run uses, the recursion need not converge to a point. The standard intuition is that the iterate instead stays in a step-size-dependent

neighborhood of w^* , which would let the controller re-adapt after a shift, but making that precise requires constant-step tracking arguments that we do not develop. Section 6.2 assesses re-adaptation under reward-scale shift empirically on the decaying schedule, together with the reset-on-shift variant described next.

The *reset-on-shift* variant of the controller watches the usage of the most recent episodes. When that usage persistently misses the budget after the step size has already decayed below half its initial value, it restarts the episode counter t in the step schedule of (9) at zero, so that the step size jumps back to η_0 and the controller can move quickly again. Resetting the decay clock mid-run in this way breaks the trajectory into a sequence of Robbins–Monro-style segments stitched together by a change-detection heuristic, rather than a single non-increasing step schedule. The proposition therefore does not cover the reset-on-shift variant when that variant is treated as a single run.

The proposition’s scope has one further limit. It is conditional on a single, sign-consistent crossing of B . It is silent when U crosses B more than once, which the discussion after Proposition 5 shows to be a real possibility, since the usage staircase need not be monotone.

U.7 Relationship to Constrained Planning and Experimental Design

Our weight w is calibrated within the Planning+IG family so that expected sensing usage meets a stated target, and it is not the Lagrange multiplier of the usage cap in problem (6) (Section 4.1). We do not claim novelty for the underlying idea of adjusting a scalar price until a resource quantity meets a target. That idea is classical in constrained MDPs [1], rational inattention [38, 56], and SAC temperature auto-tuning [25]. Sims [56] models agents with finite Shannon capacity, and Matějka and McKay [38] derive discrete-choice logits from costly information acquisition. Both take an information-processing constraint as the primitive. We take an operational sensing budget as the primitive and recover the Planning+IG weight that implements it. Kim et al. [30] extend constrained dynamic programming to partial observability directly, showing that optimal constrained POMDP policies can require randomization and giving point-based dynamic programming over belief and admissible-cost pairs. Their method is not a Lagrangian one. It enforces the cost bound inside the point-based pruning of dominated value vectors at each collected belief point, exactly with a minimax quadratically constrained program or approximately with a linear program, rather than pricing it. Their constraints are exogenous costs enforced offline. The usage target here is also stated on a non-epistemic quantity, the count of sensing actions or their cumulative cost, rather than on information gained. What differs is that the price w on belief-computed information gain is set offline by the grid solve of Section 4.4 or adjusted online by the controller of Section 4.5 until expected sensing usage meets the target.

The single-price receding-horizon family is the set of policies that run one scalar weight w throughout an episode. It includes the per-episode mixture of Definition 2, which randomizes between the two bracket endpoints from one episode

to the next. This family attains the usage budget in expectation. We do not claim it is reward-optimal among all budget-feasible policies. An exact solver in the style of Kim et al. [30] can randomize belief-dependently and may attain strictly higher constrained reward, particularly at gap budgets. We do not have such a solver, so Section 6.7 benchmarks against a near-optimal constrained-POMDP reference on Tiger, Diagnosis, and Bandit, the feasible envelope of a SARSOP usage-penalty sweep evaluated by Monte Carlo simulation (Appendix W.5). It is a feasible-policy estimate rather than a certified optimum, so up to its own sampling error and the upward selection bias of maximizing over noisy points it estimates the constrained optimum from below, and any measured gap to it understates the gap to the optimum. The estimated gap between the EFE agent at $w=1$ and this reference is zero on Tiger and small relative to the estimates’ standard errors on Diagnosis and Bandit.

No constrained reference is available for RockSample. Applying the same per-observation penalty λ over the 12-point log-spaced grid of Appendix W.5 to this project’s depth-3 tree search on RockSample[7,8] (`experiments/run_rocksample_cpomdp_reference.py`) gives a planner that checks zero or one rock at every λ , at reward 9.5 or 12.3, far from EFE’s 9.84 checks. A usage penalty credits a check only through what a 3-step lookahead sees beyond it, whereas the information weight credits each check’s information gain at the node where it occurs, so the penalty construction depends on SARSOP’s effectively unbounded horizon. A deeper RockSample solver, or a usage-cap construction that values information locally, would be needed.

Among the classical dualizations listed above, SAC is the one from which the update rule itself, as distinct from the budgeting idea, is adapted. Soft Actor-Critic’s applications paper [25] automatically tunes the entropy temperature by dual gradient descent on a target-entropy constraint, and our update (9) mirrors that construction with sensing usage in place of policy entropy.

Adaptive test selection is also the subject of sequential Bayesian experimental design (SBED), which chooses the next experiment to maximize expected information gain about a latent parameter, then updates and repeats [36]. Recent work amortizes this loop with a learned design policy so that near-optimal adaptive experiments run in real time [16]. SBED’s design objective is expected information gain about the latent alone. Ours is the scalarized $R + wI$ objective, task reward plus information gain at weight w , with w chosen inside the single-price receding-horizon family so that expected sensing usage equals a stated budget. Information gain is the quantity the weight prices, and sensing usage, in observation count or cumulative cost, is the quantity the budget controls. Reward at the calibrated weight is then a measured outcome, not a design target. The two objectives coincide only in the limit of a weight large enough that information gain dominates both sensing cost and task reward in the scalarized objective. In that limit the test chosen is the one expected information gain alone would select. Otherwise the budgeted objective declines an informative test that an SBED policy would take, because the decision variable here is how much to test, not only which test to run next.

Two contemporaneous 2024–2026 papers make an even closer structural comparison available. Kouw [31] derive expected free energy itself as the stationary point of a constrained Bethe-Lagrangian optimization, the Bethe free energy being a variational free energy defined on a factor graph. In that optimization a Karush–Kuhn–Tucker multiplier, the multiplier that constrained optimization attaches to a constraint, sits on an information constraint. Standard EFE is recovered at one specific value of that multiplier as the constraint moves through inactive (not binding), interior, and saturated (binding at its limit) regimes. This is a Lagrangian reading of EFE’s information term, and it is the Bethe-side counterpart of this paper’s own information-floor reading of w in Section 4.1. The two floors bound different quantities. Kouw’s floor is a constraint internal to the Bethe free energy, inside the constrained optimization whose stationary point is EFE, whereas the floor that w prices here is on $I(\pi)$, the cumulative expected information gain over the trajectories a policy generates, a constraint on the planner’s choice of policy. Neither floor is the usage cap in (6), and Kouw’s treatment does not address scale equivariance, crossing brackets and endpoint mixtures at usage gaps, or online tracking under budget or reward-scale shift. Stocco et al. [60] apply Lagrangian-guided Monte Carlo tree search with recursive, history-dependent dual ascent inside constrained POMDP planning itself, the same dual-ascent mechanism as our online controller (Proposition 6), but tracking per-node multipliers within a single tree search rather than a single scalar w across episodes toward a stated sensing budget.

The budgeted formulation carries one cost. Estimating the usage curve $U(w)$ well enough to solve $U(w)=B$ offline requires sweeping w over a grid and running episodes at each point, which is at least as much sampling as the direct per-task weight tuning this paper criticizes. The budgeted formulation does not save that sampling. What it buys instead is a design primitive a practitioner can state (a sensing budget in observation count or cost) rather than a weight they must guess, and an online controller (Proposition 6) that skips the offline curve and, under that proposition’s conditions, converges from interaction alone to the budget crossing point. Scale equivariance (Proposition 4) lets a count-usage curve computed at one reward scale transfer across pure reward-and-cost rescalings without recomputation, and a cost-usage curve transfer after rescaling by the same factor. That is a property of the planner and its weight, not of how the weight was chosen, and it carries a directly tuned weight w to αw under the same rescaling and tie-break assumptions, on a weight grid rescaled with the rewards (Section 3.3). The contribution is thus not generic Lagrangian duality, which is prior art in all of these literatures, but its concrete operationalization for budgeted ρ -POMDP sensing.

V Extended Experimental Specifications and Protocols

This appendix gives the full environment definitions and evaluation protocols summarized in Section 5. Tiger, Diagnosis, Bandit, and Tileworld follow the observe-then-commit structure of Section 3, in which a variable-length run of

observation actions ends in a single terminal commit action and the hidden state never changes. RockSample and Structural Inspection are factored observation POMDPs in the sense of Proposition 3, whose observation, navigation, and exploitation actions interleave within an episode, which is why the paper also calls them interleaved observe-act settings.

We call Tiger, Diagnosis, and Bandit the core environments. They run at the full protocol of 1,000 episodes per seed across 5 random seeds ($\{42, 123, 456, 789, 1024\}$) for a total of 5,000 episodes. They anchor the main results table, and they are the three environments on which the near-optimal SARSOP reference of Section 6.6 is computed. Tileworld and Structural Inspection use 500 episodes per seed, and each RockSample instance uses the protocol stated in its table caption. The same agent on the same environment is also evaluated in other batteries, among them the Pareto sweep, the SARSOP and CPOMDP studies, and the shadow-price atlas, each under the episode count and seeding rule its own section states, so its usage and reward differ slightly from one battery to the next. Tiger EFE, for example, averages 4.20 observations in Table 5 and 4.32 in Table 2. Where a comparison does cross batteries, as the committed-reference column of Table 34 does, the text says so.

The rest of the protocol concerns how results are aggregated and tested. Each reported reward, success rate, and observation count is the mean over all episodes of all seeds. Pairwise statistical comparisons are seed-level Welch t -tests, which aggregate each seed’s episodes to one mean and then test across the 5 seeds, with Holm–Bonferroni correction for multiple comparisons (Appendix I). Bootstrap confidence intervals for the core-environment means, computed under the protocol given in the same appendix, are in the file `results_bootstrap_ci.csv` of the code repository. Comparison against POMCP is in Appendix S. That appendix includes a scaling analysis over POMCP’s simulation budget, meaning the number of simulated rollouts it runs per decision, a quantity distinct from the sensing budget B of Section 4, with wall-clock timing at 500–5,000 simulations per decision. Full specifications are in Appendix E, and all environments are implemented as Gymnasium [64] environments, Gymnasium being a widely used Python interface for reinforcement-learning environments.

Tiger [29]. A tiger waits behind one of two doors, so the hidden state takes two values. The single observation action, listen, reports the tiger’s door correctly with probability 0.85. The two commit actions open one door or the other and end the episode. A correct commit pays +10, an incorrect one −100, and each listen costs −1. Because there is only one sensor, Tiger poses no choice of what to observe, only whether to observe again.

Sequential diagnosis (Diagnosis in tables and figures). Exactly one of $N=4$ hidden conditions is present, and the agent’s task is to name it. There are $K=2$ binary tests (accuracy 0.80). Each test splits the four conditions into two groups along its own partition and reports, with that accuracy, which group contains the true condition, so the two tests together can identify any condition. There are N diagnose actions, one per condition, each of which commits and ends the

episode. A correct diagnosis pays +10, an incorrect one −50, and each test costs −1. Unlike Tiger, the agent must choose which test to run.

Structured bandit (Bandit in tables and figures). One of $K=4$ arms is the best, and which one it is forms the hidden state. The bandit is structured in the sense that this single hidden variable, rather than an independent unknown payoff per arm, ties the arms together, so evidence about one arm is evidence about the others. Each of the K inspect actions (accuracy 0.80, cost −0.5) targets one arm and returns a noisy yes/no signal about whether that arm is the best. Each of the K pull actions commits to an arm and ends the episode. Pulling the best arm pays +10, and pulling any other arm pays +1. As in Diagnosis, the agent must choose which arm to inspect.

Tileworld. Tileworld is a spatial generalization of Diagnosis. An $N \times N$ grid ($N=6$, $|S|=36$) hides a single target tile, and the hidden state is the cell that tile occupies. Each scan action asks a yes/no question about one binary digit of the target’s row index or of its column index, and that question splits the grid in two, into the cells whose digit is 0 and the cells whose digit is 1. Three binary digits suffice to write any of the six row indices, and likewise for the six column indices. Three scans for the row digits and three for the column digits therefore give the $K=6$ scan actions, which together pin down the cell. Each scan alone tells the agent only which side of its own split the target lies on, and each returns its answer as a noisy binary signal (accuracy 0.80, cost −1). Each of the N^2 commit actions declares one cell as the target’s location and ends the episode (correct +10, incorrect −50). Appendix E and Figure 17 label this action collect, following the implementation. Because the belief lives on the grid itself, its evolution over an episode can be drawn directly, as shown later in Figure 25.

RockSample [58]. RockSample is the first environment in the suite where sensing is interleaved with movement. An $N \times N$ grid holds K rocks at known positions, each with a hidden binary quality (good or bad), and the agent’s own position is fully observable. Move actions change the agent’s position. Check actions produce noisy observations of a chosen rock’s quality, with an accuracy that decays with the agent’s distance from that rock. Sampling collects the rock at the current position (good +10, bad −10). Exiting ends the episode from wherever the agent stands and gives +10. All actions cost −0.5. We evaluate four instances, written $RS[N,K]$ for an $N \times N$ grid with K rocks, namely $RS[5,3]$, $RS[7,4]$, $RS[7,8]$, and $RS[11,11]$. Unlike the above environments, RockSample has interleaved observe-act dynamics with state transitions, the setting addressed by Proposition 3.

This is a cost-modified variant of Smith and Simmons [58]’s benchmark, not the original, and it differs in six respects that a reader comparing against published RockSample numbers must know. First, every action including checks carries the −0.5 cost. The original charges for neither moves nor checks. It pays +10 for sampling a good rock, −10 for sampling a bad one, and +10 for reaching the exit, with all other moves free, so the only thing penalizing a long path there

is the discount factor. Second, rock positions are fixed per instance, as in the original, but are our own choice rather than the specific maps used in the published instances. Third, returns are reported undiscounted, with episodes truncating at the N^2+10K step cap stated in Appendix E, whereas the original relies on its discount factor instead.

The next two differences concern the dynamics. Fourth, the exit action grants its reward from any cell rather than only at the east edge. This makes exit an unconditional outside option, that is, a fallback of known value the agent can take at any moment from wherever it stands. The POMCP analysis in Appendix M turns on this fact. POMCP’s search tree values each action by the mean of the returns of the simulations that pass through it (a mean backup). A guaranteed exit payoff available at every cell is the fixed alternative against which every one of those averaged estimates is measured. Fifth, the hidden rock-quality vector stays fixed when a rock is sampled. In the original, sampling a good rock turns it bad. Here, which rocks have already been sampled is tracked outside the hidden state, and the agent can recover that list from its own action history and its fully observable position. A re-sample is therefore recognized without any change to the hidden state. Re-sampling a collected rock therefore earns zero sampling reward rather than the bad-rock penalty, although the action still pays the -0.5 cost that every action pays. This fixity is what Proposition 3’s application to RockSample relies on.

Sixth, the check sensor differs in both its close-range accuracy and its tail. Efficiency is the original paper’s name for the sensor’s reliability as a function of range. The original sensor is exact at range zero, blends toward chance as its efficiency decays with Euclidean distance, and never reaches chance at any finite range. The accuracy of our check sensor starts at 0.95, and it is the accuracy value itself that halves with every 4 cells of distance and is then floored at 0.5 (`rho_aif/environments/rocksample.py`). The decay meets that floor at a range of 3.7 cells, a little short of one full halving distance because the accuracy starts at 0.95 rather than at 1. Since a binary observation with accuracy 0.5 is a coin flip, our sensor is exactly uninformative beyond 3.7 cells rather than asymptotically so.

The undiscounted, check-charging reward model is what makes sensing usage directly interpretable as a budget in Section 4, which is why we adopt it. But it means the absolute returns reported here are not comparable to RockSample values in the literature. We use $|S|$ throughout for the number of hidden-state configurations (2^K rock-quality vectors), excluding the fully observable agent position, so our $|S|=2,048$ for RS[11,11] is not the convention under which the literature quotes larger state counts for the same instance.

Structural inspection. In Structural Inspection, N components sit at known spatial locations on a grid, and each has a hidden binary state (nominal/faulty, prior $p_{\text{fault}}=0.3$) that the agent is tasked with declaring. There are $K=2$ non-destructive test types, visual (accuracy 0.70, cost -0.5) and detailed (accuracy 0.90, cost -2). The agent navigates between components, runs tests, and declares a diagnosis for one component at a time, in whatever order its own navigation

takes it, so a completed episode carries one declaration per component rather than a single terminal commit. An episode ends once every component has been declared, or earlier at a step cap. A correct nominal declaration pays +2, a correct fault declaration pays +5, a missed fault costs −50, a false alarm costs −5, and each move costs −0.5. Neither tests nor moves change fault states, so this is a factored observation POMDP. The environment maps directly to industrial inspection, medical screening, and fault detection domains. We evaluate $N=8$ ($|S|=256$) and $N=16$ ($|S|=65,536$).

W Extended Budget Evidence

The following studies retain the detailed protocols, auxiliary comparisons, and robustness analyses behind Section 6. The main text carries the primary calibration results and their principal limitations.

W.1 Curve Collapse across Reward Scales

This subsection gives the protocol and breadth table behind Section 6.1’s test of the first corollary of Proposition 4, that usage plotted against w/α is one and the same curve at every reward scale α . The weight grid at scale $\alpha \in \{0.1, 1, 10\}$ is the base grid multiplied by α , so each grid point at one scale has a partner with the same w/α at the other two and the curves can be compared point by point. Table 33 extends the Diagnosis and Bandit check to Tiger and Tileworld-6×6.

Every environment shows bit-exact collapse, and every crossing bracket coincides across scales down to floating-point noise ($\leq 10^{-15}$). This is the strongest form the check can take, and it is available only because environment streams are seeded per episode. Tiger passes trivially because instrumental listening already saturates usage across the tested range, so the budget formulation has no usable dial in this window. Proposition 4 guarantees collapse of whatever curve there is, not that every environment has an informative curve to collapse.

W.2 Proposition 2 Onset on Positive-Threshold Two-State Instances

This subsection checks that the closed-form onset threshold w_{thresh} of Proposition 2, the weight above which observing first switches on, is where the measured usage staircase takes its first upward step, as Corollary 1 predicts. Standard Tiger and Testbed parameters yield negative w_{thresh} , so any $w \geq 0$ observes and the check is vacuous there. We therefore construct two positive-threshold instances of the two-state Testbed of Appendix C (the `InfoSeekingEnv` class of the implementation, standard parameters in Appendix E), which differ from the standard Testbed only in a costlier, less accurate observation ($p \in \{0.60, 0.58\}$, $c=0.3$, $R_{\pm} = \pm 1$, where $R_{\pm}$ are the correct-commit reward and the incorrect-commit penalty). Here α is the proposition’s reward asymmetry ratio $|R^-|/R^+$,

Table 33. Curve-collapse breadth. Fraction of matched w/α points across $\alpha \in \{0.1, 1, 10\}$ within $2 \cdot \text{SE}$, and whether the B -crossing bracket coincides across scales (in w/α units).

Environment	Within $2 \cdot \text{SE}$	Max spread	Bracket at B
Diagnosis ($B=8$)	100%	0.00 obs. (bit-exact)	$(0.141, 0.323]$ at every α
Bandit ($B=8$)	100%	0.00 obs. (bit-exact)	
Tiger ($B=4$)	100% (trivial)	0.00 obs. (bit-exact)	unbracketed (B below observed U at every α)
Tileworld-6 $\times$ 6 ($B=15$)	100%	0.00 obs. (bit-exact)	$(5.80, 20.0]$ at every α

not the reward scale that Section 6.1 writes with the same letter. These symmetric instances have $\alpha=1$, the boundary of the proposition’s strict asymmetry hypothesis, which the threshold formula itself does not require. Their closed-form thresholds are $w_{\text{thresh}} \approx 3.44$ and 7.55 in the implementation’s bit units, the convention the swept weight grid uses, equivalently 4.97 and 10.89 reward units per nat. A refined weight grid around the threshold shows $U=0$ at and below w_{thresh} , with onset in $(w_{\text{thresh}}, 1.03 \cdot w_{\text{thresh}}]$ on both environments (Figure 24), matching Corollary 1. The measured onset never falls below the closed-form threshold, and the bracket is one refined grid step (3%) wide. Negative-threshold sanity checks on standard Testbed and Tiger remain consistent ($U(0) > 0$ when $w_{\text{thresh}} < 0$).

W.3 Multi-Seed Online Dual Control under Reward Rescale

Section 6.2 summarizes this study. Proposition 4 carries an offline count-budget bracket across a known reward-and-cost rescaling, but not across an unannounced one, so a deployed controller must then re-adapt online. The setting is Diagnosis with target $B=8$, learning rate $\eta_0=0.05$, decay $\delta=0.02$, and a $\times 10$ reward-and-cost rescale at episode 200 of a 400-episode run, with 10 independent controller seeds for each of two variants. The decay-only variant runs the update of Equation 9 with its step decaying from the first episode and never intervenes. The reset-on-shift variant adds the change-detection heuristic of Section 4.5. It watches a rolling window of per-episode usage (window 20), and once the step has decayed below half its initial value it declares a shift when the window mean departs from B by more than the larger of k standard errors of the window mean ($k=3$) and an absolute floor of 0.5 observations. A declared shift restarts the step-size clock of Section 4.5, so the step is large again.

Re-adaptation time is measured on a rolling mean of usage that, at each post-rescale episode, averages per-episode usage over the previous 20 episodes and is computed from post-rescale episodes only. Re-adaptation time is the number of episodes after the rescale until that rolling mean stays within ± 1 of B for 20

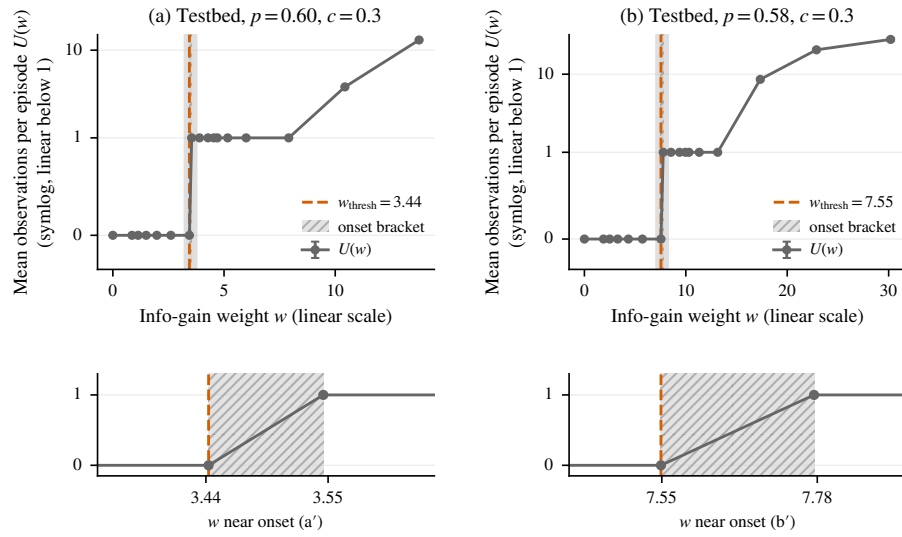


Fig. 24. Onset of observing on positive-threshold testbeds. The vermilion dashed vertical line marks the $H=1$ closed-form threshold (Proposition 2) and the hatched gray band is the empirical onset bracket after grid refinement, magnified in the zoom row beneath each main panel, with the shaded strip on the main panel marking exactly the region the zoom row below it shows.

consecutive episodes. Restricting the window to post-rescale episodes matters. A window allowed to straddle the rescale point reports a spuriously fast recovery by mixing in already-converged pre-rescale usage (`experiments/run_price_of_information.py:readaptation_episodes`).

Resetting the step-size clock re-adapts roughly $3\times$ faster than decay alone. The comparison of record uses one estimand for every seed, the restricted recovery time $\min(T, 180)$, where T is the post-rescale-only recovery time and 180 is the last index at which the 20-episode hold can still be checked inside the 200 post-rescale episodes, with a seed that never completes the hold scored at 180. On that estimand decay-only averages 157.3 episodes and reset-on-shift 51.2, a ratio of 3.07. Because the two variants share their ten controller seeds, the paired difference is 106.1 episodes with a Student- t 95% interval of [96.8, 115.4], and every one of the ten paired differences is positive, the smallest being 89 (`results_price_dual_multiseed_metrics.csv`, column `readapt_restricted`, and the summary JSON’s `dual_ms_readapt_restricted` block). For descriptive context only, decay-only recovery averages 154.8 episodes, 95% CI [142.7, 166.9], over the 9/10 seeds that recover inside the window. Two of those nine recover only at episode 180, and the tenth seed’s rolling mean lies within tolerance over the final episodes without having held there for 20 consecutive episodes, so the decay-only sample is right-censored at 180 and its restricted mean is a lower bound on recovery time for these ten runs, not a population inference. Reset-on-shift recovers on 10/10 seeds, each triggering exactly one step-size reset, so its restricted and conditional means coincide (51.2, CI [46.1, 56.3]). Post-rescale steady-state $|U - B|$, the absolute gap between the budget and mean usage over a fixed block of episodes at the end of the run, is 0.06 ([0.024, 0.096]) under reset versus 0.39 ([0.118, 0.666]) under decay-only. Pre-rescale steady-state error over the block immediately before the rescale is 0.10 ([0.051, 0.149]) for both variants, which diverge only after the rescale, well inside the ± 1 recovery tolerance. Every CI here other than the paired interval is a normal-approximation CI over the controller seeds that enter it, and the separate intervals of the two conditional means are not an interval for their ratio. The battery runs under per-episode seeding, which covers the environment constructed at the rescale. Figure 6 shows the median trajectory with interquartile band for both variants.

Per seed, reset-on-shift reaches a tighter post-rescale steady state than decay-only on five of the ten seeds, ties on four, and is looser on one. The mid-run rescale makes the environment nonstationary and the reset-on-shift variant restarts its schedule, so neither variant is covered by Proposition 6’s stationary guarantee, and what is measured here is nonstationary re-adaptation. The mechanism is the residual step size that the discussion after Proposition 6 identifies. At the rescale the decayed step is still large enough to move the controlled weight w_t , which is why decay-only recovers at all, but only slowly. Reset-on-shift restarts the schedule as soon as the detector declares the shift, and since the reset is the only thing that distinguishes the two variants, that restored step is where its speed advantage comes from.

W.4 SARSOP as a Near-Optimal Offline Reference

This subsection gives the mechanics and archives behind Section 6.6, which compares the canonical $w=1$ agent’s reward and sensing usage with SARSOP [32] on the three core observe-then-commit environments. The benchmarks are exported to the Cassandra `.pomdp` format, the plain-text model file that offline POMDP solvers read, through one export path that reads the hidden-state, action-layout, and observation-model definitions the agents already use (Appendix T.1). The solver stops once the gap between its upper and lower bounds on the value of the initial belief falls below 10^{-3} , in under one second per environment at $\gamma=0.999$.

What the solver returns is an alpha-vector policy, a finite set of vectors over states, each of which defines a linear function of the belief. The value of a belief is the largest of these linear functions evaluated at that belief, and the vector that attains the maximum names the action to take. We evaluate that policy in-simulator through the same episode runner used for the other observe-then-commit agents, so the agents compared in Table 2 share mechanics and metrics exactly. Table 2 reports the full protocol (5 seeds $\times$ 500 episodes).

On Tiger, SARSOP and EFE select identical actions on every episode. Tiger offers only one observation action, so both policies face the same listen-or-commit choice at every belief. On that choice the near-optimal policy and the EFE policy coincide exactly. On Diagnosis and Bandit the two show no statistically significant reward difference at our sample sizes, with the Diagnosis means differing by about 0.6 SE, and their usage differs by less than 0.1 observations (Table 2).

A non-significant difference is not evidence of equivalence, since a small sample can fail to reject any difference at all. We therefore formalize the comparison with a fixed-margin two one-sided tests (TOST) procedure [53], which tests two null hypotheses, that the true reward difference is at least the margin in the positive direction and that it is at least the margin in the negative direction, so that rejecting both places the true difference inside the margin. The margin was set equal to the cost of one sensing action in each environment, which is ± 1.0 reward for Tiger and Diagnosis and ± 0.5 for Bandit, a value fixed by each environment’s own cost parameter. The test itself was added after an earlier comparison of the same two policies on the same canonical seeds, one without any equivalence test, had been inspected, so the margin was chosen with that comparison known and the five-seed test is retrospective rather than prospectively predeclared. The fifteen seeds that the $n=20$ robustness study below adds to the canonical five were fixed before that study was run, after the margin, and are the prospective part of this evidence. It is an operational margin for this domain, not a universal equivalence standard.

Two choices in the test mechanics need stating, the unit of analysis and the decision not to pair. Each of the two one-sided tests is an unpaired Welch t -test on per-seed means ($n=5$ seeds, 500 episodes per seed) rather than on pooled episode-level values, so degrees of freedom reflect independent seeds. We do not pair by seed index even though EFE and SARSOP share a nominal seed list. The reason is not that shared seeds leave the trajectories uncorrelated, which does not follow in general, since $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2 \text{Cov}(X, Y)$ and a shared seed

can induce positive covariance. On Tiger the two policies take identical actions on every episode, so their trajectories are perfectly correlated and a paired test would report a zero difference with zero variance. On Diagnosis and Bandit the two policies diverge on many episodes and consume the environment’s random draws in different orders, so the covariance is partial and unknown. Treating the two samples as independent is the conservative choice when the covariance is positive, since it can only overstate the variance of the difference and so make the equivalence test harder to pass, and it is what the reported p -values assume. Pairing would be the sharper design where the randomness is genuinely shared, and a paired analysis of the per-seed differences is reported below as a sensitivity check, next to the $n=20$ robustness study.

All three environments reject both one-sided nulls at $\alpha=0.05$, where the reported TOST p -value, written p_{TOST} below, is the larger of the two one-sided p -values (Tiger $p=0.0010$, Diagnosis $p=0.0458$, Bandit $p=0.0130$). The same fact viewed as an interval is that the 90% CI of each mean difference lies entirely inside the margin (Tiger $[-0.41, +0.41]$, Diagnosis $[-0.51, +0.98]$, Bandit $[-0.35, +0.31]$). The interval is 90% because two one-sided tests at level α together correspond to a two-sided interval at confidence $1 - 2\alpha$. The three tests survive Holm–Bonferroni correction as a family. EFE at the canonical weight is therefore statistically equivalent to the SARSOP reference within the fixed margin on this suite (`results_tost_sarsop.csv`).

The five-seed sample is small. We reran the identical procedure at $n=20$ seeds, fixed before running rather than added post hoc, reusing the same SARSOP policies (`results_tost_sarsop_n20_robustness.csv`, per-seed means in `results_tost_sarsop_n20_robustness_per_seed.csv`). All three environments remain equivalent, and p_{TOST} tightens by roughly six, five, and eight orders of magnitude (Tiger 4.5×10^{-10} , Diagnosis 6.8×10^{-7} , Bandit 6.0×10^{-11}). The intervals narrow accordingly, with Tiger’s 90% CI shrinking from $[-0.41, +0.41]$ to $[-0.21, +0.21]$. The paired sensitivity check promised above runs the same two one-sided tests on the five per-seed differences (`results_tost_sarsop.csv`, columns `p_tost_paired` and companions, per-seed means in `results_tost_sarsop_per_seed.csv`). On Tiger the differences are identically zero, so the paired test is degenerate and equivalence holds trivially. On Diagnosis the paired p_{TOST} is 0.016 against the unpaired 0.046, and on Bandit it is 3.6×10^{-5} against 0.013, with the paired standard errors of the difference (0.24 and 0.03) well below the unpaired ones (0.38 and 0.18). The shared-seed covariance is therefore positive on both environments, so the unpaired analysis reported above (`results_tost_sarsop.csv`) is the conservative one and the equivalence conclusion does not depend on the choice.

Seed-paired differences are still not episode-matched, because the two policies consume each seed’s random stream differently. A predeclared episode-matched check removes that limitation (`run_tost_sarsop.py -episode-seeding`, `results_tost_sarsop_episode_paired.csv` and its per-seed file). It resets episode e of seed s at $10^4 s + e$, the budget-frontier convention, so both policies face the same hidden states and the same observation draws for as long as their

actions agree, and it uses the same twenty seeds, 500 episodes per seed, and margins as the $n=20$ study. Equivalence holds on all three environments. Tiger’s policies act identically. Diagnosis gives a paired difference of +0.11 with SE 0.09 and paired $p_{\text{TOST}} = 3.2 \times 10^{-9}$, and Bandit gives +0.04 with SE 0.03 and $p_{\text{TOST}} = 4.6 \times 10^{-12}$. The unpaired versions on the same runs give 3.8×10^{-5} and 1.8×10^{-8} .

We do not report a usage-matched comparison with SARSOP, because on Tiger it is ill-posed. SARSOP’s usage of 4.32 sits just below the Planning+IG usage curve’s floor of 4.37 (within one seed-level SE, Table 30), so the offline usage curve has no crossing at that target, and any higher target falls in its single wide step $w \in (19.3, 43.9]$. The usage-matched row in `results_sarsop_baseline.json`, labeled Planning+IG ($w=0$), is the resulting low-end grid fallback.

W.5 A Near-Optimal Constrained-POMDP Reference

Section 6.6 compared the family against a reference free to spend whatever sensing it likes. The budgeted framing of Section 4 calls for a second reference that is near-optimal subject to a charge on sensing. A constrained POMDP is a POMDP solved for the best reward attainable under a bound on an expected cost, here sensing usage. For the three environments of Section 6.6, we extend the same SARSOP export with a literal per-unit usage penalty, which adds $\lambda \geq 0$ to each observation action’s own cost cost_j , so its reward becomes $-(\text{cost}_j + \lambda)$. Solving to SARSOP’s precision (10^{-3}) then gives, for that λ , a near-optimal solution to $\max_{\pi} \mathbb{E}_{\pi}[R] - \lambda \mathbb{E}_{\pi}[U]$ over the full space of POMDP policies (not the additive Planning+IG family). This penalized problem is the literal Lagrangian of the budgeted problem (6) (Section 4.1), the relaxation that penalizes usage rather than rewarding information. An exact maximizer at rate λ cannot be beaten in reward by any policy that spends no more sensing than it does, by the mirror image of the two-line argument Section 4.1 gives for the information floor, so it is optimal under a cap set at its own usage. Sweeping λ therefore collects constrained references across a range of budgets. The reference is near-optimal rather than certified exact, because SARSOP is a point-based approximate solver and the reported reward and usage are Monte Carlo estimates from simulated episodes.

Two definitions clarify what a penalty sweep can and cannot certify. The achievable region is the set of pairs of expected usage and expected reward that some POMDP policy attains, and its frontier is the curve giving the best reward attainable at each usage level. Strong Lagrangian duality holds for a finite discounted constrained MDP [1], but the belief MDP here (the Markov decision process over beliefs that the POMDP is equivalent to) has a continuous state space, so we do not assume a zero duality gap carries over. Under strong duality every budget’s constrained optimum would be attained by the penalized problem at some λ , so a fine enough sweep would recover the whole frontier. Under a duality gap some budgets have a constrained optimum that no λ picks out, and the sweep misses those points. Either way, the swept points are supported points

of the achievable region, points that a linear trade-off between reward and usage picks out, each attained by an actually solved policy. They therefore estimate the frontier from below, each up to the sampling error of its own reward and usage rather than as a certified population bound, so any duality gap makes the reference conservative about frontier achievability. For an optimality comparison the effect is the opposite. A reference that undershoots the constrained optimum makes our agents look closer to optimal than they are, so the measured gap to the reference is a lower bound on the true gap.

With the caveat stated, the protocol is a grid sweep over the penalty. Sweeping λ over a log-spaced 12-point grid ($\lambda \in [0, 100]$, $\gamma=0.999$, 5 seeds $\times$ 300 episodes per λ) traces an estimated upper concave envelope of the achievable ($\mathbb{E}[U], \mathbb{E}[R]$) region, the frontier of the previous paragraph once mixtures of policies are allowed, sampled at these 12 supported points rather than continuously. A per-episode mixture of two adjacent swept policies approximates any intermediate budget, the same mixture argument used for the Planning+IG shadow price (Definition 2), though the approximation inherits whatever slack the constituent near-optimal solutions carry. $\lambda=0$ recovers the ordinary reward-only SARSOP solve of Table 2 as a sanity check.

The usage-matched Planning+IG grid weights evaluated in Table 3 are 0.00, 0.45, and 1.07 for Tiger, Diagnosis, and Bandit. They are nearest grid points selected on a lighter usage curve rather than crossing brackets. Each lies on a run of consecutive sampled weights with constant usage that also straddles $w=1$, namely $[0, 19.3]$, $[0.316, 19.3]$, and $[0.72, 1.64]$ respectively in `results_price_usage_curves.csv`. The corresponding rewards in `results_cpomdp_baseline.csv`, columns `R_PIG` and `R_EFE`, agree to the precision stored. This observed agreement is not a consequence of Proposition 1 at weights other than 1.

Table 3 reports the comparison. The reference at B_{EFE} is the estimated feasible envelope of the sampled reference policies, the best reward attainable by a per-episode mixture of sampled policies whose expected usage is at most B_{EFE} (`rho_aif/budget.py`, `feasible_envelope`, shared with Section 6.8). On all three environments the envelope selects a single sampled policy, the $\lambda=0$ policy on Tiger and Bandit and the $\lambda=0.05$ policy on Diagnosis. Its usage equals B_{EFE} on Tiger and lies below it on Bandit and Diagnosis, where B_{EFE} exceeds every sampled usage, so the usage cap does not bind. The estimated gap is zero on Tiger and -0.08% on Bandit, smaller in magnitude than the reward change between adjacent swept points, and -3.27% on Diagnosis, where the reference value is slightly below R_{EFE} . The population constrained optimum is at least the expected reward of any feasible policy, so a negative estimated gap can reflect sampling error on either side or approximation error in the reference rather than a violation of that bound. R_{ref} and R_{EFE} are 1,500-episode Monte Carlo estimates with standard errors (± 0.12 – 0.39) several times the gap, and the best feasible sampled Diagnosis policy uses 9.785 observations against EFE’s 9.824, so a finer λ grid near 0 and more episodes per point would sharpen the estimate.

The SARSOP comparison concerns reward when sensing is unconstrained, while this reference compares rewards at the sensing level EFE actually uses, where EFE’s measured reward, and that of the usage-matched Planning+IG weights, is empirically close to the estimated best (`results_cpomdp_baseline.csv`, `results_cpomdp_frontier.csv`, `experiments/run_cpomdp_baseline.py`). The reference does not extend to RockSample or Structural Inspection, where SARSOP is no longer a practical solver at the sizes we study (Section 4.6 and Appendix O), so an optimality-gap estimate there remains future work.

W.6 Calibration-Derived Target Budgets and What Calibration Buys

This subsection gives the full design and results behind Section 6.8. Its target budgets are set by a predeclared rule from each calibration curve’s range and largest jump, independent of the weight-one usage though not application-specified, and evaluated on seeds the calibration never saw. The study was predeclared in the producer’s docstring before any result was inspected (`experiments/run_budget_frontier.py`), and Table 34 reports it.

Section 6.8 states the calibration protocol and the five budgets per environment. The gap budget, at the midpoint of the curve’s largest jump, is one that no single tested weight attains, and the budget at half of $U_{\min}$ is unattainable and must be declared rather than approximated. The crossing bracket and its mixture probability q come from Definition 2, and the best feasible single member is the grid weight with the highest calibration reward among those whose calibration usage is at or below B . Two further policies were added after the first results had been inspected and are disclosed as such in the producer’s docstring. The best target mixture is the reward-maximizing mixture of calibration-grid weights whose calibration usage equals B exactly, and the best feasible mixture is the same program with calibration usage at most B . Each is a linear program over the calibration grid with at most two support weights, each is selected on the calibration data alone, and each is then run on the held-out seeds like the others. Neither says anything about weights or mixtures the grid does not sample. Each selected policy is then run on five held-out seeds, $\{7, 8, 9, 10, 11\}$, for 100 episodes each. The reference is the estimated feasible envelope at B of the constrained-POMDP reference of Section 6.7, a linear program over the committed Lagrangian frontier whose solution mixes at most two sampled points, and which is flat at the largest sampled reward once B exceeds the largest sampled reference usage. On Tiger every attainable budget lies above the reference’s largest sampled usage of 4.32, so the envelope is 5.02 at all four budgets. On Bandit the three interior budgets are slack, with envelope 6.31, and the gap budget mixes two reference points to 6.23. On Diagnosis the envelope is -2.00 , -1.35 , -1.27 , and -1.95 at the 0.25, 0.5, 0.75, and gap budgets. The reference is at once the constrained reference and the direct usage-penalty method for these domains, since it is built from a Lagrangian penalty on SARSOP. All three unattainable budgets were declared. Tiger’s gap budget coincides with its 0.5 budget, since

its curve has a single jump. Bandit’s calibration curve is not monotone, so the last-crossing convention of Definition 2 applies there. It changes the reported bracket at the 0.25 budget of 5.51, which the curve crosses twice, once between $w=0.06$ and $w=0.14$ and again between $w=1.64$ and $w=3.73$ after dipping to a usage of 5.11 on the plateau $w \in [0.72, 1.64]$, and the reported bracket is the later one, $(1.64, 3.73]$. At the gap budget of 4.71 the curve crosses only once, since the plateau’s usage stays above the budget, so the bracket $(0.06, 0.14]$ is the only candidate.

Section 6.8 reports the held-out usage results. By environment, the endpoint mixture’s usage errors are 0.03 to 0.08 on Tiger, 0.02 to 0.12 on Diagnosis, and 0.09 to 0.12 on Bandit.

Section 6.8 gives the reward comparison on Diagnosis’s three smaller budgets, on Tiger, and for the best target mixture at Bandit’s gap budget. The remaining rows complete it. On Bandit’s gap budget 4.71 the mixture earns 5.62 against the best feasible member’s 5.51, and at $B=7.79$ and 10.07 the best feasible member earns 6.16 against the mixture’s 5.70 and 4.95. Like Bandit’s two larger interior budgets, Diagnosis at $B=11.35$ and Bandit at $B=5.51$ lie above the usage that reward alone justifies, and there the mixture sits nominally below the best feasible member (-2.31 against -2.22 , and 6.06 against 6.16) by less than one held-out standard error. The best feasible mixture over the calibration grid coincides with the best feasible member wherever the usage constraint is slack, which is every Tiger budget, the three interior Bandit budgets, and Diagnosis at $B=11.35$, and with the selected endpoint mixture at the other three Diagnosis budgets, where the constraint binds. At Bandit’s gap budget, where it also binds, it coincides with the best target mixture. The best target mixture is the selected endpoint mixture itself on eleven of the twelve budgets, up to endpoints that sit on the same usage plateau and so run the same policy. The exception is Bandit’s gap budget, where it mixes $w=0$ and $w=0.72$. Against the reference envelope, the mixture’s reward sits 0.14, 0.76, 1.05, and 0.10 below it on Diagnosis at the 0.25, 0.5, 0.75, and gap budgets, which is 0.3, 1.1, 2.3, and 0.2 held-out standard errors of the mixture, and 0.26, 0.62, 1.36, and 0.61 below it on Bandit, about 4%, 10%, 22%, and 10% of the reference reward. On Tiger the mixture’s reward is 0.34 above the envelope at the smallest budget and 0.05 and 0.36 below it at the two larger ones. These comparisons are descriptive rather than tests, and they compare estimates from different evaluation streams, since the reference was evaluated on the canonical seeds with once-per-seed seeding and the family on the held-out seeds with per-episode seeding. The stream effect is not small. Re-evaluated on the held-out stream, Tiger’s unpenalized reference policy scores 5.68 rather than 5.02, the same reward as the best feasible member, which runs the reward-only policy. Section 6.8 describes the same-stream comparison, added after the results above had been inspected. It re-evaluates every sampled reference policy on the held-out seeds under the family’s per-episode seeding, after checking that each saved policy reproduces the committed reference to 10^{-9} on the canonical stream, and that section states the nominal pointwise plug-in qualification of its paired intervals (the Same-stream reference

and Paired gap columns of Table 34, `experiments/run_frontier_reference_heldout.py`, `results_budget_frontier_heldout_reference.csv`). The three intervals that include zero are Diagnosis’s 0.25 and gap budgets and Bandit’s 0.25 budget (-0.03 ± 0.28 , $+0.00 \pm 0.22$, and -0.21 ± 0.12). For the Tiger rare-event qualification stated there, every held-out Tiger per-seed reward is 10 minus the observation count (`results_budget_frontier.csv`, `results_cpomdp_frontier_heldout.csv`), and the canonical stream’s reference error rate of 0.6 percent (success 0.994, `results_cpomdp_frontier.csv`) is worth about 0.66 in expected reward, nearly all of the 0.67 difference between the reference’s canonical and held-out rewards. Tiger’s nominal excess at its smallest budget was therefore the stream effect, not a family reward above the reference. The same-stream envelope is still a maximum over noisy sampled points, so it removes the stream mismatch but not the envelope’s selection bias, and the reference remains near-optimal and estimated rather than exact.

A predeclared fresh-seed replication measures how far the same-stream paired intervals above depend on the five held-out seeds (`experiments/run_frontier_fresh_seed_replication.py`, `results_budget_frontier_fresh_seed_replication.csv`). It first replays all twelve held-out paired gaps to 10^{-9} , then keeps every committed mixture and reference support fixed and re-evaluates both on seeds 12 to 21, 100 episodes per seed, with intervals on 9 degrees of freedom. The count replicates. Intervals exclude zero at nine of the twelve rows, again eight of the eleven distinct budgets, and eleven of the twelve paired means are negative. Tiger’s three shortfalls and Bandit’s three shortfalls replicate, and Bandit’s smallest target again includes zero. The Tiger rare-event qualification above still applies. Diagnosis’s rows replicate only in part. Its 0.75 shortfall replicates at -0.82 ± 0.28 , but its 0.5 paired gap of -0.76 becomes $+0.02 \pm 0.29$, while its smallest target, which included zero on the held-out seeds, now excludes it at -0.56 ± 0.24 , and its gap target includes zero only narrowly, at -0.46 ± 0.20 . The 0.5 mixture plays $w=0.316$ with probability 0.94, on a calibration plateau from $w=0.316$ to 19.3 that contains $w=1$, against a reference that plays the unpenalized SARSOP policy with probability 0.91. Its held-out shortfall therefore sat uneasily with the $n=20$ equivalence of $w=1$ and that policy in Section 6.6. On those five held-out seeds the plateau policy trails the unpenalized policy by 1.01 ± 0.18 on matched episodes, whereas the twenty-seed episode-matched check of Appendix W.4 gives $+0.11 \pm 0.09$. Scaled from 500 to 100 episodes per seed, that check’s per-seed spread implies a five-seed SE near 0.40, which places the held-out value about 2.8 such SEs from the twenty-seed estimate, unusual but within the range of seed-set variation. A wrong diagnosis costs 60 relative to a correct one, so a few errors can move a five-seed mean, and the replication, in which the 0.25 row moves the opposite way, is consistent with seed-set variation as the source of the discrepancy. That one Diagnosis pair had been evaluated on these seeds before the replication was specified, so the seeds are not blind for it. The plug-in intervals therefore support the aggregate shortfall count, Bandit’s rows, and Tiger’s largest-budget row, not the identity of the Diagnosis budgets that fall short.

The cap reference is flat at the unpenalized reward wherever B exceeds that policy’s usage, which holds at seven of the eleven distinct budgets, so its shortfalls there include the cost of spending B . A predeclared target-matched reference removes that component (`experiments/run_frontier_target_reference.py`, `results_cpomdp_frontier_subsidy.csv`, `results_budget_frontier_target_reference.csv`). SARSOP is solved at eleven subsidized penalties $\lambda = -cs$, with c the observation cost and s from 0.1 to 0.98, so net observation reward stays negative. The larger subsidies raise held-out usage above the unpenalized level, to at most 7.17 on Tiger, 16.58 on Diagnosis, and 11.59 on Bandit, while the smaller ones leave it unchanged or slightly lower. Every sampled policy is evaluated on the held-out and fresh streams, after the committed policies reproduce their held-out rewards. At each budget the reference is the reward-maximizing mixture of sampled points whose held-out usage equals B , fitted on the held-out seeds and frozen for the fresh ones. At the four binding budgets it coincides with the cap reference. At the seven slack budgets its support includes a subsidized policy, and no interval excludes zero on either stream. The held-out target gaps there run from -0.16 ± 0.12 (Bandit, $B=5.51$) to $+0.08 \pm 0.04$ (Tiger, $B=4.72$), and Diagnosis’s largest target moves from -1.10 ± 0.22 against the cap reference to -0.05 ± 0.16 . On the held-out seeds, the Tiger family and reference-support policies earn 10 minus their usage, so the held-out target gaps equal the mixture’s usage shortfall below B and measure no reward efficiency. More generally, the mixture meets B only within its usage error, which a target gap also carries. A post hoc sensitivity analysis, run without new episodes, refits the reference at each mixture’s realized held-out usage instead (`-usage-matched`, `results_budget_frontier_target_reference_usage_matched.csv`). Every held-out Tiger mean gap becomes zero, Diagnosis still falls short at $B=9.53$ on the held-out seeds and at $B=7.72$ on the fresh ones, and Bandit’s gap-budget interval reaches $+0.001$ on the held-out seeds while still excluding zero on the fresh ones. In the predeclared comparison at B , the two held-out shortfalls are Diagnosis at $B=9.53$ and Bandit’s gap budget. On the fresh seeds they are Diagnosis at $B=7.72$ (-0.56 ± 0.24) and Bandit’s gap budget (-0.50 ± 0.11), so only Bandit’s recurs. The reference is still a maximum over a finite set of approximately solved, noisily evaluated policies, so it estimates rather than bounds the best policy meeting each target.

W.7 Distractor Robustness under Reward-Irrelevant Sensing

This subsection gives the full distractor study behind Section 6.9. A sensing budget fixes how much sensing is bought and leaves open what that sensing is about. `DistractorDiagnosisEnv` is an 8-joint-state Diagnosis variant whose hidden state pairs a condition (the variable the commit action must identify, as in Diagnosis) with an independent binary nuisance bit. Commit rewards depend only on the 4-way condition, and one added test, the distractor test, is informative only about the nuisance bit. Every observation model in the environment factors through exactly one of the two independent components, so the distractor test carries exactly zero information about the condition marginal (the agent’s belief

Table 34. The calibrated family at calibration-derived target budgets, set by a predeclared rule from each calibration curve and evaluated on five held-out seeds it was not calibrated on (100 episodes per seed, seed-level SE). Budgets are set from each calibration curve’s attainable range $[U_{\min}, U_{\max}]$ at three fractions of the range, at the midpoint of the largest jump (gap), and below $U_{\min}$ (declared unattainable, not run). The mixture plays the bracket’s upper endpoint with probability q and its lower endpoint otherwise, drawn afresh each episode. Usage is the observation count. See `results_budget_frontier.csv` and `results_budget_frontier_heldout_reference.csv`.

Env.	Budget	B	q	Mixture usage	Mixture reward
Tiger	0.25 of range	4.72	0.25	4.64 ± 0.10	5.36 ± 0.10
Tiger	0.50 of range	5.07	0.50	5.04 ± 0.14	4.96 ± 0.14
Tiger	0.75 of range	5.42	0.75	5.35 ± 0.14	4.65 ± 0.14
Tiger	gap	5.07	0.50	5.04 ± 0.14	4.96 ± 0.14
Tiger	below $U_{\min}$	2.18	–	declared unattainable	
Diagnosis	0.25 of range	7.72	0.47	7.70 ± 0.25	-2.14 ± 0.48
Diagnosis	0.50 of range	9.53	0.94	9.47 ± 0.20	-2.11 ± 0.67
Diagnosis	0.75 of range	11.35	0.46	11.47 ± 0.37	-2.31 ± 0.45
Diagnosis	gap	7.84	0.50	7.86 ± 0.25	-2.06 ± 0.47
Diagnosis	below $U_{\min}$	2.95	–	declared unattainable	
Bandit	0.25 of range	5.51	0.18	5.62 ± 0.21	6.06 ± 0.26
Bandit	0.50 of range	7.79	0.29	7.67 ± 0.37	5.70 ± 0.25
Bandit	0.75 of range	10.07	0.17	9.95 ± 0.36	4.95 ± 0.22
Bandit	gap	4.71	0.50	4.62 ± 0.07	5.62 ± 0.21
Bandit	below $U_{\min}$	1.62	–	declared unattainable	

over the condition alone), a property that Appendix T.2 states and verifies by unit test.

Planning+IG is swept over a sixteen-point log weight grid (5 seeds $\times$ 100 episodes) with extra points inserted inside the one coarse step on which distractor spending first appears, so the onset, the first weight at which the agent buys any distractor test, is located to a fraction of that step. The distractor fraction, the share of total sensing usage (all observation actions in an episode) that goes to the distractor test, is exactly 0 for $w \leq 3.5$, is 0.103 ± 0.001 at $w=4$, reaches 0.233 ± 0.003 by $w=5$, and saturates in the 0.330–0.332 range for $w \geq 31.6$ (seed-level SE 0.004–0.005, Figure 10). The onset therefore lies in the bracket (3.5, 4.0].

Ordinary information gain does not distinguish reward-relevant from reward-irrelevant uncertainty, so once the weight makes the distractor’s nonzero but task-useless information gain worth its cost, Planning+IG spends budget on it. EFE at the canonical $w=1$ stays at zero distractor fraction on this instance only because the onset bracket (3.5, 4.0] lies above 1, not because EFE is structurally immune. The root cause is that $I(b)$ is computed over the full joint belief rather than over the quantities the pending commit decision depends on.

Table 34. The calibrated family (continued). Held-out reward comparisons within the calibration grid. Best member is the calibration-grid weight with the highest calibration reward among those whose calibration usage is at or below B . Target mixture is the reward-maximizing mixture of calibration-grid weights whose calibration usage equals B , and feasible mixture the same with usage at most B , each a linear program with at most two support weights, selected on the calibration data and run on the held-out seeds.

Env.	Budget	Best member	Target mixture	Feasible mixture
Tiger	0.25 of range	5.68 ± 0.08	5.36 ± 0.10	5.68 ± 0.08
Tiger	0.50 of range	5.68 ± 0.08	4.96 ± 0.14	5.68 ± 0.08
Tiger	0.75 of range	5.68 ± 0.08	4.65 ± 0.14	5.68 ± 0.08
Tiger	gap	5.68 ± 0.08	4.96 ± 0.14	5.68 ± 0.08
Tiger	below $U_{\min}$	declared unattainable		
Diagnosis	0.25 of range	-2.86 ± 0.43	-2.14 ± 0.48	-2.14 ± 0.48
Diagnosis	0.50 of range	-2.86 ± 0.43	-2.11 ± 0.67	-2.11 ± 0.67
Diagnosis	0.75 of range	-2.22 ± 0.68	-2.31 ± 0.45	-2.22 ± 0.68
Diagnosis	gap	-2.86 ± 0.43	-2.06 ± 0.47	-2.06 ± 0.47
Diagnosis	below $U_{\min}$	declared unattainable		
Bandit	0.25 of range	6.16 ± 0.23	6.06 ± 0.26	6.16 ± 0.23
Bandit	0.50 of range	6.16 ± 0.23	5.70 ± 0.25	6.16 ± 0.23
Bandit	0.75 of range	6.16 ± 0.23	4.95 ± 0.22	6.16 ± 0.23
Bandit	gap	5.51 ± 0.10	5.94 ± 0.19	5.94 ± 0.19
Bandit	below $U_{\min}$	declared unattainable		

The repair scores information gain on the reward-relevant marginal, $\tilde{I}_a(b) = \mathbb{E}_{o \sim Q(\cdot|b,a)} D_{\text{KL}}[Q(s_{\text{rel}} | b, o, a) \| Q(s_{\text{rel}} | b, a)]$, restricted to the state components s_{rel} the commit reward depends on. This is the ordinary expected information gain $I_a(b)$ of Section 3 evaluated on the belief over s_{rel} instead of the full joint belief, so the only change is which belief the epistemic term is scored on.

In the implementation, s_{rel} is the reward-equivalence class of the state. Two states are reward-equivalent when every commit action pays the same in both, so the partition into classes is read directly off the commit reward matrix the agent is given (Section 3), and information gain is scored on the class marginal. On `DistractorDiagnosisEnv` the two states that share a condition and differ only in the nuisance bit have the same commit-reward column, so the four classes are exactly the four conditions. The finite-horizon lookahead still propagates the full joint posterior, and on an environment where no two states are reward-equivalent the variant reduces exactly to ordinary information gain.

Sweeping both agents over the same refined grid makes the effect a paired curve (Figure 10). The relevance-weighted agent takes zero distractor tests at every swept weight, against 6.46 tests and 33.2% of total usage for ordinary information gain at $w=100$, and recovers the reward that ordinary information gain gives up by buying them. Listing ordinary information gain’s mean reward

Table 34. The calibrated family (continued). Constrained-reference comparisons. The reference is the estimated feasible envelope of the near-optimal constrained-POMDP reference of Section 6.7, the largest reward attainable by mixing its sampled points subject to expected usage at most B , flat at the largest sampled reward where B exceeds the largest sampled usage. That reference was evaluated on the canonical seeds with once-per-seed seeding. Same-stream reference is the same envelope with every sampled reference policy re-evaluated on the held-out seeds under the family’s per-episode seeding, and paired gap is the mixture’s reward minus that envelope, per seed, with its seed-level SE. Reference support and mixture weights are fitted on those same held-out seed means and held fixed for the SE. The nominal pointwise plug-in intervals discussed in the text omit uncertainty from that fit and cap feasibility, and make no multiplicity adjustment. Target reference is the best mixture of sampled reference policies, including subsidized ones with negative usage penalties, whose held-out usage equals B , and target gap is the paired held-out gap against it (`results_budget_frontier_target_reference.csv`).

Env.	Budget	Reference	Same-stream reference	Paired gap	Target reference	Target gap
Tiger	0.25 of range	5.02	5.68	-0.33 ± 0.04	5.28	0.08 ± 0.04
Tiger	0.50 of range	5.02	5.68	-0.72 ± 0.06	4.93	0.03 ± 0.06
Tiger	0.75 of range	5.02	5.68	-1.03 ± 0.06	4.58	0.07 ± 0.06
Tiger	gap	5.02	5.68	-0.72 ± 0.06	4.93	0.03 ± 0.06
Tiger	below $U_{\min}$	declared unattainable				
Diagnosis	0.25 of range	-2.00	-2.11	-0.03 ± 0.28	-2.11	-0.03 ± 0.28
Diagnosis	0.50 of range	-1.35	-1.35	-0.76 ± 0.15	-1.35	-0.76 ± 0.15
Diagnosis	0.75 of range	-1.27	-1.21	-1.10 ± 0.22	-2.26	-0.05 ± 0.16
Diagnosis	gap	-1.95	-2.06	0.00 ± 0.22	-2.06	0.00 ± 0.22
Diagnosis	below $U_{\min}$	declared unattainable				
Bandit	0.25 of range	6.31	6.27	-0.21 ± 0.12	6.22	-0.16 ± 0.12
Bandit	0.50 of range	6.31	6.27	-0.57 ± 0.09	5.78	-0.08 ± 0.06
Bandit	0.75 of range	6.31	6.27	-1.32 ± 0.08	4.90	0.05 ± 0.04
Bandit	gap	6.23	6.21	-0.59 ± 0.20	6.21	-0.59 ± 0.20
Bandit	below $U_{\min}$	declared unattainable				

first, the pairs are -2.45 against -1.45 at $w=4$, -4.31 against -1.45 at both $w=5$ and $w=10$, and -10.48 ± 0.11 against -3.64 ± 0.36 at $w=100$ (seed-level SE). Success rates are identical through $w=31.6$, and the variant’s is nominally higher at $w=100$ (99.2% against 98.4%, not significant). Seed-level Welch tests with Holm–Bonferroni correction over the weight-by-metric family (every swept weight and reported metric, corrected together) confirm the usage and distractor-count differences at every weight past onset and the reward difference at $w=31.6$ and $w=100$ ($p=1.2 \times 10^{-5}$ and 1.5×10^{-5}), but not the smaller reward differences at $w=4$, 5, and 10.

Information-directed sampling (IDS) [48], which is explicitly reward-aware through its information-ratio numerator, the squared expected regret of taking the observation action (Appendix U.4), is not immune either, for an implemen-

tation reason. Its denominator, information about the optimal commit (`info_astar` in the implementation), is correctly zero for the distractor action, and the ratio is then undefined. The implementation falls back to a raw state-entropy reduction (`info_state`) whenever `info_astar` is at or near zero, which lets the distractor’s nuisance-bit informativeness through. IDS’s distractor fraction is 0.305 ± 0.005 (seed-level SE), measurably below Planning+IG’s saturated 0.330–0.332 (seed-level SE 0.004–0.005, roughly a 3-SE gap) but well above zero. This is evidence against that implementation pattern, which reintroduces the reward-blindness the reward-aware measure was meant to avoid, not against reward-aware information measures in general. Neither ordinary information gain nor this IDS adaptation is a clean immune baseline against reward-irrelevant sensing on this benchmark.

The relevance-weighted variant has no such fallback. It scores ordinary information gain on a coarsened state space, so a test uninformative about the reward-equivalence classes earns exactly zero. Appendix N reports the IDS baseline in full on the observe-then-commit environments.

X Extended EFE Comparisons

These analyses retain the full per-environment comparisons and behavioral qualifications summarized in Section 7.

X.1 Core Environments

Table 5 compares four of the six core agents of Table 1 on Tiger, Diagnosis, and Bandit, the EFE agent at its derived weight $w=1$, the reward-only baselines Myopic ($H=1$) and same-horizon Planning, and Planning+IG at its success-tuned weight w_{succ}^* . Together they test whether EFE’s untuned epistemic term buys anything over reward-only planning at the same depth, and whether it gives anything up against a tuned information bonus.

Table 5 uses seed-level reward SE, whereas Appendix B reports pooled episode-level SE. The committed CSVs carry both. Pooled SE is smaller for some cells and larger for others, depending on within-seed and between-seed variation. The full tables omit untuned Info Gain rows that are bit-identical to Myopic and Tiger’s pymdp-AIF row, which is bit-identical to Posterior-vote on all reported metrics (`results_tiger.csv`).

On Tiger (single observation action), EFE and same-horizon Planning coincide exactly, a measured fact rather than a structural necessity. At $w=1$ the epistemic term is in reward units, one reward unit per bit (Section 3.3), and because Tiger’s hidden state takes only two values one listen gains at most one bit, so it moves an action’s value by at most one reward unit. At Tiger’s stakes (+10, −100) that margin never flips a decision that reward does not already dictate in these runs, and since both agents run the same finite-horizon recursion over the same belief with the same fixed tie-break rule, they select the same actions. The trajectories are bit-identical (Table 5, confirmed by `results_tiger_stats.csv`,

where the difference is 0 with $p=1$ and $d=0$ on every metric). With no weight selection at all, EFE reaches within a fraction of a percentage point of the success rate that Planning+IG attains only after a weight search.

The Epistemic-only ablation (Table 1) commits immediately rather than over-exploring. On Tiger, Diagnosis, and Bandit alike it takes zero observations and succeeds at each environment’s chance level (49.7%, 24.7%, and 24.7% respectively), and on Tileworld 6×6 it likewise commits immediately and matches Myopic’s chance-level performance exactly (2.9%, Appendix B). Pure information gain without reward alignment produces degenerate non-exploration here, not the catastrophic over-exploration a naive information-seeking ablation might suggest. Either way, the pragmatic term is essential.

On Diagnosis and Bandit, EFE Pareto-dominates same-horizon Planning with the seed-level tests of Section 7, while success-tuned Planning+IG ($w^*=50$ on Diagnosis, $w^*=20$ on Bandit) reaches near-ceiling success at a reward cost from over-exploration (Table 5). EFE is the competitive untuned trade-off rather than uniquely best on every metric.

Posterior-vote, which draws 100 states from the belief and takes the majority vote over each sample’s optimal action with no information-gain term, is the closest competitor among agents that do not tie EFE exactly. On Bandit it is not significantly different from EFE on any metric (reward +6.32 against +6.27, seed-level Welch $p=0.58$, success 87.0% against 86.9%, $p=0.97$, inspections 5.01 against 5.11, $p=0.30$), and on Testbed it ties reward-only Planning exactly and beats EFE. It does not transfer. On Diagnosis it tracks Planning rather than EFE (−2.78 against EFE’s −1.37), and on Tiger it is well behind (+3.79 against +5.19). Bootstrap 95% CIs (10,000 resamples over 5,000 episodes, Appendix I) confirm non-overlapping Bandit reward intervals between EFE ([6.18, 6.37]) and Planning ([5.64, 5.86]).

Effect sizes clarify where the differences lie, using the conventional interpretation bands listed in Table 17. On reward, Cohen’s d between EFE and same-horizon Planning is negligible on Tiger, Diagnosis, and Bandit, exactly 0 on Tiger, where the two agents tie, and $|d| < 0.15$ at the pooled episode level on Diagnosis and Bandit, where both agents already achieve reasonable returns once they explore enough. The gap that remains is whether an agent gathers enough evidence before it commits, and it shows in success rate, where d is around 0.33 on Diagnosis and 0.40 on Bandit (Appendix I), values in the small band. EFE raises success rate, the objective that Planning under-serves, without sacrificing reward, the Pareto dominance above in effect-size terms.

Two statistical conventions govern how we read those numbers. First, with a balanced design, meaning equal episodes per seed for both agents in a comparison, the seed-level and pooled point estimates of the mean difference coincide exactly, so agreement in sign is guaranteed by construction rather than an independent check. What the seed-level test adds is a significance decision computed from only 5 independent replications instead of 5,000 pseudo-replications. On that stricter test, among the EFE against Planning comparisons on the three core environments, every pooled comparison that was significant after Holm–

Bonferroni correction, which is Diagnosis and Bandit on both metrics, remains significant at the seed level. The one negligible pooled comparison in that set, Tiger EFE against Planning, is non-significant at both levels.

Second, seed-level Cohen’s d magnitudes are not comparable to the pooled values, since they are computed from the standard deviation of only 5 seed means per group. That much smaller and noisier denominator mechanically inflates $|d|$ by roughly one to two orders of magnitude, factors of about 6 to 90 across the committed comparisons (Appendix I and the committed `*_stats.csv` files, which carry a seed-level d column for completeness). We therefore do not interpret seed-level d magnitudes and rely on the seed-level p -values and confidence intervals above.

Large reward effects ($|d| > 0.8$) appear where Planning fails to explore sufficiently. The clearest case is the 8×8 Tileworld grid of Section 7.3, where at the lookahead horizon used there reward-only Planning takes no scans at all. Tileworld 8×8 gives pooled episode-level $|d| = 1.13$ (`results_tileworld_8x8_stats.csv`). Against the over-exploring success-tuned baselines, Info Gain and Planning+IG at their tuned weights, the largest reward effect across the tested environments is medium, $d=0.69$ on Testbed (Table 17). At the 6×6 grid that Table 35 reports, the same EFE against Planning comparison is negligible at $d \approx 0.01$, since Planning and EFE scan similarly often at that smaller scale.

X.2 Pareto Analysis and the Information-Unit Weight $w=1$

By Proposition 1, EFE is exactly Planning+IG with $w=1$, so $w=1$ is a derived default rather than a tuned choice. This subsection extends the eleven-weight sweep of Section 7.2 on Tiger, Diagnosis, Bandit, Tileworld, and the two-state Testbed of Appendix C (Figure 11), where that section identifies the reward-maximizing tied runs and the two exceptions. RockSample and Structural Inspection are not part of the sweep and are discussed in Sections 7.4 and 7.5.

The success-maximizing weight lies at the sweep’s upper endpoint, $w=200$, on all five environments, so $w=1$, although near-optimal for reward on three of them, is not near-optimal for success rate, and agents requiring near-certain accuracy (e.g., safety-critical applications) would benefit from higher weights at a reward cost. On Tiger and Diagnosis, moving to $w=50$, the success-tuned Planning+IG weight of Table 5, buys at most 1.8pp of accuracy while costing 1.07 and 2.41 reward, all of it at the single grid step to $w=50$. The grid does not sample between adjacent tested weights, so a cheaper untested weight is not ruled out. Bandit trades more evenly, gaining 9.8pp success for 0.41 reward at $w=5$, and Tileworld’s reward peaks at $w=20$, inside the swept range, and falls again at larger weights. On Testbed the tied reward-maximizing run sits at the sweep’s low-weight boundary, where $w=0$ Planning matches it. Within the same eleven-weight grid, the reward-maximizing run therefore moves from at most 0.5 on Testbed to 20 on Tileworld. On the three environments where it attains the swept reward maximum, the derived weight replaces that grid search.

X.3 Spatial Epistemic Foraging on Tileworld

To test whether EFE’s advantage extends to larger state spaces, we turn to Tileworld, the spatial generalization of Diagnosis defined in Section 5, and vary its grid size N below. Tileworld projects the Diagnosis partition structure onto an $N \times N$ grid ($|S|=N^2$), producing spatially interpretable belief evolution (Figure 25, with step-by-step belief strips in Appendix D.4).

Table 35. Tileworld 6×6 (2,500 episodes at 500 per seed $\times$ 5 seeds, $H=2$). Tuned weight $w_{\text{succ}}^*=50$. Reward uncertainty is seed-level SE. Bold marks EFE and any agent not significantly different from it at the seed level (Welch $p > 0.05$, uncorrected), not the single best value per column.

Agent	Scans	Success	Reward
Myopic	0.00	2.9%	-48.25 ± 0.33
Planning ($H=2$)	15.64	73.8%	-21.39 ± 0.14
Planning+IG ($w=50$)	32.25	97.3%	-23.85 ± 0.19
EFE ($H=2$)	14.75	72.0%	-21.53 ± 0.57

EFE and reward-only Planning are not significantly different on reward (-21.53 vs. -21.39 , seed-level Welch $p=0.82$) or success (72.0% vs. 73.8%, $p=0.17$), failures to reject rather than a TOST-backed equivalence claim as in Section 6.6, but EFE uses significantly fewer scans (14.75 vs. 15.64, seed-level $p=8.8 \times 10^{-5}$, pooled $p < 10^{-13}$). Planning+IG’s 32.25 scans, more than double EFE’s, erode reward despite 97.3% success, the over-exploration pattern of Diagnosis and Bandit at a larger scale. Figure 25 shows the mechanism. EFE concentrates belief by partition-based narrowing and commits once confident, Planning lacks scan-selection guidance, and InfoGain-Tuned keeps scanning past the point of diminishing returns. InfoGain-Tuned, the figure’s bottom row, is the myopic Info Gain agent of Table 1 at its own success-tuned weight, a different agent from the tree-search Planning+IG row of Table 35, which adds the same bonus at every node of its lookahead tree.

Spatial scaling. We scale the grid from 4×4 to 8×8 with five agents, Myopic, Planning, InfoGain-Tuned, Planning+IG, and EFE (Figure 12). The sweep is a separate run from the seven-agent Tileworld battery of Table 35, at the smaller episode count stated in the figure’s caption, and it omits that battery’s untuned Info Gain ($w=1$) and Epistemic-only agents, whose full seven-agent tables at 6×6 and 8×8 are in Appendix B. At 8×8 ($|S|=64$), reward-only Planning collapses to $1.4\% \pm 0.3\text{pp}$ success while EFE maintains $69.4\% \pm 1.0\text{pp}$ (seed-level SE). There reward-only Planning takes zero scans and is bit-identical to Myopic on every outcome metric (0.00 scans, 1.4% success, -49.16 reward for both). A depth-2 lookahead sees at most two scans before it must value a commit at its leaf, and after two binary scans a belief that began uniform over 64 cells is still spread over many cells, so the commit valued there stays close to chance and its

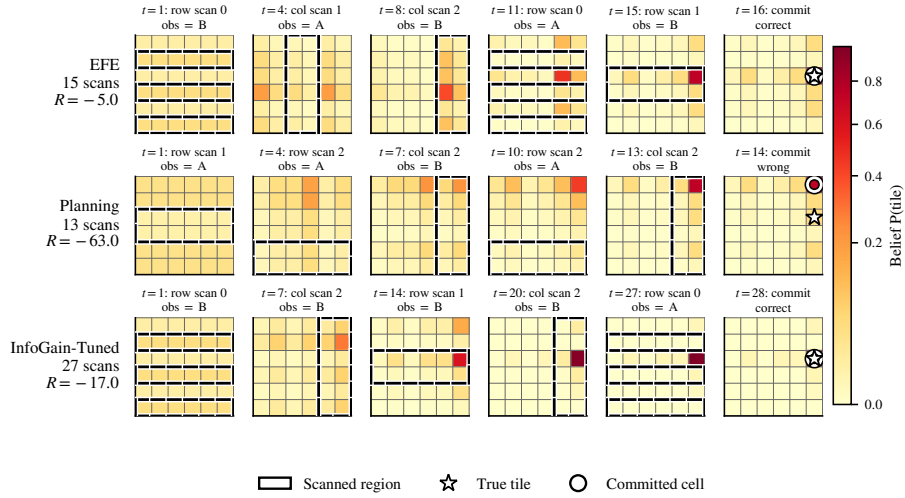


Fig. 25. Agent comparison on the same 6×6 Tileworld episode. All agents start from the identical uniform prior, stated here rather than drawn once per row. Belief probability is encoded by the shared colorbar on a square-root color scale. Black outlines mark the cell group queried by each scan, each panel’s title reports that scan’s noisy binary readout as obs=A or obs=B naming which half of the queried group the signal favors, the black-edged white star marks the true tile, and the black-stroked white ring marks the committed cell. Scan panels are subsampled evenly within each row, so columns are not time-aligned across rows and each panel’s t gives the true step. EFE (top) commits correctly with efficient scanning. Planning (middle) commits on the wrong cell after 13 scans. InfoGain-Tuned (bottom) over-scans well past the point of diminishing returns.

expected improvement over committing at once is smaller than the scans cost. The comparison at this scale is therefore EFE against a non-scanning agent, a joint horizon-and-scale effect at the fixed $H=2$ of this sweep, not evidence about reward-only planning at deeper horizons, where the scan would begin to pay for itself.

Across scales, the gap between EFE and Planning is a threshold at 8×8 rather than a gradual widening. At 4×4 and 6×6 , EFE’s reward and success are within sampling error of reward-only Planning’s, within 1 SE at both scales, with Planning nominally slightly ahead on both. No formal test was run for this specific sweep. EFE’s advantage opens up only at 8×8 , exactly where Planning’s success collapses.

Figure 12’s two weighted series are not tuned the way the main tables’ weighted agents are. Planning+IG at $w=100$ achieves 97.8% success but at substantial reward cost from extensive scanning. The scaling sweep fixes $w=100$ for both weighted agents, InfoGain-Tuned and Planning+IG, rather than tuning each within its own agent class as the main tables do, so the figure is a fixed-weight comparison, and its InfoGain-Tuned series keeps that label from the main battery only as a series name.

Observation structure sensitivity. To test whether EFE’s Tileworld result depends on the bit-level partition of scan actions, we replace the deterministic row/column splits with random and overlapping partitions at the 6×6 scale of Table 35. At that scale the advantage is the scan-count saving, so scan counts are the primary readout, with reward and success reported alongside. Under random partitions each scan assigns the cells at random to two equal halves, so the six splits no longer correspond to separate index bits and their answers are no longer guaranteed to combine into a unique cell. Under overlapping partitions each scan draws random weights for the row and column indices, scores every cell by the weighted sum of its two normalized coordinates, and splits the cells at a random threshold, so each scan cuts the grid along a random tilted line and different scans cut off overlapping regions, producing correlated and partially redundant observations (construction in `rho_aif/environments/tileworld.py`).

No partition mode separates EFE from Planning on reward or on success, while EFE uses fewer scans than Planning under all three. This battery is a separate run at 200 episodes per seed, so its bitwise figures differ slightly from Table 35 (6×6 , $H=2$, 5 seeds, `results_partition_sensitivity_6x6.csv`, `experiments/run_tileworld.py partition`). EFE against Planning, in reward $\pm$ seed-level SE with success in parentheses, is -21.62 ± 0.87 (71.9%) vs. -21.41 ± 0.64 (73.6%) under the bitwise partition, -29.31 ± 0.43 (56.1%) vs. -30.32 ± 0.66 (55.1%) under the random partition, and -47.53 ± 0.73 (8.5%) vs. -47.66 ± 0.68 (8.4%) under the overlapping partition. The largest reward gap is 1.01 under the random partition, at 1.28 seed-level SE of the difference (Welch $p=0.24$), and the nominal ordering changes direction across modes. On success the two are within sampling error under all three modes. On scans EFE uses fewer than Planning under all three (14.76 vs. 15.57 bitwise, 12.97 vs. 13.38 random, 2.63 vs. 2.70 overlapping).

The nominal reward gaps here are within sampling error and are not read as signal. What the battery does establish is a large and consistent effect of observation structure on every agent. Both alternative modes cut absolute performance sharply, from roughly -21 reward under the bitwise partition to -30 under random scans and -48 under overlapping ones, because less complementary scans leave less recoverable information for any agent to exploit. We therefore treat the Tileworld result as scoped to the structured partition it was measured on, and we do not claim this experiment establishes EFE’s advantage is model-independent.

X.4 RockSample as an Interleaved Observe-Act Setting

This subsection extends Section 7.4 on the cost-modified RockSample variant of Smith and Simmons [58] specified in Appendix V. The agent moves on a grid among rocks of hidden quality, and the actions are the four cardinal moves (N/S/E/W), one check per rock, sample, and exit. Proposition 3 covers the check and navigation actions, and Remark 1 states its boundary for state-altering sensing.

The agents under comparison fall into two groups. The first is the weighted family, meaning the Planning, Planning+IG, and EFE agents (Table 6 abbreviates Planning+IG as Plan+IG), which use depth-limited belief-space tree search over factored beliefs (independent per-rock). The search depth is the d in each row label of Table 6, depth 3 or 4 on the three smaller instances and depth 2 on RS[11,11]. All three share one hand-coded leaf evaluation (`rho_aif/agents/rocksample_agents.py`) that, at the depth limit, greedily plans the remainder of the episode. That leaf rule scores each rock not yet sampled by its belief-weighted sample value, the belief that the rock is good times the good-sample payoff plus the belief that it is bad times the bad-sample penalty. It considers only the rocks whose value is positive, moves to the one whose value net of the move cost of reaching it is highest and samples it, repeats, and exits when no such rock remains. Because these three agents share that leaf value, they differ only in the information gain weight w .

The second group holds the reference agents. POMCP is a search-tree solver, run at a configuration tuned once on separate seeds and then held fixed (Appendix M). The row labeled “Rollout only” runs POMCP’s approach-then-check rule standalone, with no search tree. That rule is POMCP’s rollout policy, the default policy that plays out each simulation once it leaves the search tree, and it is a hand-coded, unbounded-horizon rule rather than a depth-limited search. It walks to the unresolved rock with the highest belief per unit of distance (the nearest one at a uniform prior), checks it at close range where accuracy is 0.95, samples if the belief crosses a threshold, writes off any rock whose belief falls below 0.15, and exits when nothing is left. The prose below calls this standalone run the heuristic. Greedy, the remaining row, samples without checking.

Across the three smaller instances the reading is the same. EFE at $w=1$ is never significantly behind the $w=5$ comparator of Table 6, is ahead of it once, and is statistically tied with or below the heuristic. On RS[5,3], EFE and Planning+IG ($w=5$) are not significantly different ($+16.47$ vs. $+16.41$, seed-level

Welch $p=0.74$, a failure to reject, not a TOST-backed equivalence claim). Both are bolded in Table 6 as statistical ties with the best row, which is the heuristic at +16.67. On RS[7,4], EFE leads the weighted family outright (+15.96 vs. +14.56 for Planning+IG, $p < 10^{-5}$) and is statistically tied with the heuristic’s +15.85 ($p=0.65$, again a failure to reject rather than demonstrated equivalence). On RS[7,8], EFE and Planning+IG ($w=5$) are again not significantly different from each other (+21.85 vs. +23.76, $p=0.09$ uncorrected, not significant after Holm–Bonferroni), while both substantially outperform Planning (+12.31, $p < 10^{-4}$). On this instance the heuristic reaches $+28.33 \pm 0.82$ on the identical protocol. Both deficits against it survive the per-metric Holm correction (Planning+IG $p=2.4 \times 10^{-3}$, EFE $p=4.6 \times 10^{-4}$), so the table bolds the heuristic alone there. Planning, Planning+IG, and EFE all sit below that +28.33, so on RS[7,8] these are relative comparisons within a family that is jointly suboptimal.

More checking beyond $w=5$ never significantly helps on these three instances and twice significantly hurts, with $w=10$ falling to +20.55 on RS[7,8] ($p=0.003$, significant under the per-metric Holm family) and to +14.60 on RS[5,3] ($p=3.5 \times 10^{-9}$), while the same step is flat on RS[7,4] ($p=0.18$), where EFE at $w=1$ had already beaten both tuned weights (Appendix M). We ran only $w \in \{5, 10\}$ as tuned comparators on RockSample, not the nine-point grid used for the observe-then-commit environments (Section 3), so on RS[7,8], where $w=5$ attains the highest reward of the three weights we ran and is not significantly different from $w=1$, this identifies the best of three sampled weights rather than a grid-searched argmax, and we make no claim that $w=5$ is the reward-maximizing weight over a denser sweep.

The Bad and Good columns of Table 6 are the mean numbers of bad and good rocks sampled per episode. On RS[5,3], RS[7,4], and RS[7,8], EFE samples 0.01 to 0.06 bad rocks per episode against Greedy’s 1.51 to 3.98, with seed-level Welch $p < 5 \times 10^{-8}$ at every instance.

EFE’s gap over $w=0$ Planning, largest on RS[7,8] among the three smaller instances, does not extend to RS[11,11] ($|S|=2,048$), where EFE and Planning achieve identical reward (+13.23 each) despite that instance having the most rocks to check. Every tree-search agent vastly outperforms Greedy (+12.9 to +13.2 against -18.9), but nothing here differentiates EFE from reward-only planning. Every tree-search agent at its reported configuration converges to the same conservative pattern of checking one or two nearby rocks, sampling if a rock looks good, and exiting, so the weight on information gain moves reward by at most 0.06. This includes POMCP at its frozen setting (Appendix M). Moving from the table’s depth 2 to depth 3 does not change this. Rerunning at depth 3 (`results_rocksample_11x11_depth3_check.csv`) costs seconds and reproduces the same pattern within 1 SE, with Planning and EFE at +13.23 and Planning+IG at +13.17.

Two explanations fit that collapse, either no exploitable slack or slack that no depth-limited search can reach, and the heuristic in Table 6 settles which holds. Bound by no search depth, and walking up to a rock before spending a check on it, it scores $+28.66 \pm 0.83$ on RS[11,11] on the table’s 5 seeds and 100

episodes, where every tree-search agent, EFE and POMCP included, sits in the $+12.94$ to $+13.23$ band, and $+28.33 \pm 0.82$ on RS[7,8] against the best tree-search agent's $+23.76 \pm 0.58$. There is therefore a great deal of slack at 11 widely spaced rocks that the depth-limited belief-space search cannot reach. The Good column shows it in behavioral terms, since on RS[11,11] the heuristic samples 5.22 good rocks per episode while every tree-search agent samples about half a rock.

The mechanism is now identifiable. Recall that check accuracy decays with the agent's distance from the rock (Section 5), so a check is informative only when taken from close range. Reaching close range means walking to the rock first. At this instance the checks that could be informative are far from the agent, and a depth-2 or depth-3 tree cannot represent the walk that would make them informative. Within its horizon only a single check taken from the agent's own starting cell is worth making, so the weight on information gain has almost nothing to act on. The shared greedy leaf rule values a rock at its belief-weighted sample reward, which is exactly zero at the uniform prior because at belief 0.5 the $+10$ good-sample payoff and the -10 bad-sample penalty cancel. So no rock passes the rule's positive-value filter, and it recommends the exit. Because the informative checks lie beyond the depths we run, depth 3 leaves reward unchanged and every weight gives the same reward to within 0.06, with the higher weights buying only about half an extra check per episode.

One further detail of that leaf deserves disclosure. Its completion estimate charges move costs for a walk to the east edge before collecting the exit reward (`rocksample_agents.py`), while the environment pays the exit from any cell. This phantom travel cost deflates every depth-capped continuation relative to the exact value of exiting immediately, a bias toward earlier exits. The immediate exit escapes the deflation because an exit taken inside the tree at the current cell is valued at its true payoff, so only continuations that run on to the leaf carry the charge. Every agent in the family uses this same leaf rule, so the reported comparisons are conditional on it. But it means the leaf's exit recommendation is overdetermined here rather than resting on the zero prior expectation alone.

POMCP corroborates the mechanism from the other side. A deeper POMCP tree gains the lookahead to plan a walk out to the distant rocks, but, on our reading of its behavioral trace (Appendix M), its in-tree checks are still taken from a distance where accuracy is low, and its reward falls monotonically from $+12.94$ at $H=10$ to -54.78 at $H=40$. Reach alone, without walking up before checking, is not enough.

This is a limitation of the tree-search family at feasible depths on this instance, not evidence about information-gain weighting, and it bounds what Rock-Sample[11,11] can be asked to show about EFE. Per-instance detail, including the standalone heuristic on every instance, is in Appendix M.

The RS[11,11] collapse, the RS[7,8] advantage, and the Tileworld scaling result read as one mechanism seen at three points. EFE's advantage over reward-only Planning tracks how much slack a problem's structure leaves between the reward-only policy's default behavior and the information-optimal behavior that the agent's search can actually represent, not problem size or observation-action

count in themselves. On Tileworld, reward-only Planning scans too little to disambiguate a large grid, so directed information gain buys a large success-rate margin. On RS[7,8], eight rocks spread across a mid-sized grid leave real ambiguity about which are worth sampling. On RS[11,11] no check beyond the agent’s own cell is worth taking within the depths we run, so the weight has almost nothing to act on. We did not design instances to stress-test this reading, so it remains a mechanistic interpretation of the four RockSample instances and three Tileworld scales already reported rather than an independently validated scaling law.

X.5 Structural Inspection

This subsection extends Section 7.5 on the synthetic Structural Inspection POMDP specified in Appendix V. Each of the N components must be declared nominal or faulty while the agent stands on it, a declaration that cannot be revised, and declarations are interleaved with the moves and tests. Tests do not alter component states, so this is a factored observation POMDP and Proposition 3 applies. In Table 7, Missed is the mean number of faulty components per episode declared nominal, and Tests is the mean number of tests run per episode. As on RockSample, all tree-search agents here share one hand-coded leaf evaluation (`rho_aif/agents/inspection_agents.py`) that greedily completes the episode from the depth limit, so they differ only in the information-gain weight.

Section 7.5 summarizes Table 7. The seed-level tests behind that summary are as follows. On $N=8$, EFE’s +18.0pp accuracy gain over Planning has seed-level Welch $p < 10^{-10}$. On $N=16$, EFE’s reward deficit to Planning (−45.80 against −45.42) is not significant ($p=0.57$), so the two reward cells are bolded together. EFE’s +12.6pp accuracy gain over Planning has $p < 10^{-7}$. Planning+IG ($w=5$) runs 38.3 ± 0.05 tests against EFE’s 28.1 ± 0.09 (seed-level SE), roughly 36% more (seed-level Welch $p=1.7 \times 10^{-11}$).

Inspection’s payoffs also allow a heuristic reading of Proposition 2, whose thresholds depend on the reward asymmetry ratio $\alpha = |R^-|/R^+$ (not the reward scale that Section 6.1 also writes as α), the observation accuracy, and the cost. Declaring nominal pays +2 when correct and −50 on a missed fault ($\alpha = 25$ for that decision), while declaring a fault is symmetric (+5 when correct, −5 on a false alarm, $\alpha = 1$). Reading the two-state thresholds onto this multi-state environment is a heuristic reduction, as with the two-state-reduced rows of Table 31. On that reading the missed-fault decision places Inspection in the large-asymmetry regime of Part (a), where a heavy wrong-commit penalty drives the lower threshold w_{thresh}^* downward and $w=1$ clears the interval’s lower endpoint once that threshold is below zero. We do not evaluate the over-observation threshold of Part (b) for Inspection, so this places Inspection in the regime where Part (a) stops binding, not $w=1$ inside the full interval.

Y Additional Interpretation and Tree-Search Controls

This appendix retains the detailed weight-transfer, near-optimality, and tree-search analyses summarized in Section 8. The comparisons use the protocols and reference files stated locally. They describe the tested settings rather than general regime boundaries.

Y.1 The Information-Unit Weight and Tuning

EFE fixes the exploration–exploitation weight at $w=1$ rather than eliminating it (Section 3.3), and $w=1$ is tuned to neither criterion, unlike the reward-maximizing weight w_{ret}^* or the success-maximizing weight w_{succ}^* (10 to 50) used for Planning+IG in the main tables. That its measured reward sits inside the reward-maximizing tied run on three of the five swept environments is an empirical finding of the Pareto sweep (Section 7.2, Appendix X.2), not a design target. Applications that require near-certain accuracy may warrant a higher weight, which must still be tuned per environment, so EFE provides a principled starting point rather than a scale-free automatic calibration. The weight’s dependence on the reward scale and on the bit-versus-nat convention is set out in the reward-to-nats calibration discussion in Appendix U.1.

A further complication for tuning is that adding planning depth to a weighted IG bonus amplifies the weight’s effect, so the tuning problem becomes harder with depth. The same weight buys more sensing at depth two than at depth one, so the map from weight to sensing usage shifts with the horizon. A weight tuned at one depth need not carry over to another. One measurement supports this directly. At $w=100$ on Bandit under the canonical 5-seed protocol, Planning+IG at $H=2$ takes 12.36 inspections against myopic Info Gain’s 11.50 (seed-level Welch $p=0.003$, `experiments/run_bandit_w100_depth_comparison.py`).

The requirement for a sufficient planning horizon also has a discount-factor form. In the discount sweep (Appendix K, Table 19), EFE’s success advantage over Planning on Diagnosis and Bandit is significant at $\gamma \in \{0.99, 1.0\}$ and not at $\gamma \in \{0.90, 0.95\}$, because heavier discounting truncates the effective planning horizon below the number of observations needed for disambiguation. The sweep covers four discount factors on three environments, so it does not establish a general threshold, but it does show that applications whose time pressure mandates discounting as heavy as $\gamma=0.95$ on Diagnosis-like problems should not expect EFE’s epistemic drive to pay off.

Y.2 Zero-Shot Weight Transfer

Whether a tuned weight survives transfer to a new environment depends on the tuning criterion. Table 36 evaluates $w=1$, reward-tuned weights w_{ret}^* , and success-tuned weights w_{succ}^* across environments, all tuned within the Planning+IG class the table evaluates. Success-tuned transfer harms reward at the top of the weight grid. $w_{\text{succ}}^*=50$, the success-tuned weight for both Tiger and Diagnosis, yields +4.25 on Tiger, where it was tuned, and -0.38 on Testbed, where it was not,

versus $w=1$ at +5.33 and +0.37. Moderate weights ($w \in \{0.5, 1\}$) transfer well within the shared reward scale. On Tiger and Diagnosis, $w=1$ ties for best with $w=0.5$, Testbed’s $w_{\text{succ}}^*=10$, and Bandit’s $w_{\text{succ}}^*=20$, because the reward surface on those two environments is flat at moderate weights, and on Bandit $w=1$ is best outright. Moderate weights therefore transfer robustly under a shared reward convention whatever their provenance, while success-tuned weights overfit once pushed to $w=50$, the top of the tested grid. In that moderate band, $w=1$ is the variational default, not a scale-free optimum.

Table 36. Zero-shot weight transfer (500 episodes $\times$ 5 seeds, reward reported as mean $\pm$ seed-level SE, `results_transfer.csv`). Rows include $w=1$, reward-tuned weights (w_{ret}^*), and success-tuned weights (w_{succ}^*), the Planning+IG-class values recorded in `results_supplementary_tuned_weights.csv`, matching the evaluated agent class. Success-tuned transfer harms reward at the top of the grid, while moderate weights transfer well. Bold entries mark the best reward per column (Tiger and Diagnosis show a four-way tie among $w \in \{0.5, 1, 10, 20\}$, bit-identical, not merely within SE).

Weight	Tiger	Diagnosis	Bandit	Testbed
$w=0.5$ (w_{ret}^*)				
Diag./Testbed	+5.33 $\pm$ 0.26	-1.26 $\pm$ 0.34	+6.05 $\pm$ 0.11	+0.48 $\pm$ 0.02
$w=1$ (EFE / w_{ret}^*)				
Tiger/Bandit	+5.33 $\pm$ 0.26	-1.26 $\pm$ 0.34	+6.24 $\pm$ 0.09	+0.37 $\pm$ 0.01
$w=10$ (w_{succ}^*)				
Testbed	+5.33 $\pm$ 0.26	-1.26 $\pm$ 0.34	+5.25 $\pm$ 0.06	+0.01 $\pm$ 0.01
$w=20$ (w_{succ}^*)				
Bandit	+5.33 $\pm$ 0.26	-1.26 $\pm$ 0.34	+5.05 $\pm$ 0.07	-0.18 $\pm$ 0.01
$w=50$ (w_{succ}^*)				
Tiger/Diag.	+4.25 $\pm$ 0.08	-3.68 $\pm$ 0.14	+4.23 $\pm$ 0.06	-0.38 $\pm$ 0.01

When is $w=1$ near-optimal? Proposition 2 bounds the near-optimal weights in the two-state case between a lower threshold w_{thresh}^* , below which the agent commits without observing at $H=1$, and an upper threshold w_{over}^* , above which it buys a second observation whose instrumental value falls short of its cost. Appendix U.2 compares Table 31 with Part (a)’s prediction that large reward asymmetry drives the lower threshold below zero, which leaves $w=1$ inside the interval whenever the upper threshold exceeds 1. The high-asymmetry environments ($\alpha \geq 5$, namely Tiger, Diagnosis, and Tileworld) place $w=1$ interior to the interval, on the genuinely two-state row and, more heuristically, on the two-state-reduced rows, while Testbed places $w=1$ near the upper threshold, consistent with over-exploration.

The proposition’s two parts live at two horizons, Part (a)’s lower threshold at $H=1$ and Part (b)’s near-optimality interval at $H=2$. Appendix J asks what happens beyond them with a Monte Carlo study over 100 random two-state environments at $H \in \{1, 2, 3\}$ (Figure 20), using an empirical reward-gap criterion

rather than the closed-form interval. The fraction of environments in which $w=1$ passes that criterion rises from 30% at $H=1$ to 58% at $H=2$ and 79% at $H=3$, so about 21% still fail it at $H=3$.

Many real-world decision problems have $\alpha \gg 1$. In medical diagnosis, fault detection, and security screening, a missed condition costs far more than another test. This is the regime where the interval analysis is most favorable to $w=1$, and the multi-step study shows the near-optimality basin widening with planning depth. The practical consequence is that in these domains the information-unit weight is a strong untuned starting default under the shared reward convention, not a guarantee of near-optimality. When the default’s induced sensing differs from a stated expected-usage target, the budgeted machinery of Section 4 selects, within the single-price receding-horizon family, the weight or endpoint mixture whose expected usage meets that target.

Y.3 Approximate Planning with MCTS-EFE

At each depth the tree branches over K observation actions and then, per action, over its $|\mathcal{O}|$ observation outcomes, so the exact tree search cost is $\mathcal{O}((K \cdot |\mathcal{O}|)^H)$, limiting practical horizons to $H=2-3$ with $K \geq 3$. To push beyond this, we implement MCTS-EFE, a UCB1 tree search whose design choices we state in turn. The tree branches over observe and commit actions and, after each observe action, enumerates its $|\mathcal{O}|$ observation outcomes as child nodes. It adds the exact per-node information gain to the in-tree reward of each observe action, so that within the tree an observation is credited for what it reveals as well as charged for what it costs. Values propagate by max-backup, each node taking the maximum over its children’s values rather than the average of the returns sampled through it. Leaves are scored by an EFE-greedy rollout, which at each step takes the observation with the best one-step EFE value and credits the better of committing now and continuing to observe. The UCB1 exploration constant is left at its default, the commit reward range (110 on Tiger). That default is a static analog of the $R_{hi} - R_{lo}$ rule of Silver and Veness [55], which sizes the constant to the spread between the highest and lowest returns seen in preliminary sample runs of the search, whereas the default here takes the spread once from the commit reward table and never revisits it. On Tiger at $H=10$, MCTS-EFE with 500 simulations achieves $97.7\% \pm 0.5\text{pp}$ success (14–16 ms per episode across the three horizons tested, `results_mcts_efe.csv`), compared to $88.1\% \pm 0.2\text{pp}$ for POMCP at the same simulation count (11–12 ms per episode across horizons, seed-level SE, seed-level Welch $p=4.4 \times 10^{-6}$). The comparison is simulation-matched, not compute-matched, since MCTS-EFE’s leaf evaluations are more expensive per simulation. In this configuration, EFE’s closed-form information valuation, acting as the in-tree observe reward and inside the leaf rollouts, outperforms POMCP’s semi-informed rollouts (Appendix S).

An ablation of the two components that distinguish this search from a UCT tree, the in-tree information gain and the max-backup (Table 28, Appendix S), finds neither detectably load-bearing at five seeds after Holm correction. What remains to carry the contribution is the EFE-greedy leaf rollout together with

the exact belief representation and the exact commit-child values, which the ablation holds fixed. A commit child is the node reached by a commit action, and the search sets its value to the belief-expected commit reward computed directly from the belief instead of estimating it from sampled returns.

Appendix S addresses the two natural objections, that the POMCP gap reflects an untuned exploration constant or POMCP’s rollout policy (Table 29). Tuning the constant narrows the Tiger gap without removing it. With both solvers’ constants selected on tuning seeds disjoint from every evaluation seed, under a rule recorded before the runs reported there and written after an exploratory sweep on the canonical seeds had been inspected, MCTS-EFE stays significantly ahead on Tiger success and reward. Where the agent must choose among tests, neither the constant nor information-gain rollouts removes more than a small part of the gap, which stays above twenty-five points on Diagnosis and above eighty points on Tileworld-6×6 at every swept configuration. Because the EFE-greedy leaf evaluation, exact beliefs, and exact commit-child values were held fixed together in the sweep, the rollout check, and the ablation, we attribute the gap to that joint configuration and not to directed observation selection in isolation. Separating them would need a further matched-compute comparison that we have not run. The claim is not that EFE-based planning exceeds state-of-the-art POMDP solvers, but that the EFE objective supplies a principled, zero-tuning in-tree reward and leaf evaluation in place of hand-designed rollouts.

The approach also scales to multiple observation actions. For single-observation environments such as Tiger, the observation-outcome enumeration (2 outcomes per action) keeps expansion costs low, and the question is whether the same holds once the agent must choose among several tests. On Diagnosis ($N=4$, $K=2$ tests), MCTS-EFE at $H=5$ achieves $94.2\% \pm 0.5\text{pp}$ success compared to POMCP’s $65.5\% \pm 2.1\text{pp}$ at the same simulation count (200 simulations, seed-level SE, seed-level Welch $p=1.1 \times 10^{-4}$, `results_mcts_efe.csv`). To exercise the approach on a larger multi-observation environment, we evaluate on Tileworld 6×6 ($|S|=36$, $K=6$ scans). MCTS-EFE saturates at $95.4\% \pm 0.6\text{pp}$ success across both simulation budgets tested (200 and 500, seed-level SE), well above the $71.9\% \pm 1.5\text{pp}$ of Exact-EFE (the full recursive EFE agent, called EFE in the earlier tables, evaluated here within the same MCTS battery), but at a reward cost (-25.02 vs. Exact-EFE’s -21.62) from scanning substantially more (32.3 vs. 14.8 observations on average). MCTS’s sampled rollouts do not value information as precisely as the exact recursion, so it trades reward for a large success-rate gain by scanning more thoroughly. On Tiger and Diagnosis there is no such trade, since MCTS-EFE sits slightly below Exact-EFE on success and further below it on reward (97.7% against 99.9% and +3.73 against +5.65 on Tiger, 94.2% against 97.1% and -3.26 against -1.44 on Diagnosis). POMCP collapses on this larger observation menu regardless of simulation budget ($2.6\% \pm 0.4\text{pp}$ at 200 simulations, $3.9\% \pm 0.5\text{pp}$ at 500, seed-level SE, with 0.09 scans per episode). MCTS-EFE’s 91–93pp success-rate gap over POMCP (seed-level Welch $p < 10^{-9}$ at both budgets) therefore shows

that the gap persists at this scale, even though the specific reward/success trade-off MCTS-EFE strikes here differs from the smaller environments. With almost no scans, POMCP commits on a nearly uninformed belief, and its success sits near the $1/36$ rate of an uninformed guess. For larger observation spaces, more efficient tree policies, such as progressive widening or double progressive widening [5, 61], techniques that limit how many children a node expands as a function of its visit count, would further improve scalability. We leave this to future work.